

CineMerge Entertainment Hub



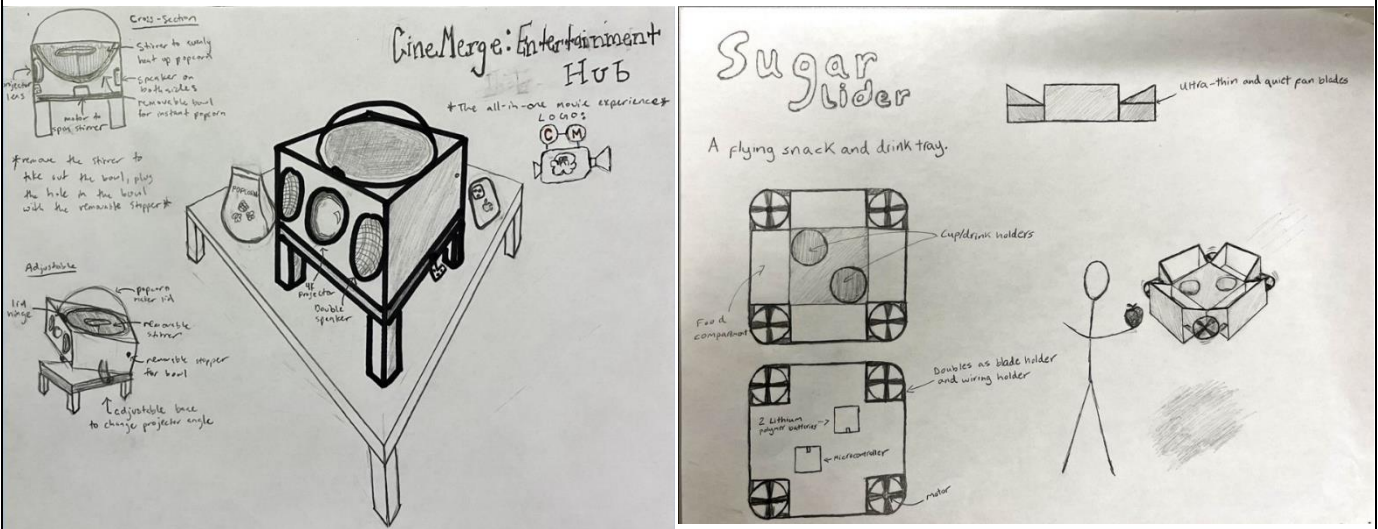
Diego Aponte
Benjamin Chamberlin
Joy D'Souza
Tallulah Selhorst
Kai Takeuchi

ME 1670
Section A
Group 10

**Original Project Proposal
and
Final Project Proposal Sketch**

ME1670

Final Project Proposal

Section / Team Number	Team 10
Team Name:	Couch Potatoes
Team Members:	Tallulah Selhorst, Joy D'Souza, Ben Chamberlin, Diego Aponte, Kai Takeuchi
Proposed Project Title:	The CineMerge Entertainment Hub
Problem Definition:	Imagine you sat down on your comfy couch after a long day at work. You get super comfortable with your blanket and are ready to watch your favorite movie, but you forgot to make your popcorn. Not to mention, you forgot that you sold your TV the day before, leaving you starving and without a TV. If only there was a way you could watch your movie and get your drinks and popcorn without leaving the comfort of your couch...  The sketches show two main components. The 'CineMerge: Entertainment Hub' is a box-shaped device with a screen on top, a handle, and four adjustable legs. It has a 'Double speaker' and a 'projector'. A 'CineMerge' logo is also present. The 'Sugar Slider' is a flying snack and drink tray. It has a 'motor', '2 lithium polymer battery', 'microcontroller', 'food compartment', 'cup/drink holders', 'beverages as blade holder and wiring holder', and 'ultra-thin and quiet fan blades'. A stick figure is shown holding the Sugar Slider.
Potential Parts List:	1. -adjustable protector stand/base 2. -4x adjustable legs

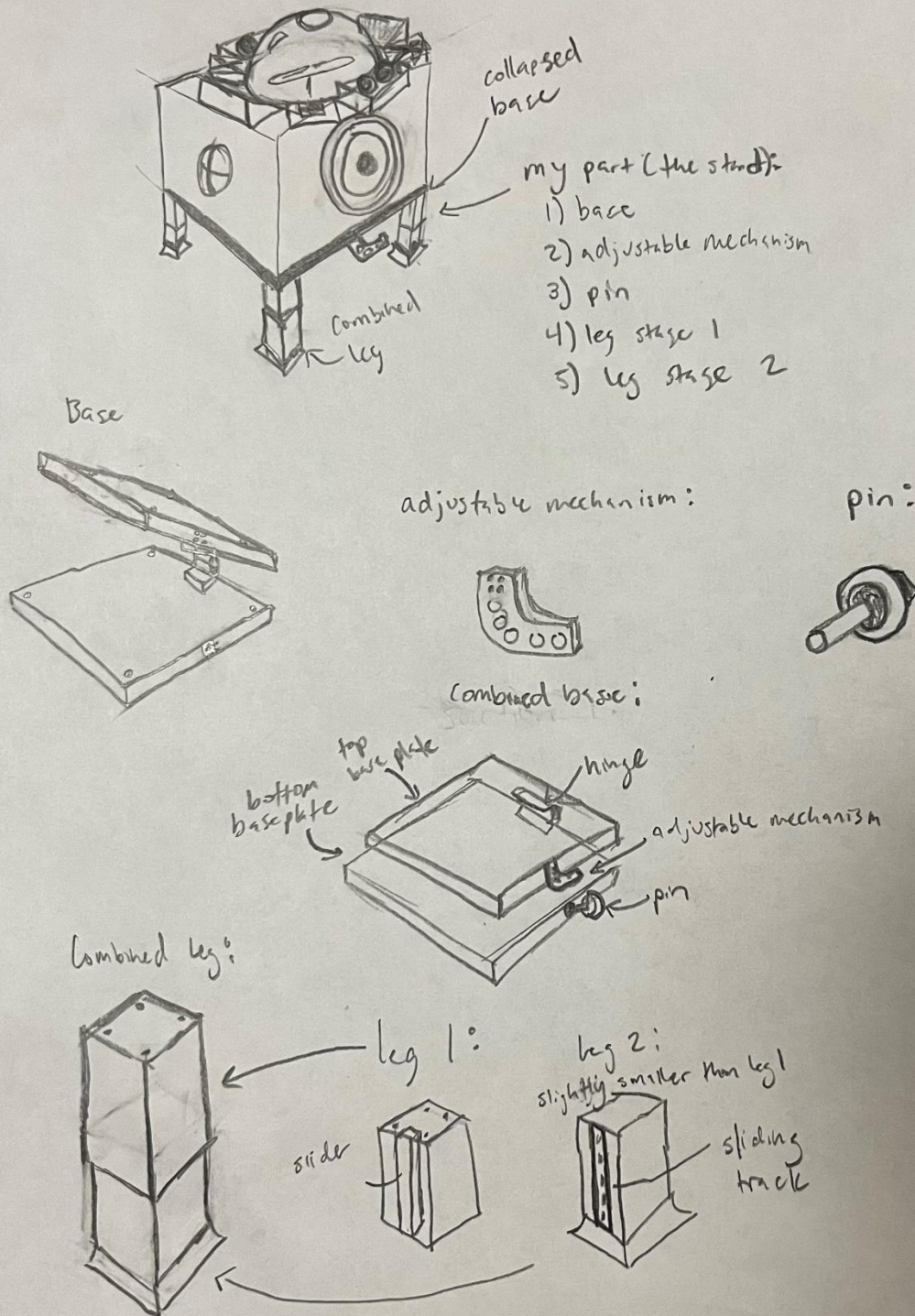
	3. -2x speakers 4. -projector lens 5. -popcorn maker motor 6. -removeable popcorn stirrer 7. -popcorn maker lid 8. -snack/food holder tray body 9. -4x fans 10. -4x fan holders 11. -2x cup holders 12. -2x covers for electronics at bottom of drone
Rationale for interest in this project topic choice:	Our group all enjoys watching movies and television shows, and we agree that getting up after you are already comfortable and settled is not ideal. Thus, the CineMerge idea was born. Not only can you save money by getting a movie experience without paying for actual movie tickets, but the drone will bring you any snack or drink you want.
Foreseeable Challenges/Difficulties in Modeling:	• Having enough complex parts for each member • The construction of putting all of the parts together • Prepping all parts for movement • Creating a sturdy enough design for the drones, extendable legs, and other adjustable parts
Argument for Level of Complexity/Difficulty of Chosen Project:	Difficulty Level: 4 Our project design has many moving parts that can attach/reattach, including the flying snack tray with a full-scale popcorn maker. While our main exterior is comprised of prisms and spheres, the interior, such as the electronics of individual parts as well as the body of the drone contains more complex shapes.

Preliminary Design

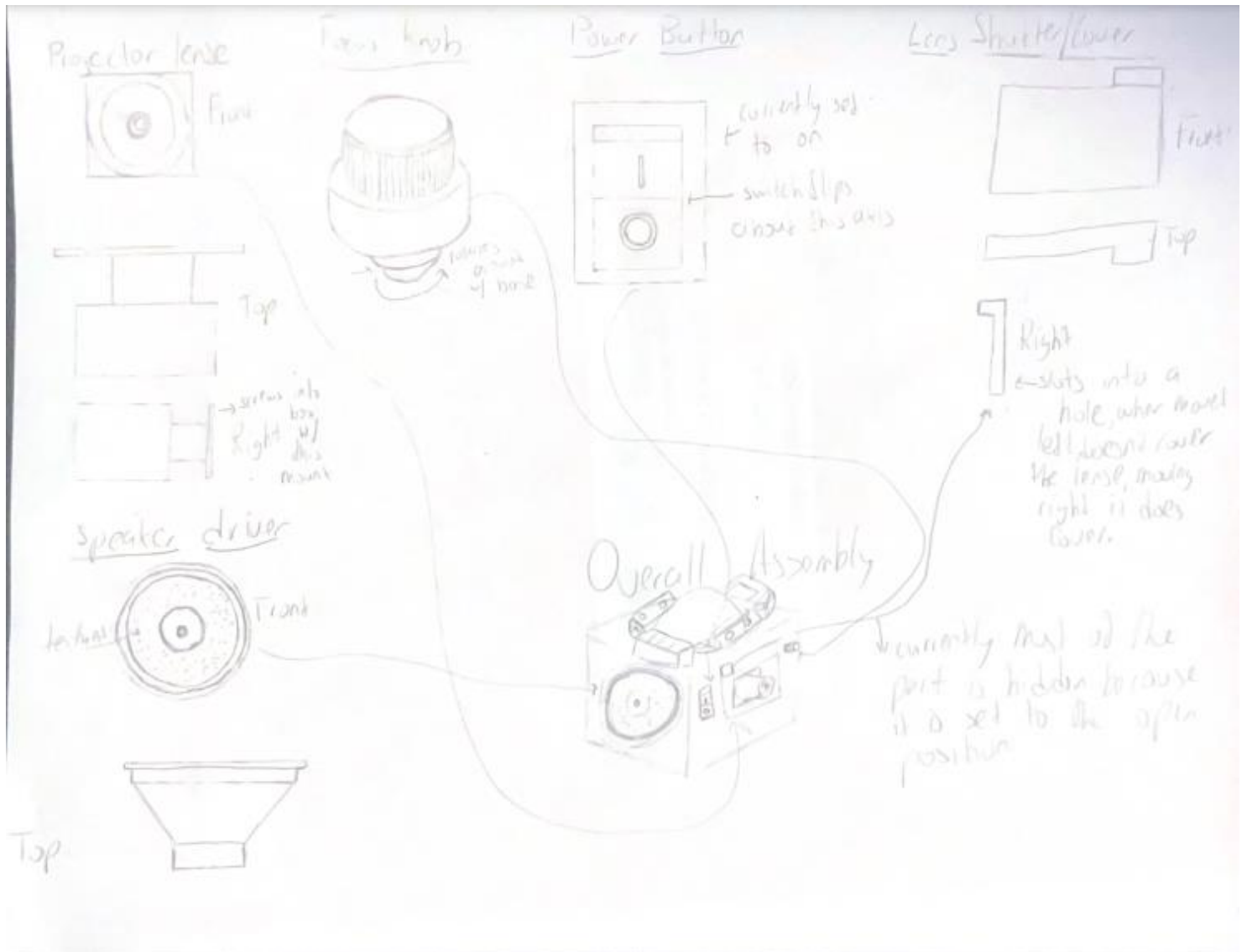
Part Sketches

Diego Aponte

CineMerge Entertainment Hub Base Close-Up:

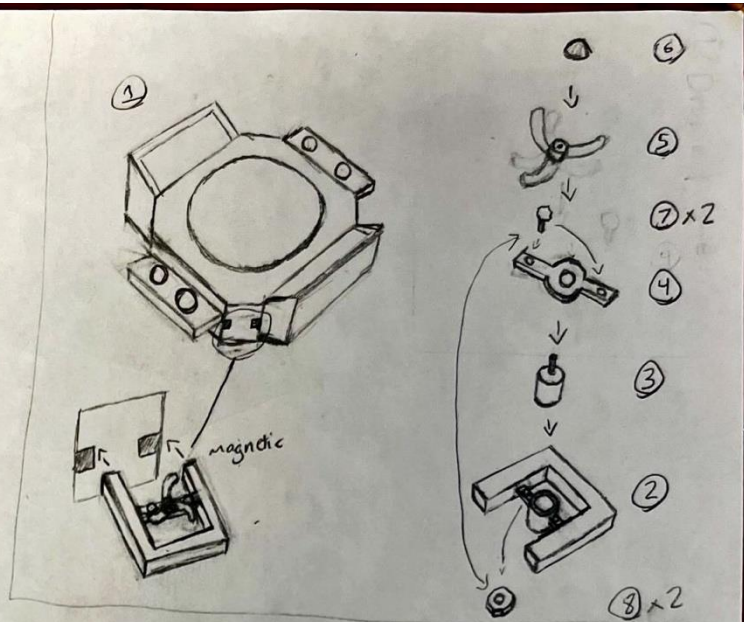
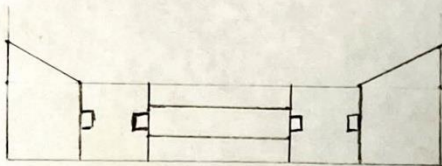
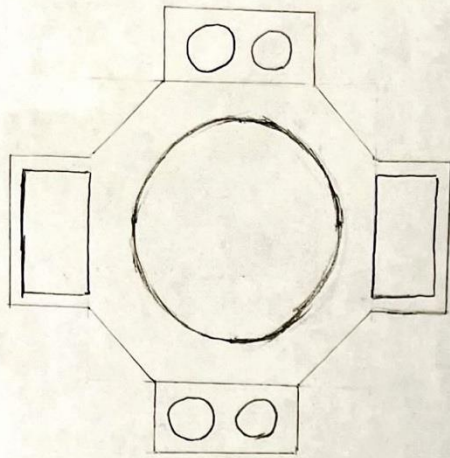


Benjamin Chamberlin

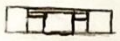


Joy D'Souza

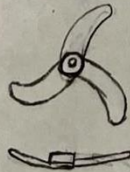
① Drone Body/Frame



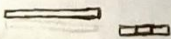
② Fan base (x4) ③ Motor (x4)



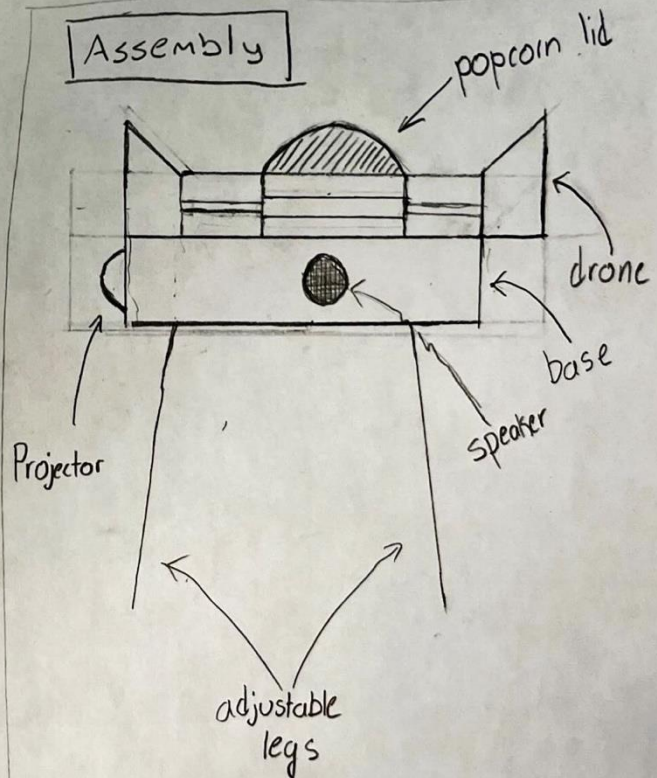
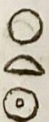
⑤ Fan blades (x4)



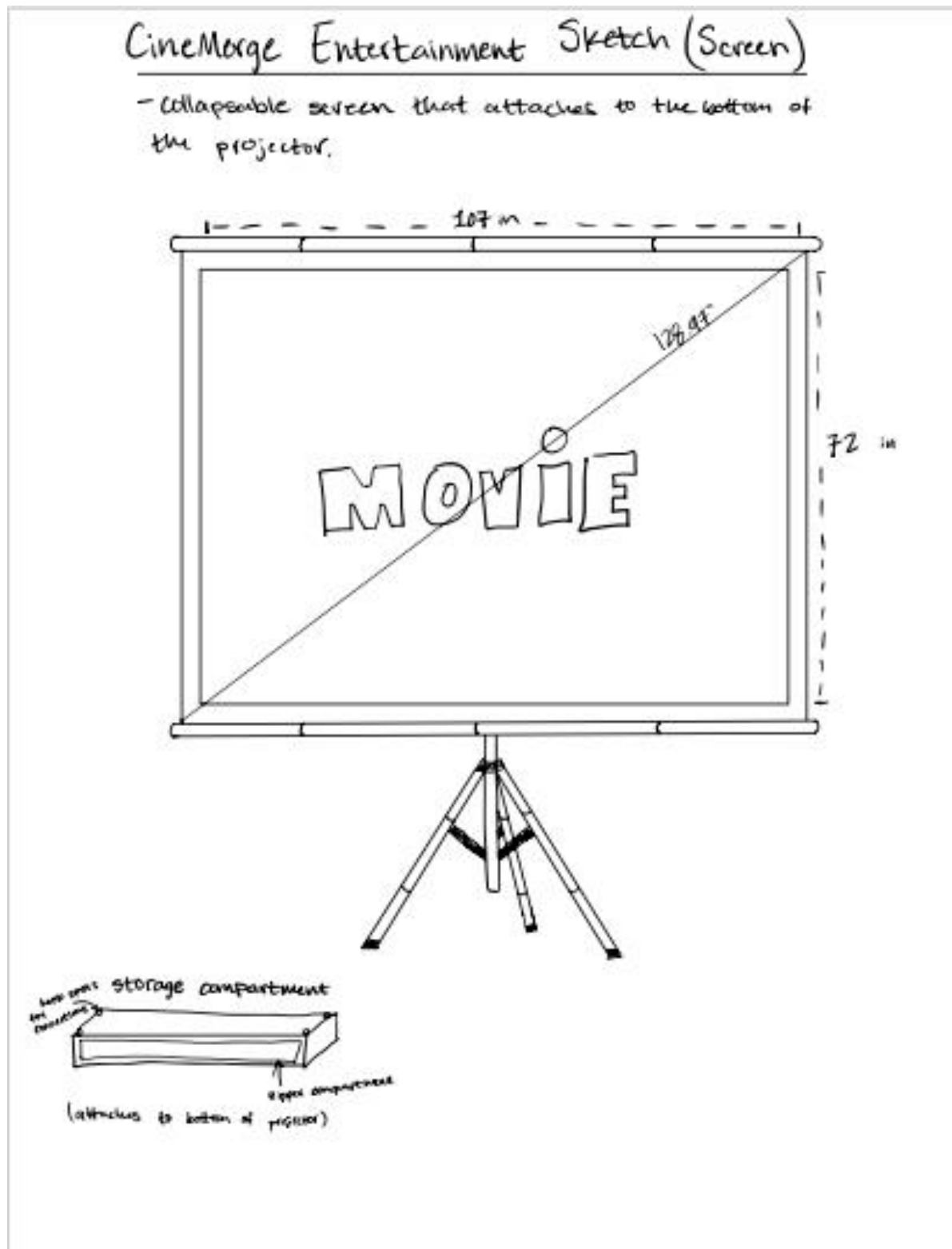
④ Motor Cap



⑥ Fan cap (x4) ⑦ Bolt (x8) ⑧ Nut (x8)



Tallulah Selhorst



Sheet screen



(just a sheet)

Storage bag



Connection point for bag to protection

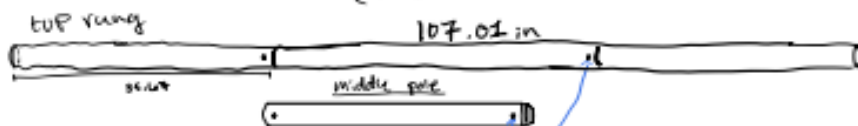
on eyelet on a vertical piece of fabric (on each corner)



Connection point on bottom of projector

top & bottom poles (collapsible)

(open)



latch holes for the poles. on the other two poles, this hole is empty if the middle pole has round extension

bottom rung (unchanged)



(on the spine)

attachment point for low rung

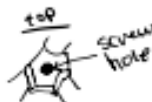
↑ attachment point for spine & stand portion

legs

adjustable



like music stand legs



screw hole

spine

(holds the sheet straight) upward 'spine'

24 in.

72 in.



screw for bottom

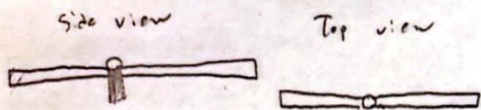
Kai Takeuchi

1. Glass Bowl/Barrier

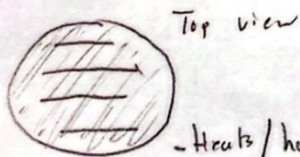


- Covers popcorn and holds from falling out.

2. Spinner

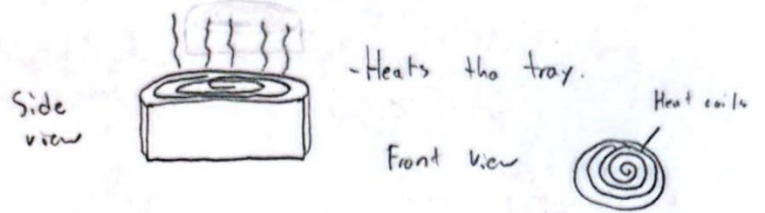


- Spins kernels to keep from burning.



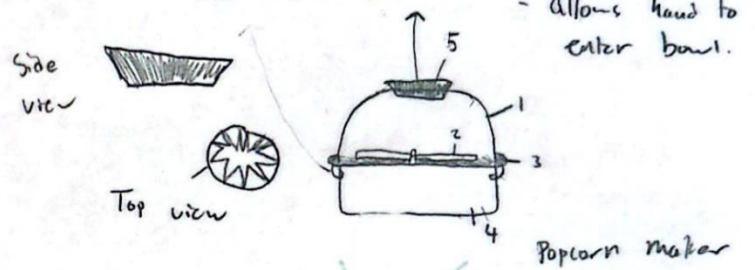
- Heats/holds pop corn.

4. Hot plate/ stove



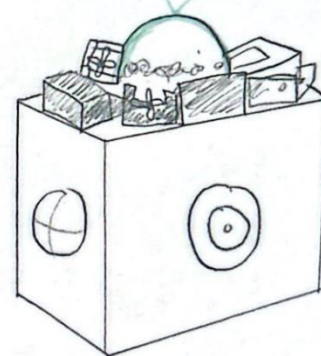
- Heats the tray.

5. Silicon Cap/hand hole



- allows hand to enter bowl.

Popcorn maker



Full view

Team Meeting Minutes

September 24th, 2024

Purpose: Discuss the initial plan for project 1 and assign action items to each group for completion by the next meeting.

Discussion

- Presented individual designs.
- Discussed pros and cons of each design. Landed on combining two designs (Joy's and Diego's) into one bigger design.
- The CineMerge Entertainment Hub will include a projector, speakers, popcorn maker, and an optional add on of a flying snack delivery device (or it can just be a snack holder added onto the hub).
- Specifics on how the snack delivery/ popcorn making would work and merge together.

To Do

- Brainstorm what parts will need to be included, so next time we can narrow it down and solidify the design
- Brainstorm movement for the projector (i.e. legs on the projectors, how to angle the lens, and how to keep popcorn maker stable)

October 3rd, 2024

Purpose: Finish up team project proposal.

Discussion

- Team members brought their lists of what parts could make up the design, and ideas on how to break up the design to have parts for each team member
- Finalized "problem definition description" and "foreseeable challenges/difficulties in modeling" sections of proposal (Joy and Kai)
- Submitted proposal (Tallulah)

To Do

- Assign parts to members

October 8th, 2024

Purpose: Clearly define how CineMerge works/is assembled.

Discussion

- Sketched overall assembly

- Focused on attachments between components (e.g. drone-popcorn maker, popcorn-maker-base, speakers/projector-base, screen-base)
- Began thinking about overall dimensions

To Do

- Determine specific dimensions
- Draw assigned part sketches (Base + Legs – Diego; Projector/Speaker – Ben; Drone – Joy; Screen – Tallulah; Popcorn Maker – Kai)

October 31st, 2024

Purpose: Dimension for the overall shape of the project.

Discussion

- Using IPS dimensioning standards (inches)
- 20in x 20 in cross-sectional
- Height yet to be determined. Dependent on the depth of the popcorn machine and projector
- Discussed dome and drone connection

To Do

- Begin modeling individual parts (based on set dimensions). Bring baseline model to show and tell.

November 20th, 2024

Purpose: Progress check in and begin working on final report.

Discussion

- Reviewed how modeling was going for each member
- Created shared OneDrive for organization; members uploaded their parts
- Created outline for final project report
- Ensuring progress would mean whole assembly was complete before Thanksgiving break at minimum

To Do

- Finish CAD parts by 21st

November 21th, 2024

Purpose: Attempt to create full assembly; work on final report.

Discussion

- Started to think about colors for parts (consistency)
- Looked at issues in the assembly and what needed to be changed

To Do

- Everyone should upload their original design sketches

November 25th, 2024

Purpose: Working on the project over break.

Discussion

- Divide remaining workload for project
- Setting timelines

To Do

- Everyone should:
 - Finish part drawings
 - Upload original design sketches if haven't already
 - Write their part's assembly instructions
- Meet Monday to finalize final report
- Meet Tuesday and/or Wednesday to practice presentation
- Tallulah will make presentation
- Kai/Diego will work on making renderings
- Joy/Ben will work on assembly drawings

December 1st, 2024

Purpose: Reviewing work done over break.

Discussion

- Presentation shared between members
- Final assembly complete + final assembly drawings

To Do

- Everyone
 - Upload pictures of their parts and subassemblies to the slides

- Think about what they want to discuss in the presentation
- Come with what materials you want your parts to be

December 2nd, 2024

Purpose: Reviewing work done over break.

Discussion

- What materials to color each part
 - Mostly metal
- Troubleshooting issues opening assembly
- What renderings to make
 - Drone flying off
 - Legs extended
 - Projecting with actual screen & angle of the base
 - Backgrounds
 - Backyard
 - Coffee table
 - Close up of projector
- Assembly talk
 - Add wall to the back of the base to prevent projector and hot plate from falling
 - Redo assembly drawings

To Do

- Everyone
 - Top priority
 - Upload pictures to slides to practice presentation on Wednesday meeting
 - Start rendering process
 - Less priority
 - Write assembly instructions if haven't yet
 - Finish and upload part and assembly drawings

December 4th, 2024

Purpose: Wrapping up presentation details

Discussion

- Look over renders
- Finalize presentation

To Do

- Diego and Kai will upload renders and possibly animation

December 5th, 2024

Purpose: Practice presentation

Discussion

- Assigning slides
- Went through 3 practice runs
- Talked about what to include or not include to fit the time

To Do

- For those able, rehearsal at 11 am before presentation tomorrow
- Meet one more time to finalize final report

December 9th, 2024

Purpose: Finalize report

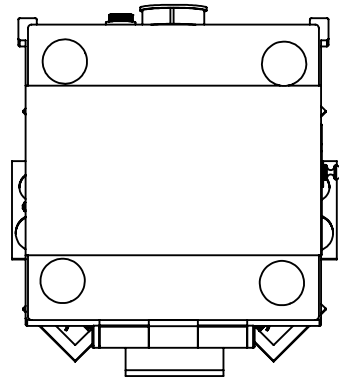
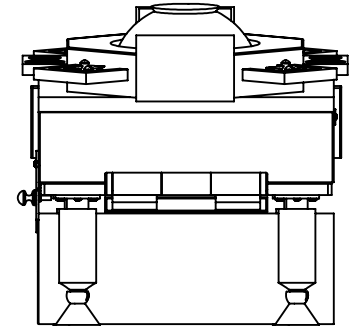
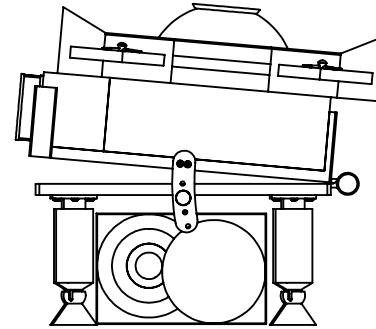
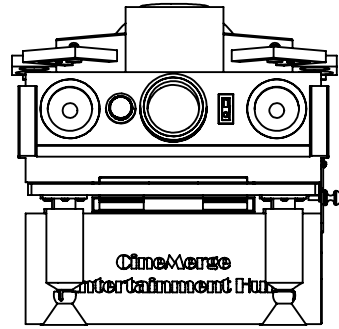
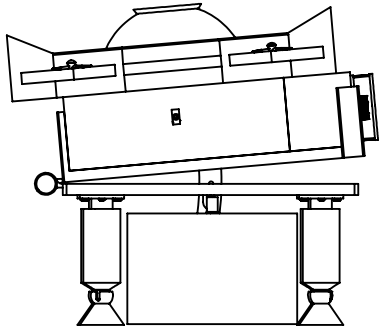
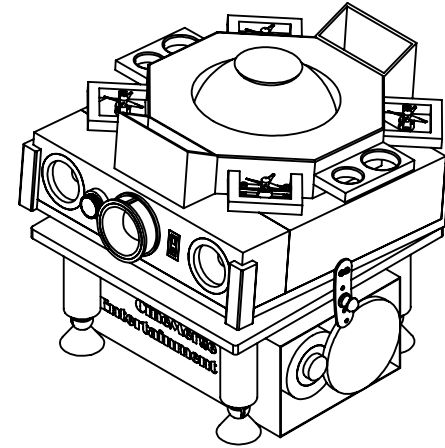
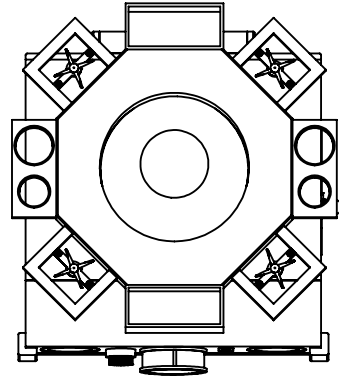
Discussion

- Combined pdfs
- Finished writing assembly instructions

To Do

- Submit final report

**Comprehensive
Assembly
Drawing**



TITLE

ASSEMBLY MULTIVIEWS

DATE

12/3/2024

NAME

GROUP 10

COURSE

ME 1670

SEMESTER

FALL 2024

SECTION

A

LAB

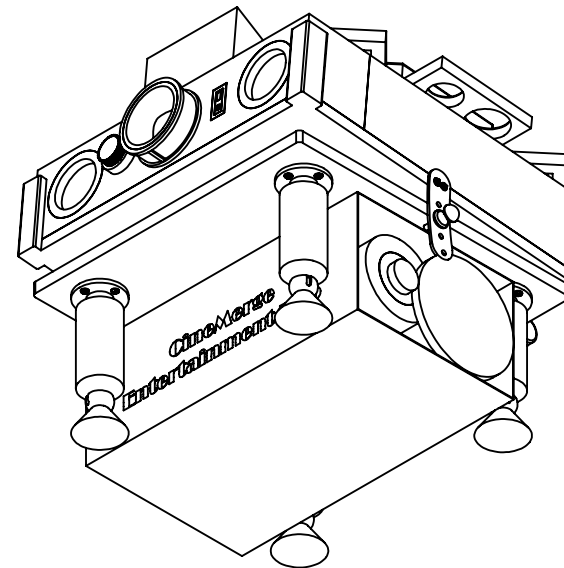
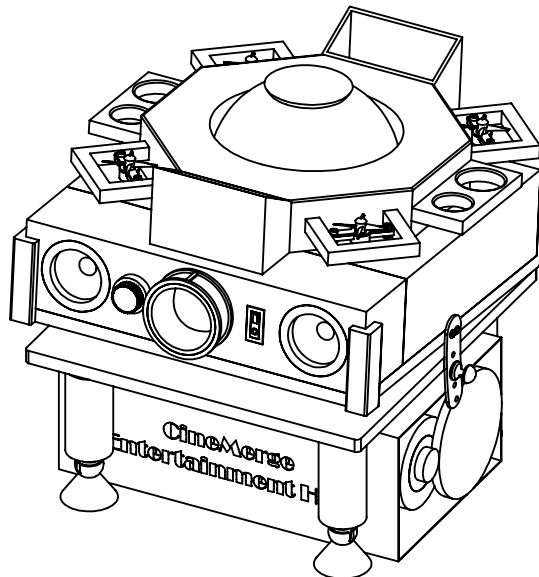
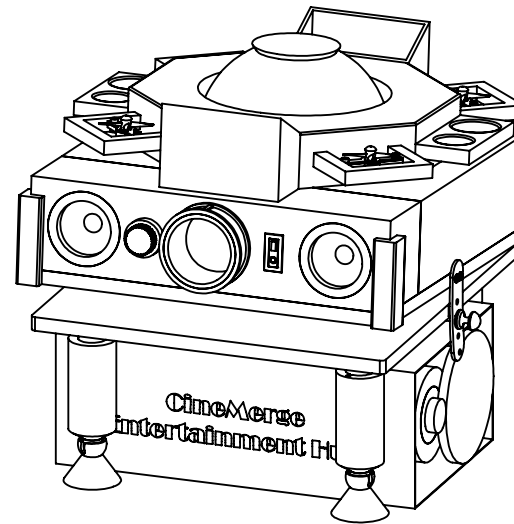
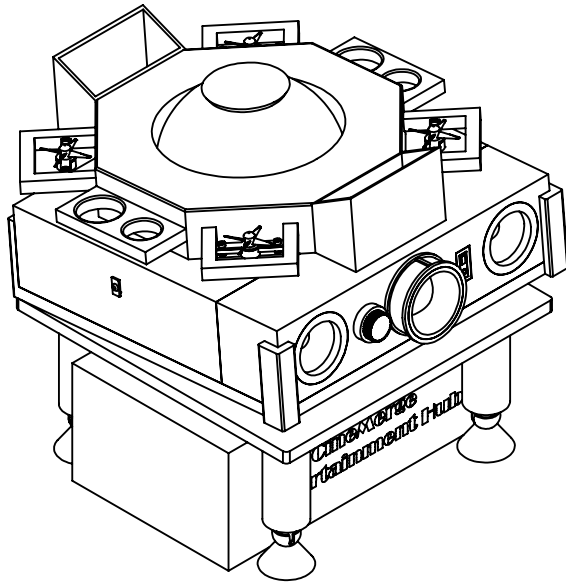
ACTIVITY

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ANGLES ARE IN DEGREES

SCALE 1:13 SHEET 1 OF 1

SIZE

A



TITLE

ASSEMBLY PICTORALS

COURSE
ME 1670

SEMESTER

FALL 2024

DATE

12/3/2024

NAME

GROUP 10

SECTION

A

LAB

ACTIVITY

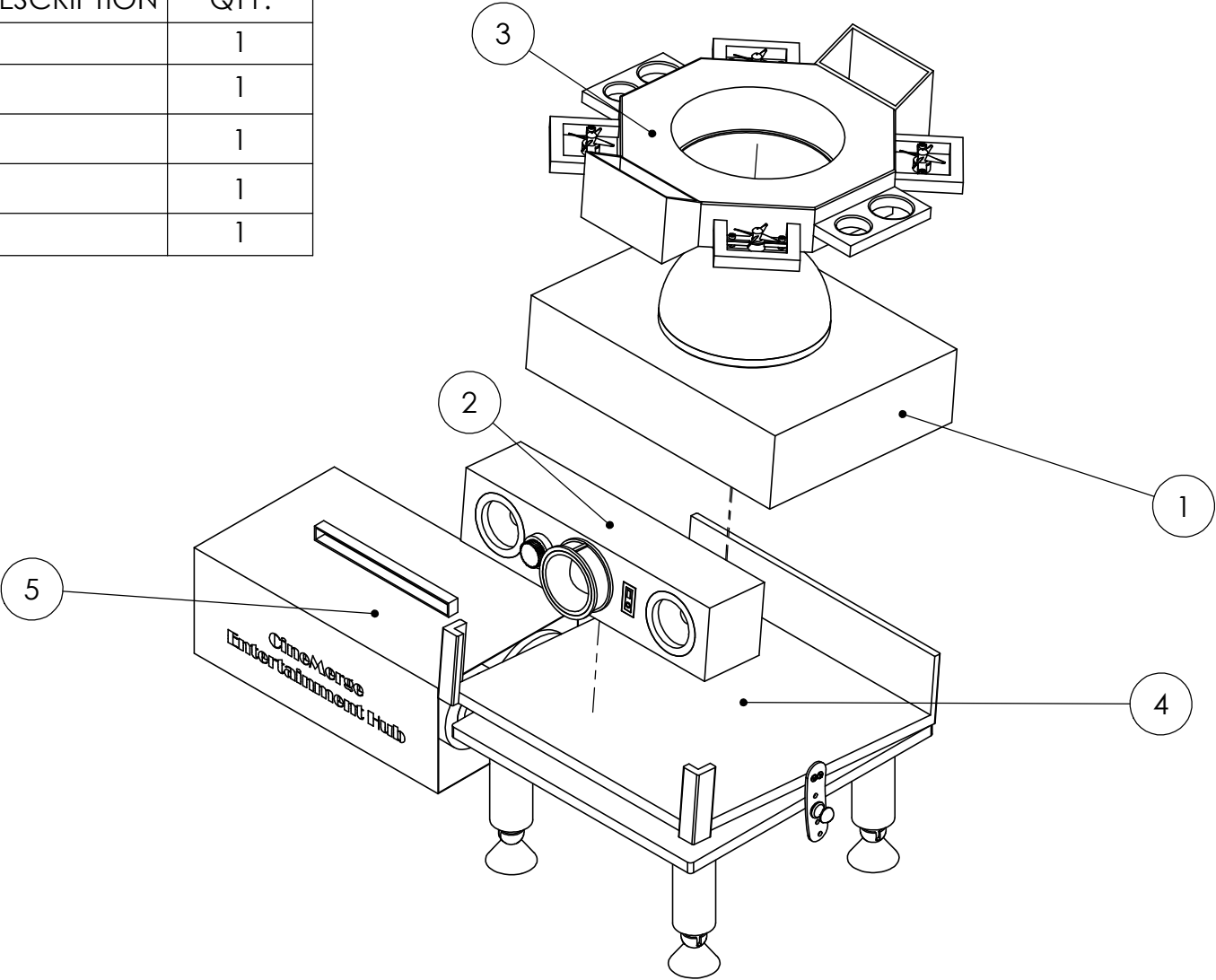
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SCALE 1:10 SHEET 1 OF 1

SIZE

A

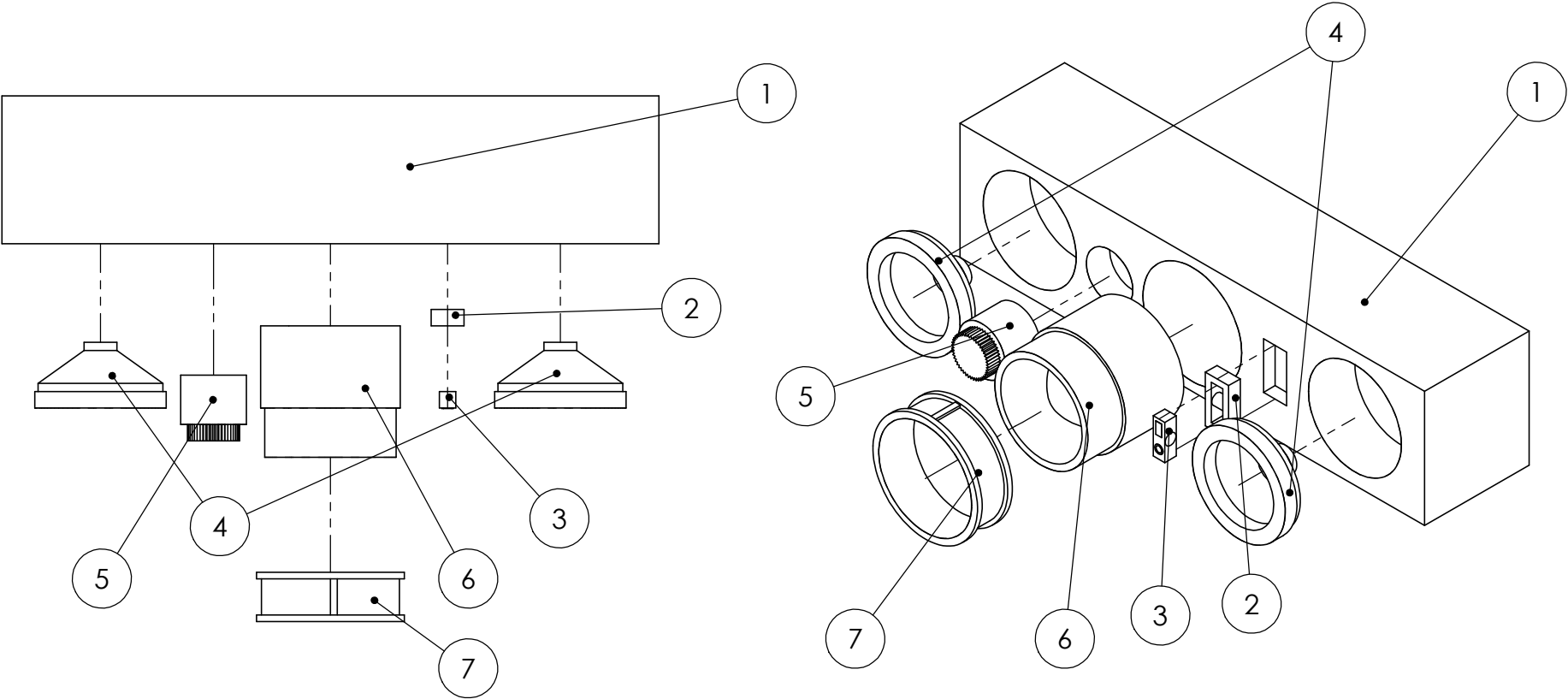
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2	Projector		1
3	Drone		1
4	Base		1
5	Screen Bag		1



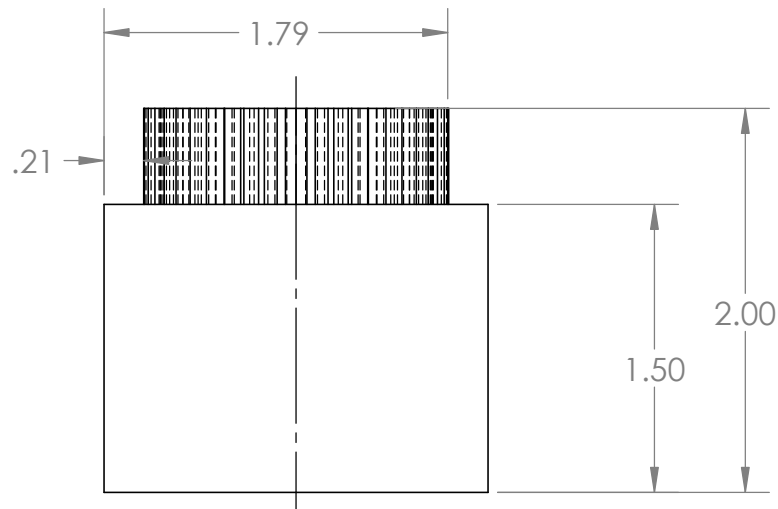
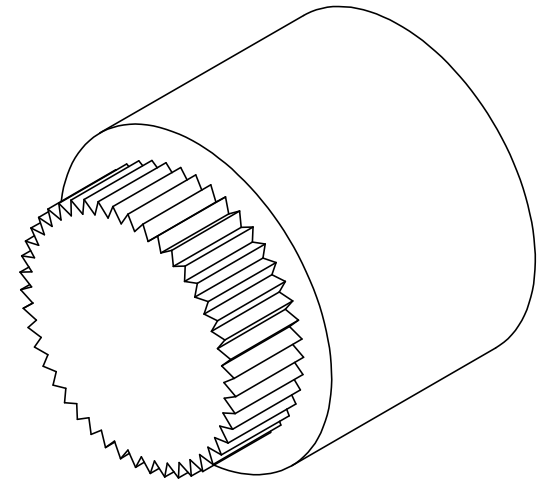
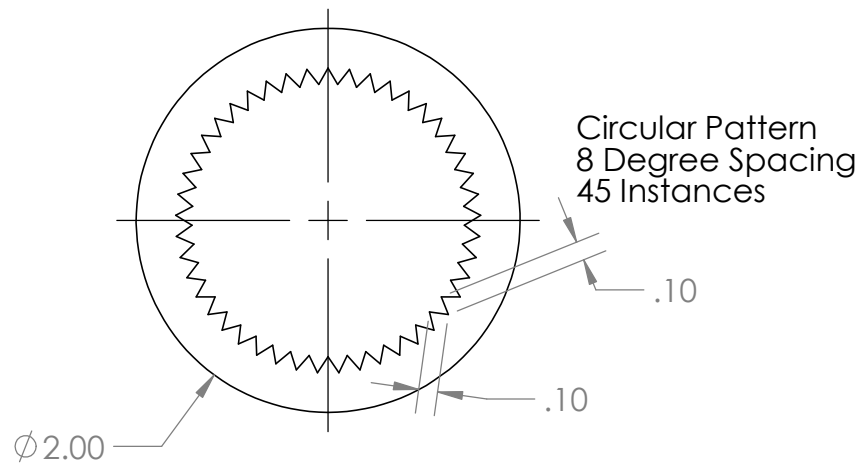
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	COURSE ME 1670	SEMESTER Fall 2024	SECTION A	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE A
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Engineering Drawings of Individual Parts

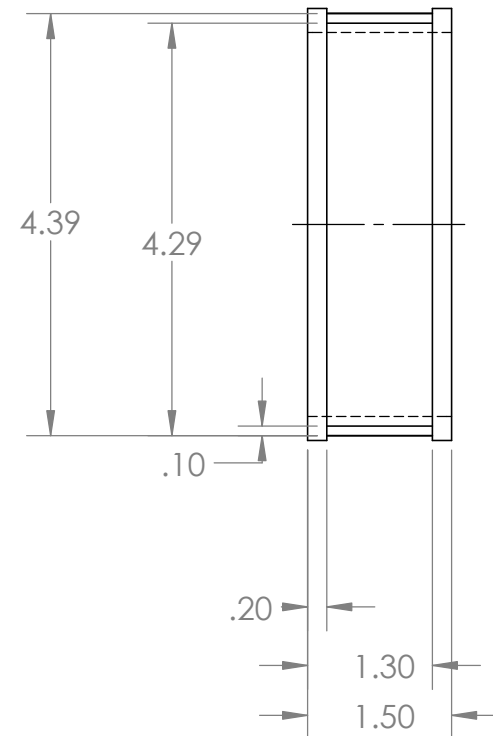
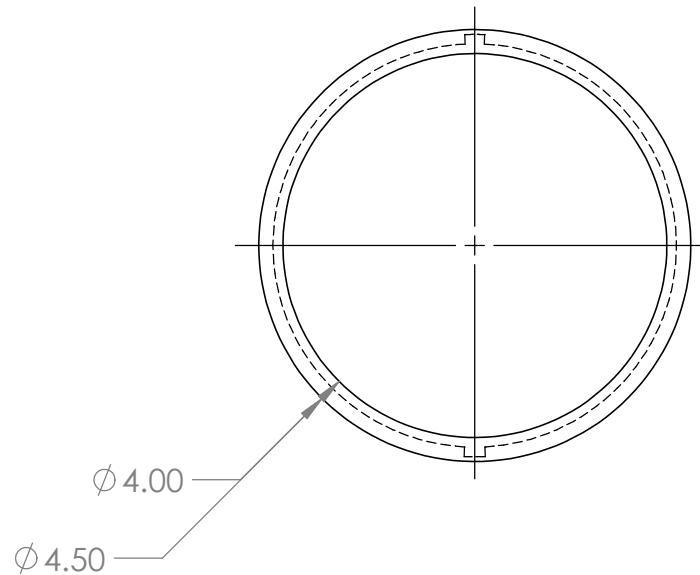
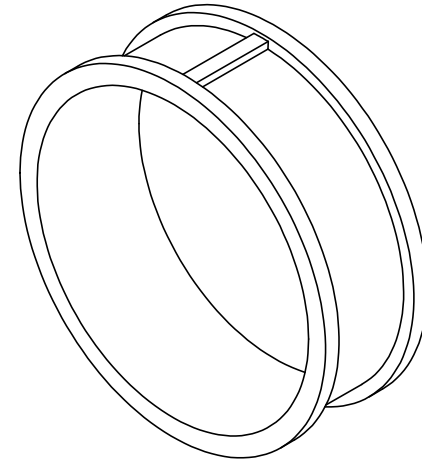
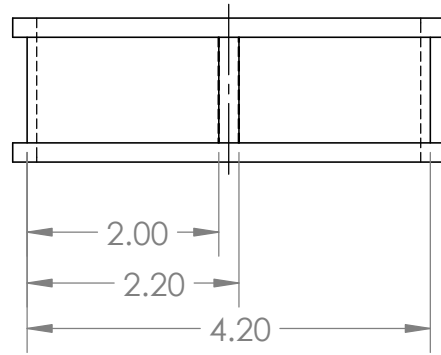
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Projector Base		1
2	Switch Case		1
3	Switch		1
4	Speaker Driver		2
5	Focus Knob		1
6	Lens		1
7	Lens Adjuster		1



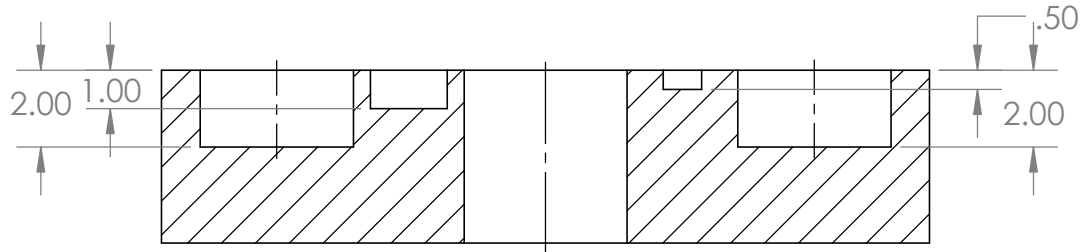
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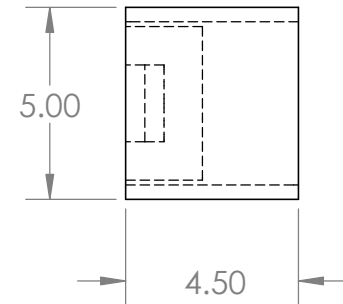
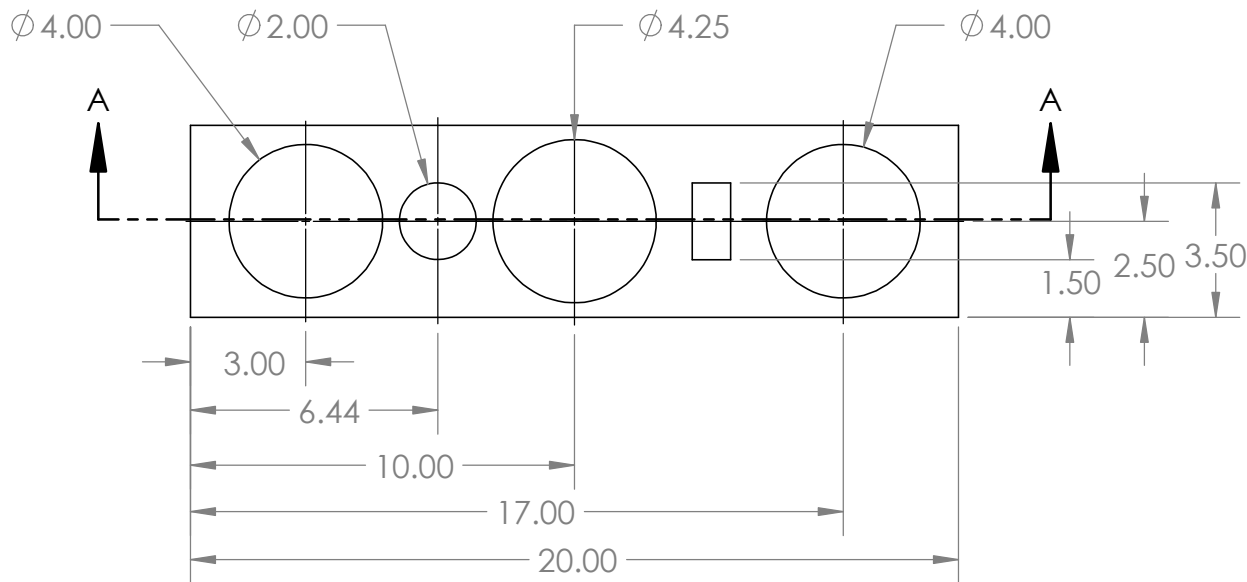
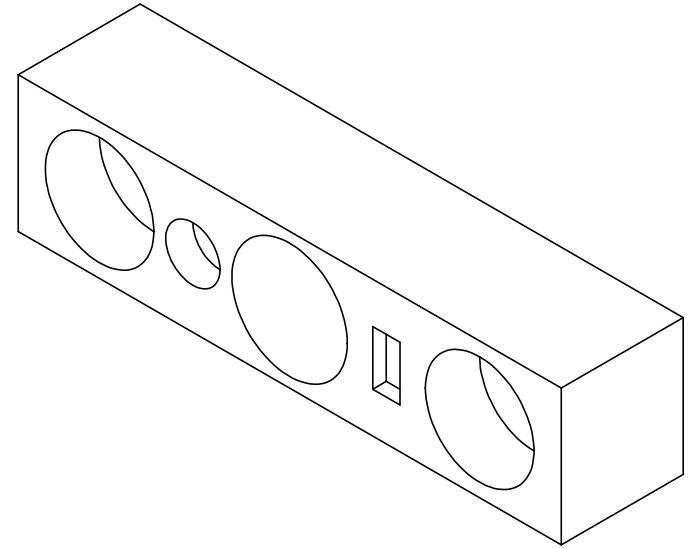
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	ME 1670	Fall 2024	A			SCALE 1:1	SHEET 1 OF 1	A



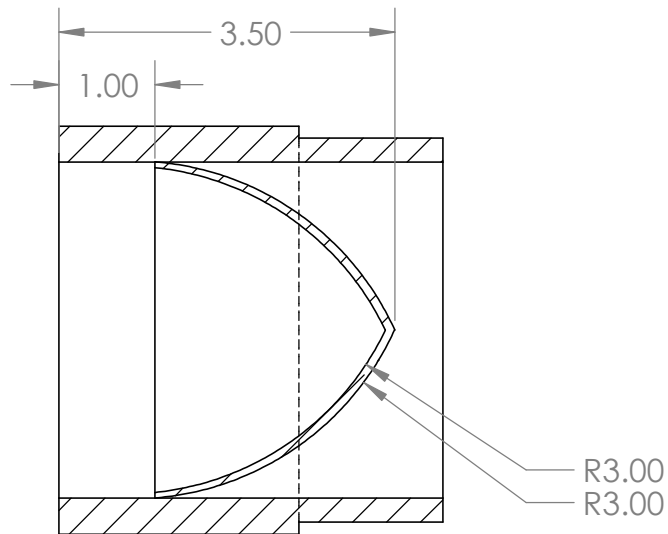
 Georgia Institute of Technology	TITLE		DATE		NAME			
	Lens Adjuster		12/8/2024		Benjamin Chamberlin			
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE
	ME 1670	Fall 2024	A			SCALE 1:2	SHEET 1 OF 1	A



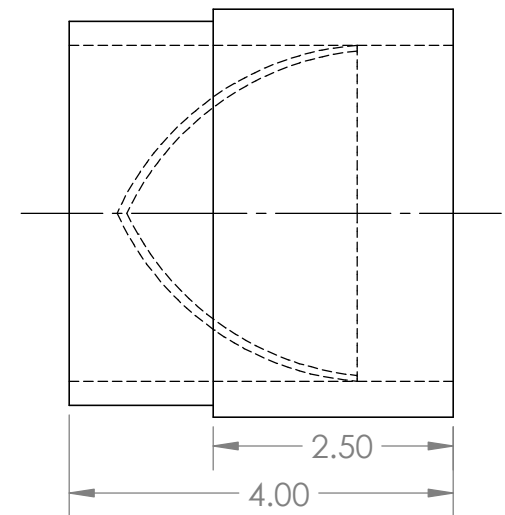
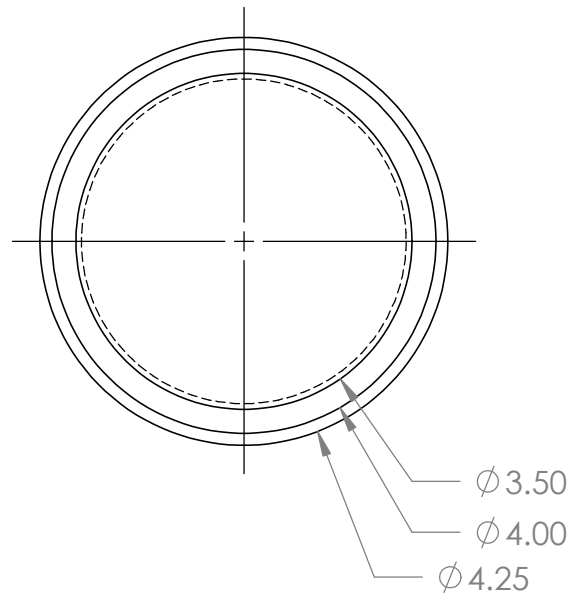
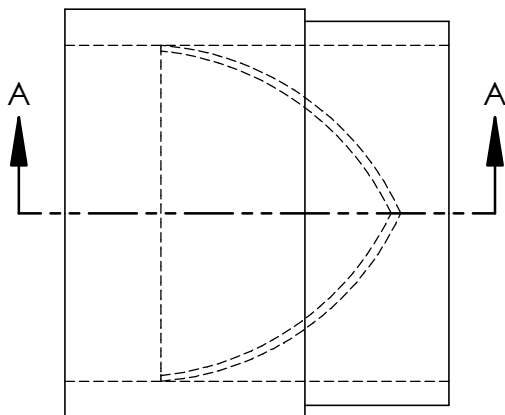
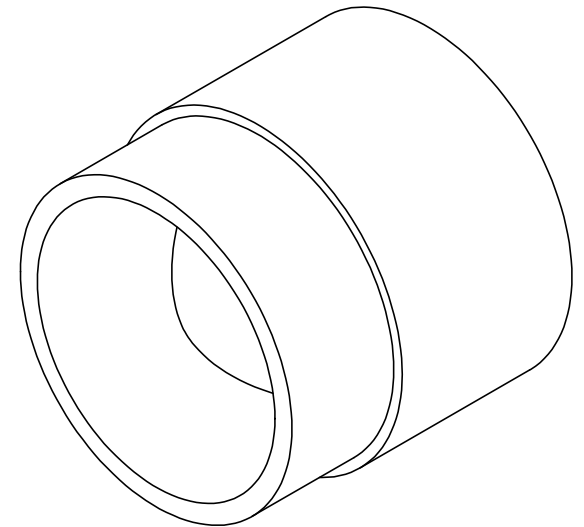
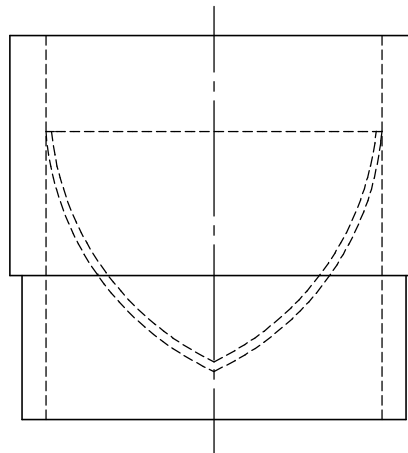
SECTION A-A

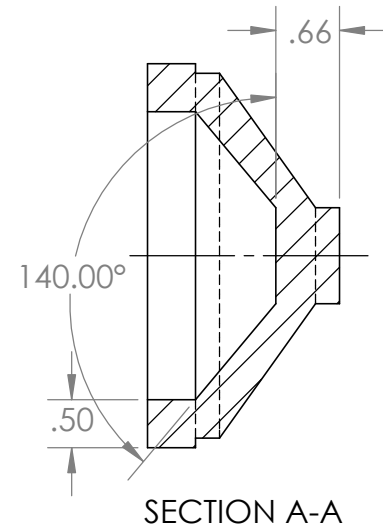
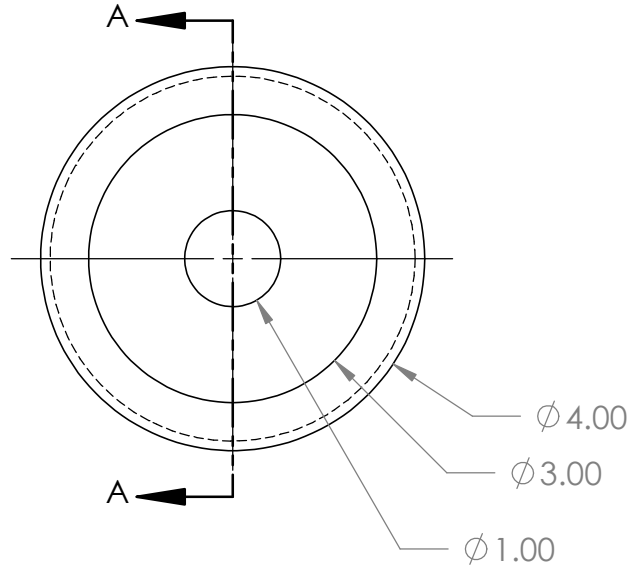
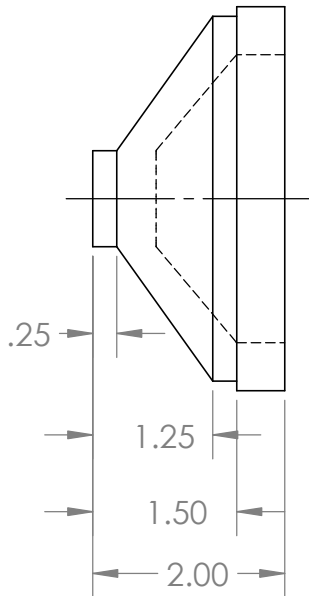
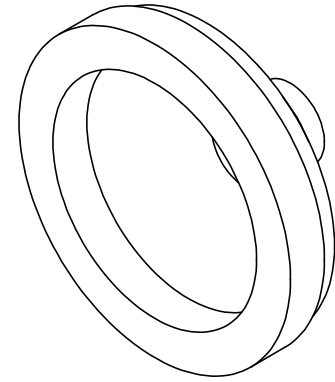
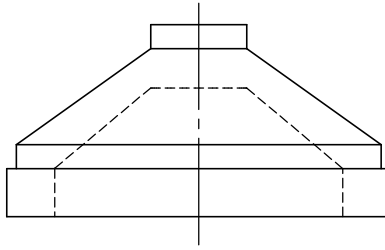


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ME 1670	Fall 2024	A				A
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SCALE 1:5 SHEET 1 OF 1						

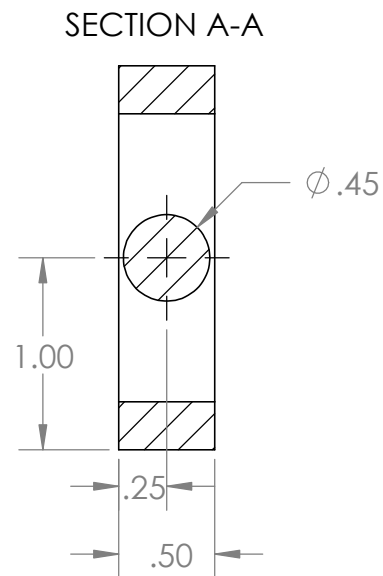
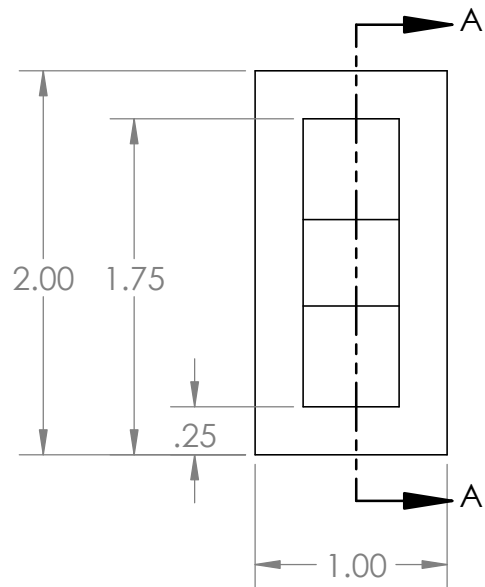
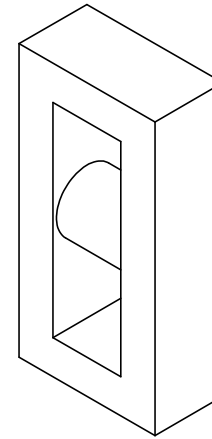


SECTION A-A

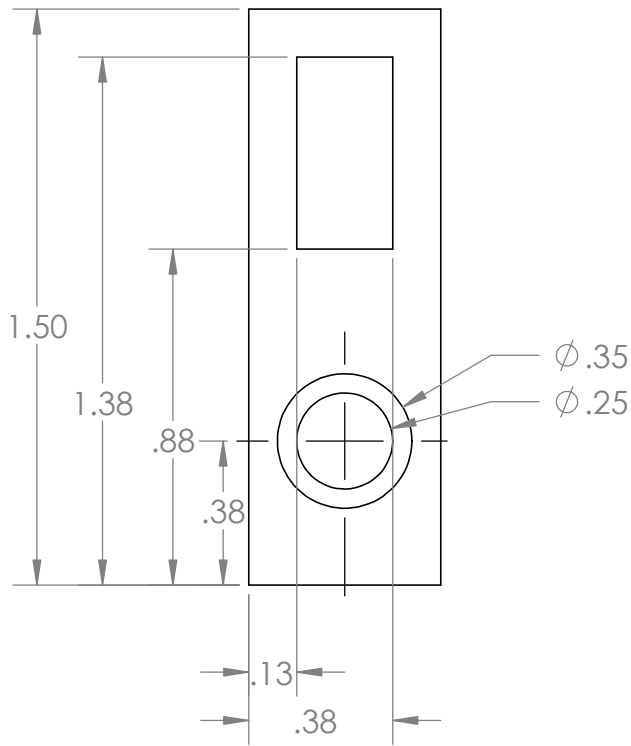
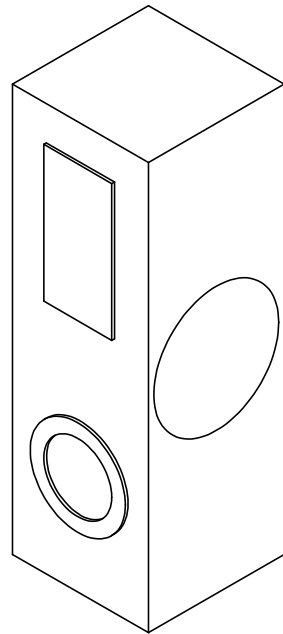
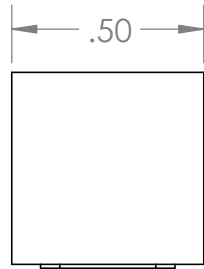
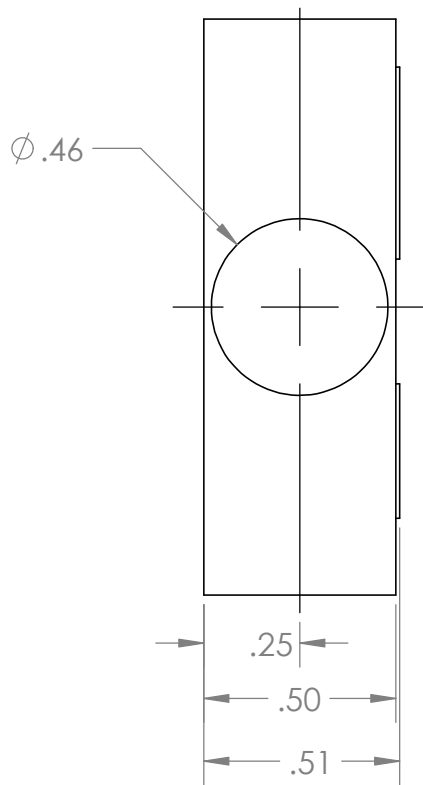




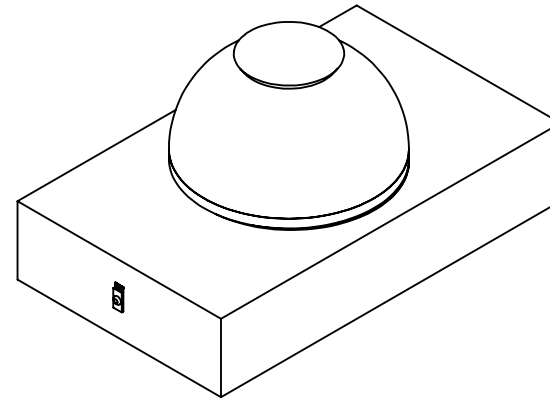
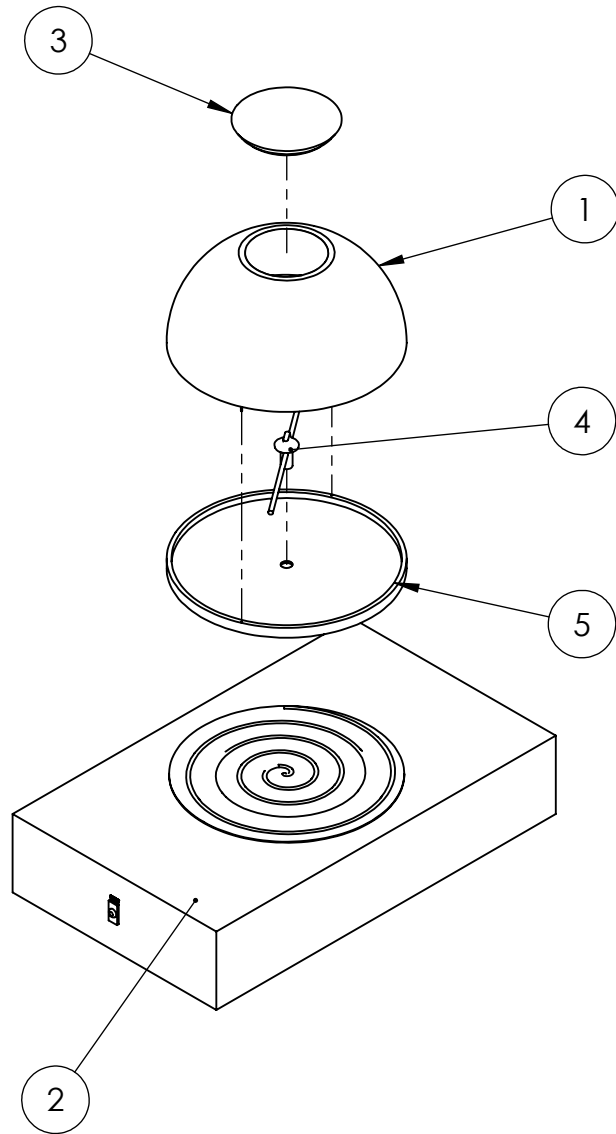
	TITLE		DATE		NAME		
	Speaker Driver		12/8/2024		Benjamin Chamberlin		
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES	
	ME 1670	Fall 2024	A			SCALE 1:2	SIZE A
						SHEET 1 OF 1	



 Georgia Institute of Technology	TITLE		DATE		NAME			
	Switch Case		12/8/2024		Benjamin Chamberlin			
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE
	ME 1670	Fall 2024	A			SCALE 1:1	SHEET 1 OF 1	A

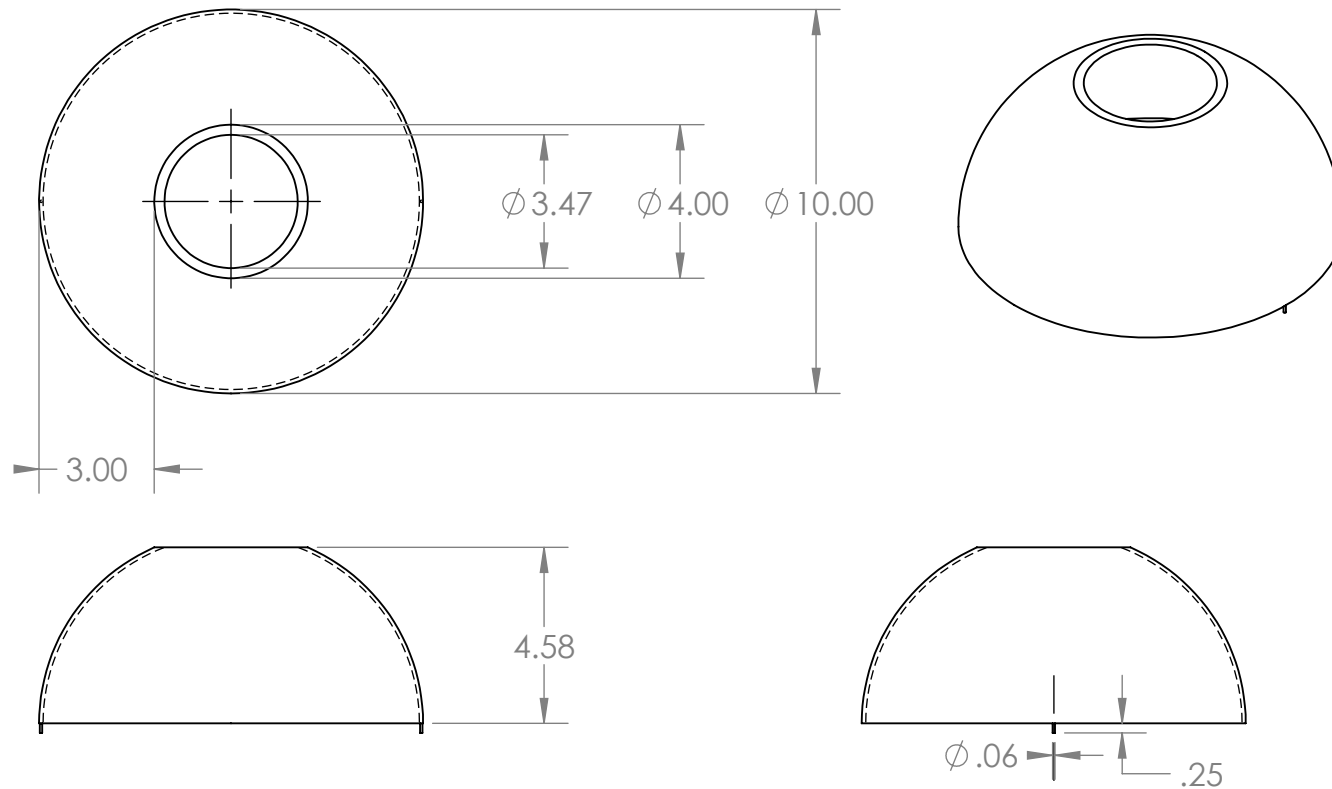


	TITLE		DATE		NAME			
	Switch		12/8/2024		Benjamin Chamberlin			
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE
	ME 1670	Fall 2024	A			SCALE 2:1	SHEET 1 OF 1	A



ITEM NO.	PART NUMBER	QTY.
1	Glass Dome	1
2	Hot Plate	1
3	Lid	1
4	Spinner	1
5	Tray	1

Thickness of Walls Assumed to be 0.1 inches unless
stated otherwise



TITLE
Glass Dome

COURSE
ME 1670

SEMESTER
FALL 2024

DATE
12/9/2024

SECTION
A

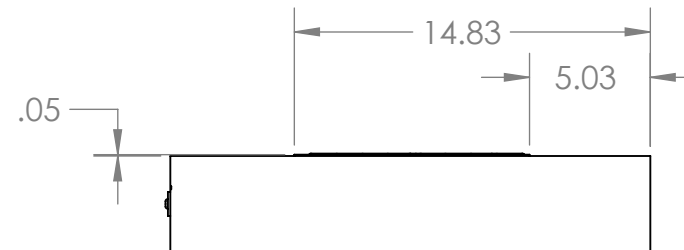
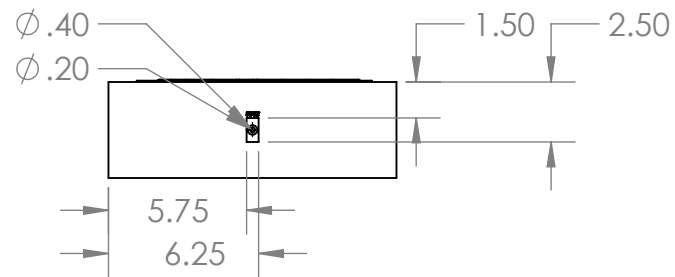
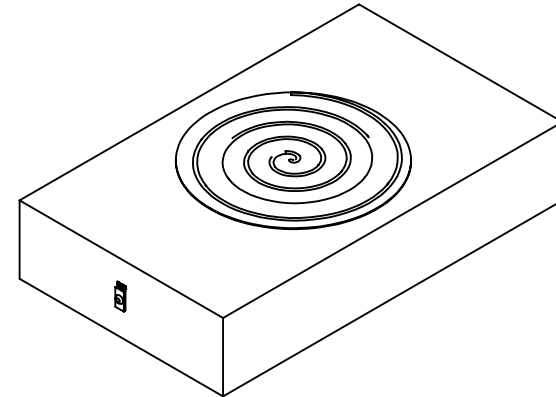
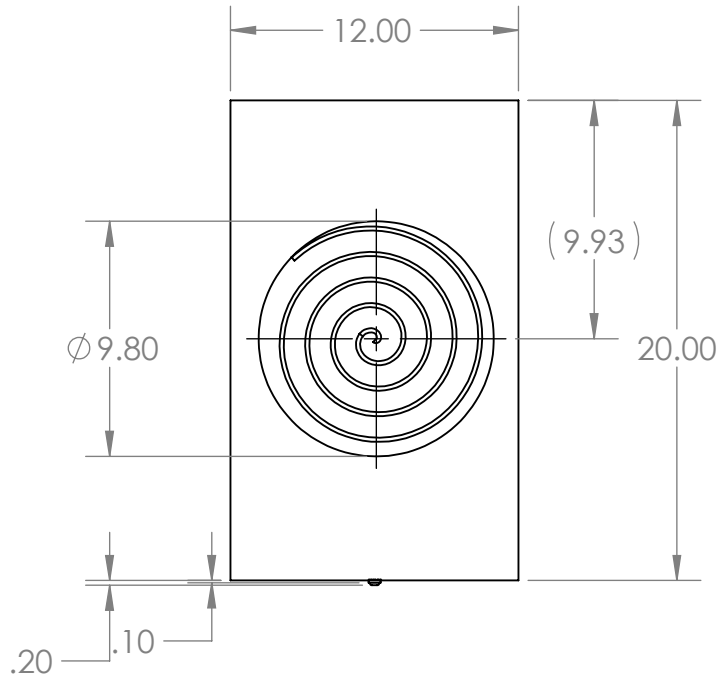
NAME
Kai Takeuchi

LAB

ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES
SCALE 1:5 SHEET 1 OF 1

SIZE
A



TITLE

Hot Plate

COURSE

ME 1670

SEMESTER

FALL 2024

DATE

12/9/2024

SECTION

A

NAME

Kai Takeuchi

LAB

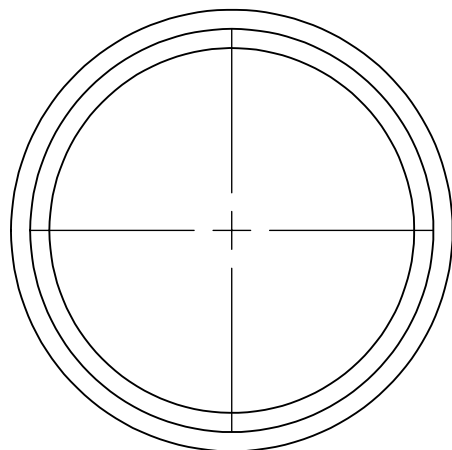
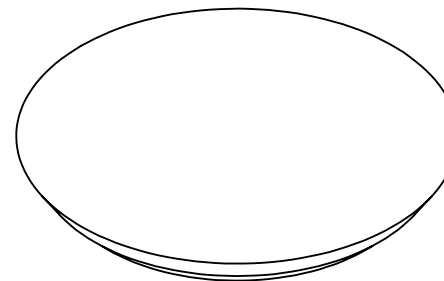
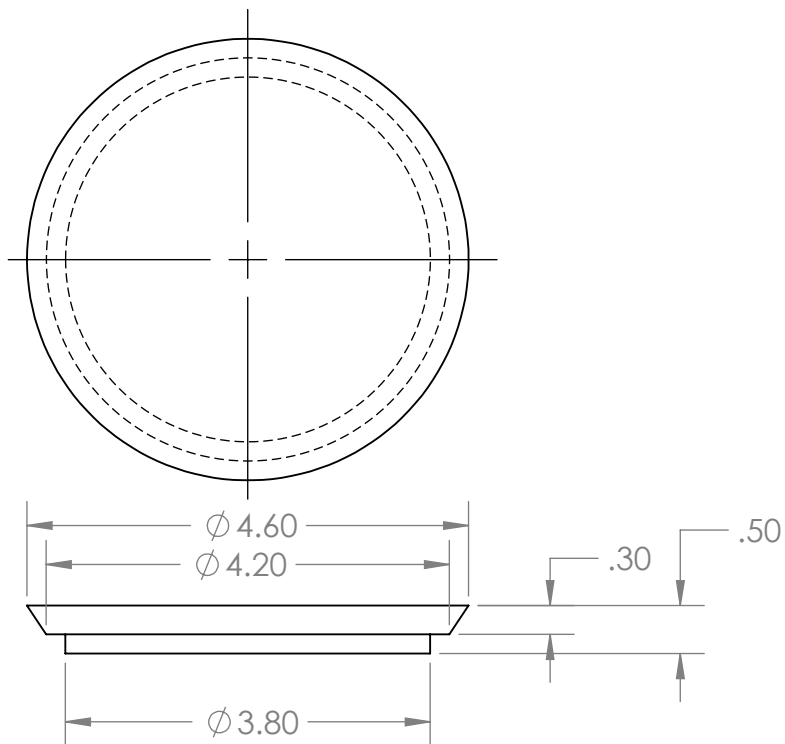
ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

SCALE 1:4 SHEET 1 OF 1

SIZE

A



**Georgia Institute
of Technology**

TITLE

Lid

COURSE

ME 1670

SEMESTER

FALL 2024

DATE

12/9/2024

SECTION

A

NAME

Kai Takeuchi

LAB

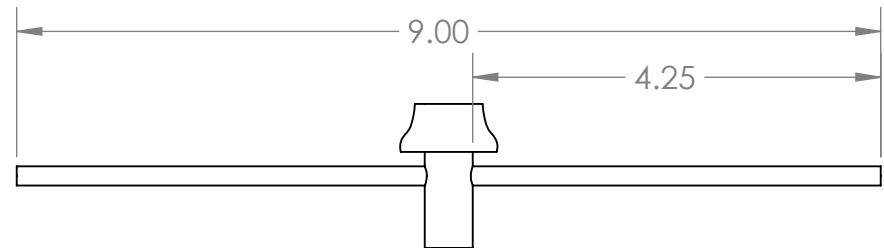
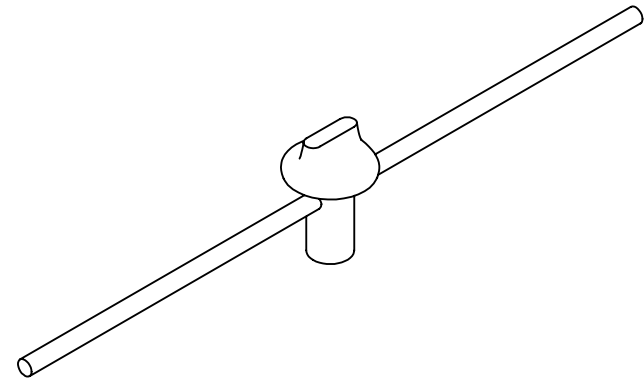
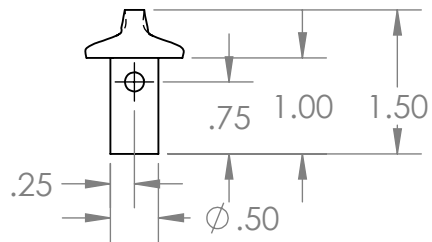
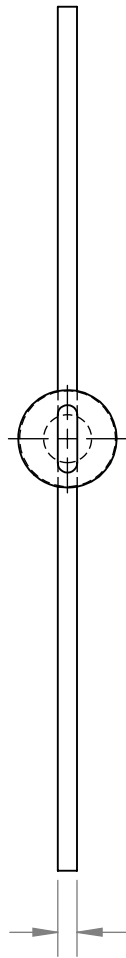
ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

SCALE 1:2 SHEET 1 OF 1

SIZE

A



TITLE
Spinner

COURSE
ME 1670

SEMESTER
FALL 2024

DATE
12/9/2024

SECTION
A

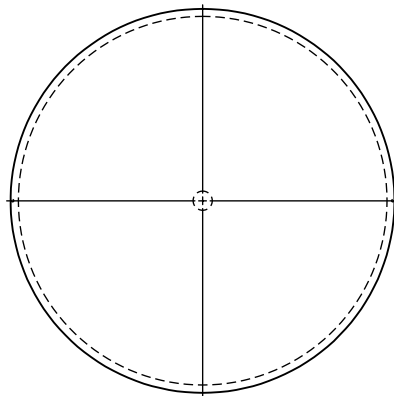
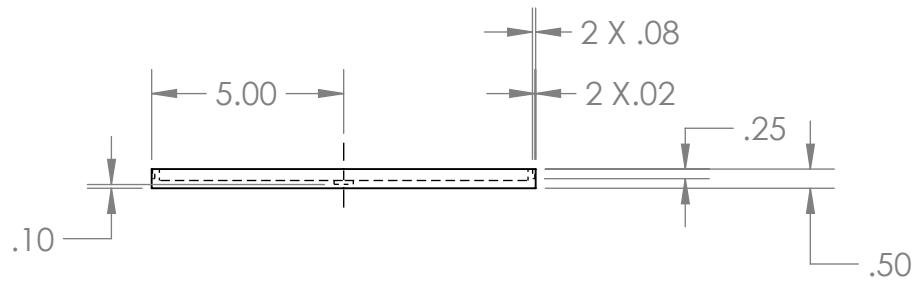
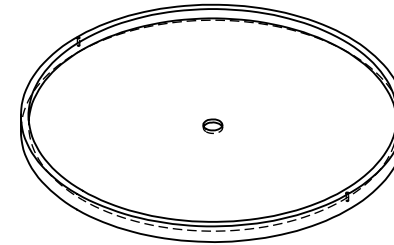
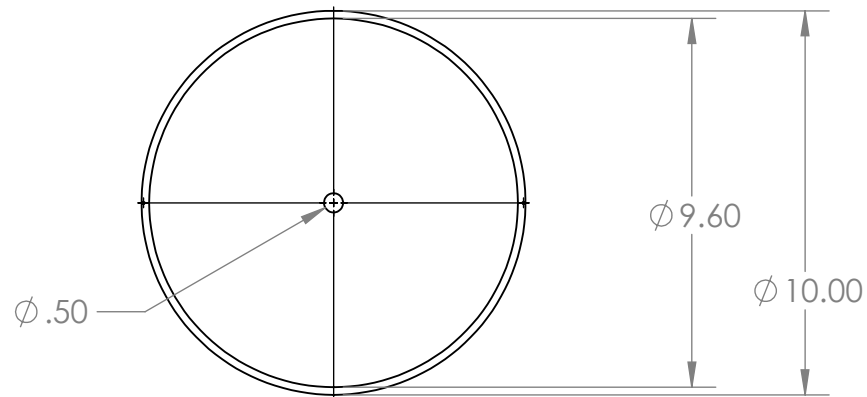
NAME
Kai Takeuchi

LAB

ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES
SCALE 1:2 SHEET 1 OF 1

SIZE
A



**Georgia Institute
of Technology**

TITLE

Tray

COURSE

ME 1670

SEMESTER

FALL 2015

DATE

12/9/2024

SECTION

A

NAME

Kai Takeuchi

LAB

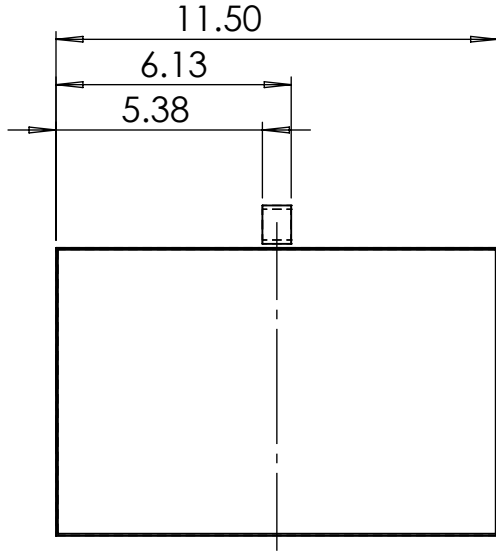
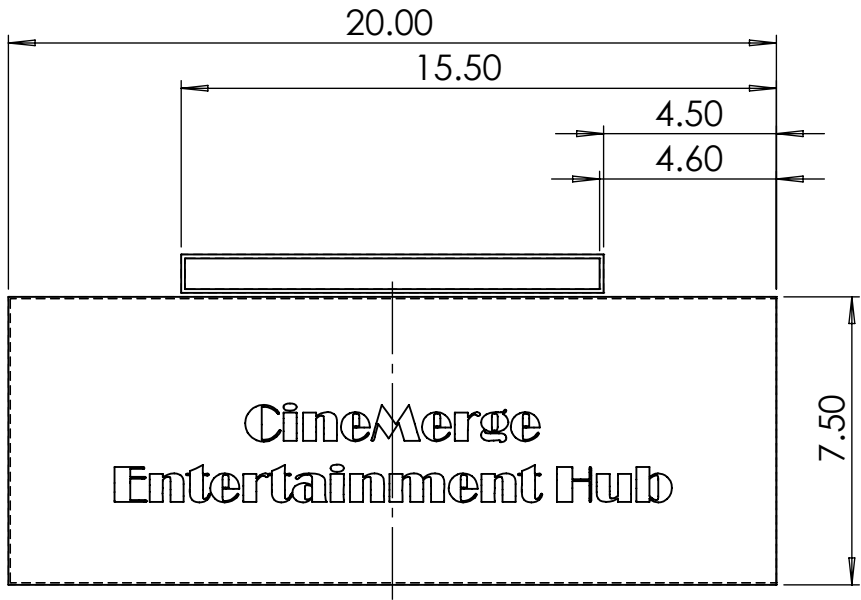
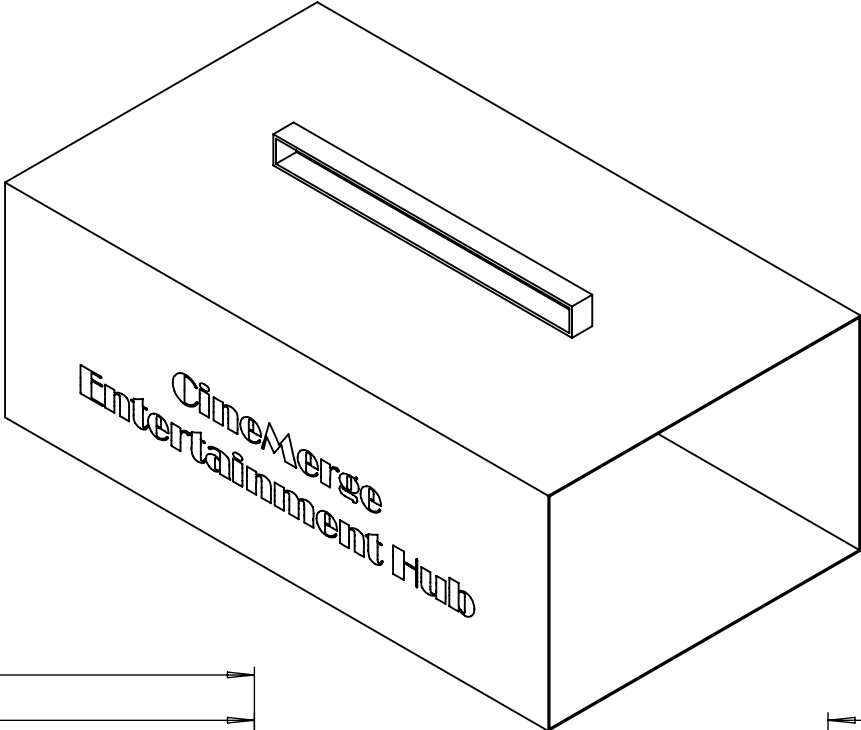
ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

SCALE 1:5 SHEET 1 OF 1

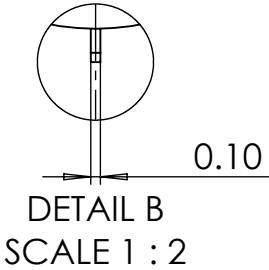
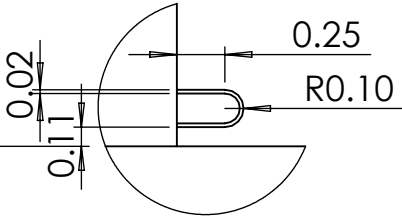
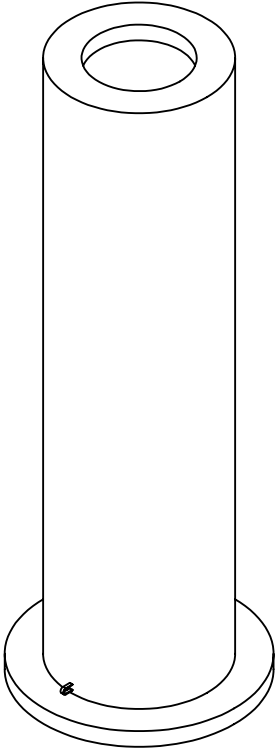
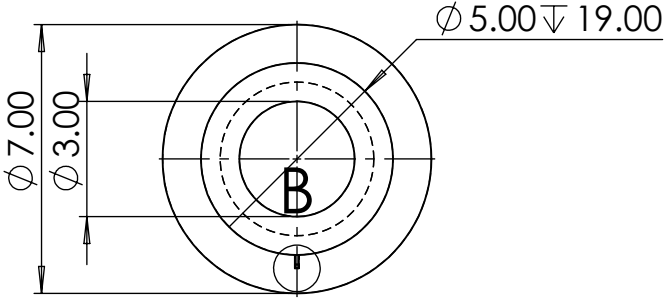
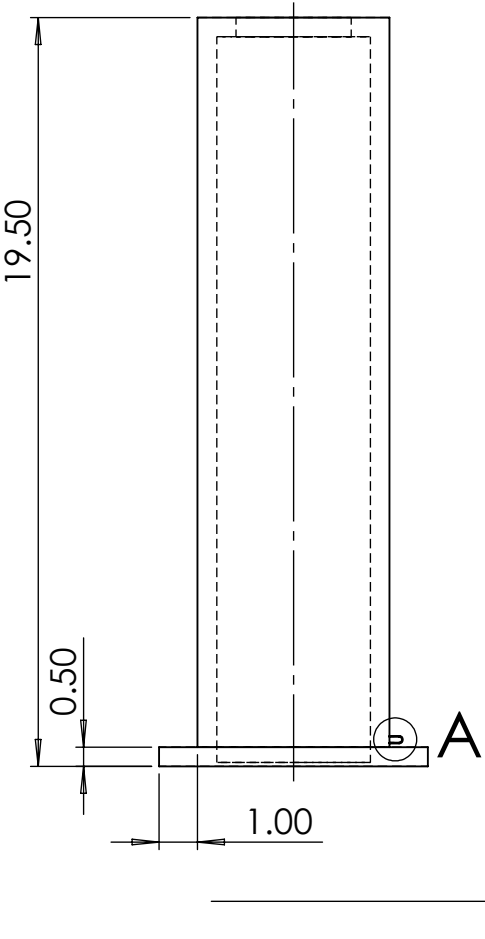
SIZE

A



	TITLE		DATE		NAME			
	Screen Bag (FP)		12/9/2024		Tallulah Selhorst			
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES		SIZE
	ME 1670	FALL 2024	A			ANGLES ARE IN DEGREES		A
						SCALE 1:5	SHEET 1 OF 12	
5	4	3		2				1

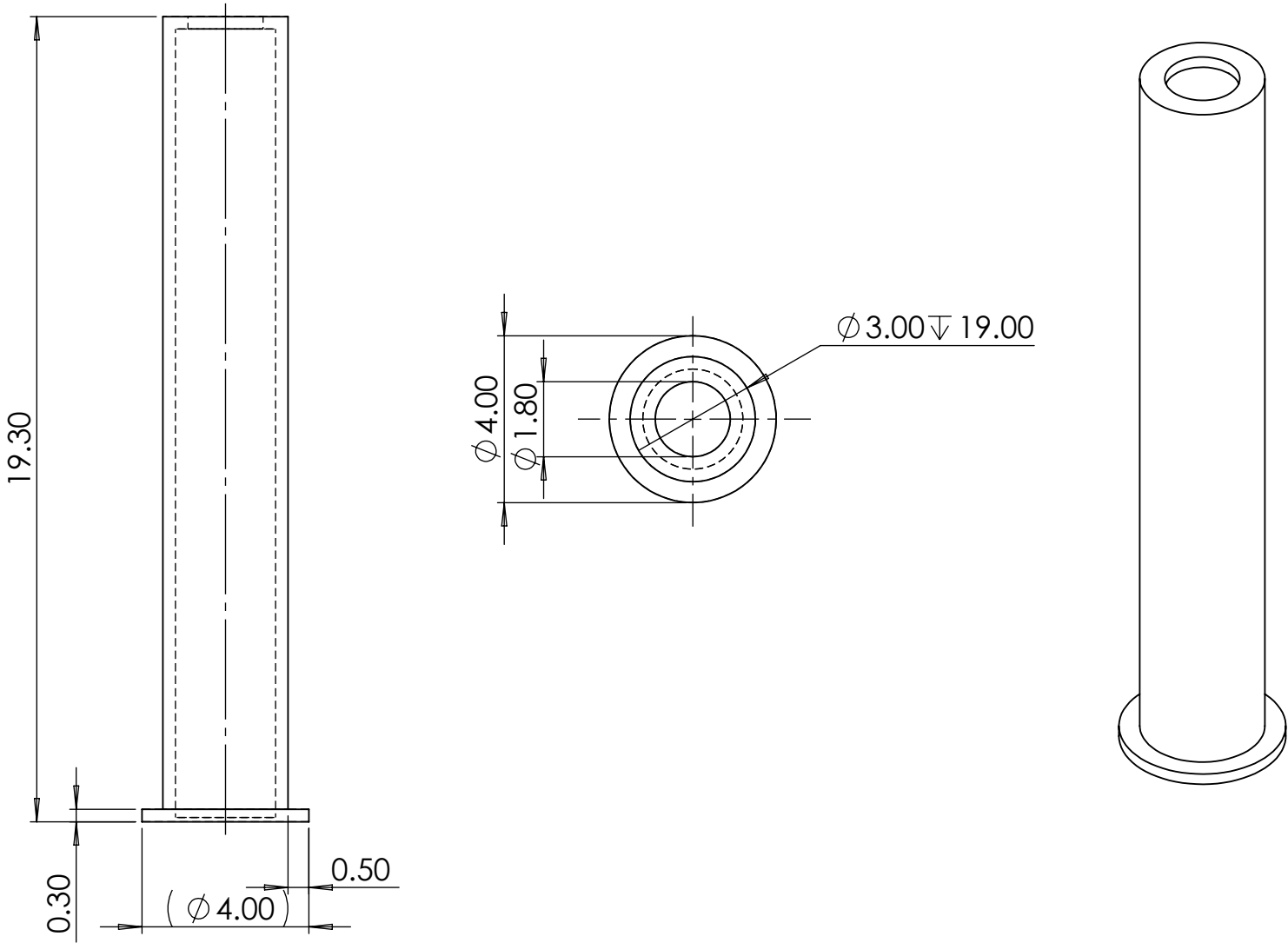
WALL THICKNESS 0.5 IN.
UNLESS OTHERWISE SPECIFIED



DETAIL A
SCALE 1 : 1

DETAIL B
SCALE 1 : 2

WALL THICKNESS 0.3 IN.
UNLESS OTHERWISE SPECIFIED



	TITLE		DATE		NAME			
	Middle Rung of Screen Leg(FP)		12/9/2024		Tallulah Selhorst			
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE
	ME 1670	FALL 2024	A			SCALE 1:5	SHEET 3 OF 12	A

5

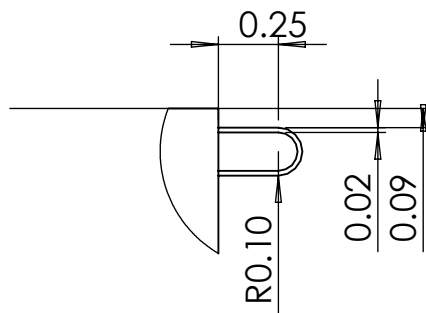
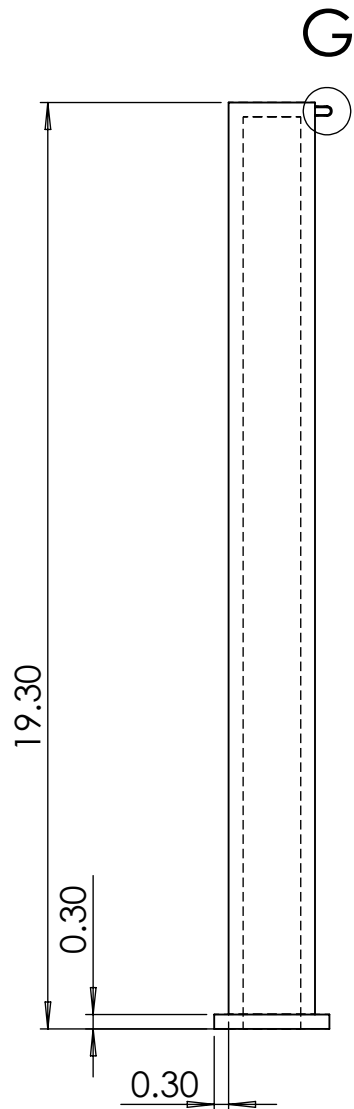
4

3

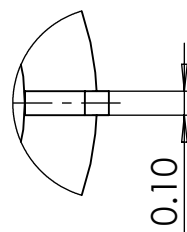
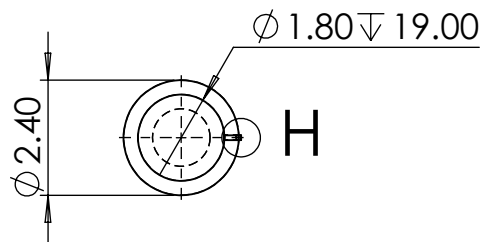
2

1

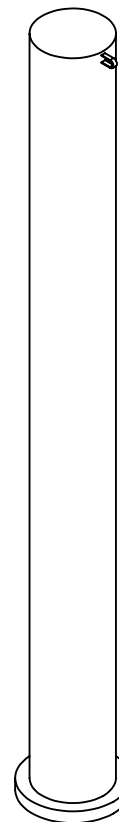
WALL THICKNESS 0.3 IN.
UNLESS OTHERWISE SPECIFIED



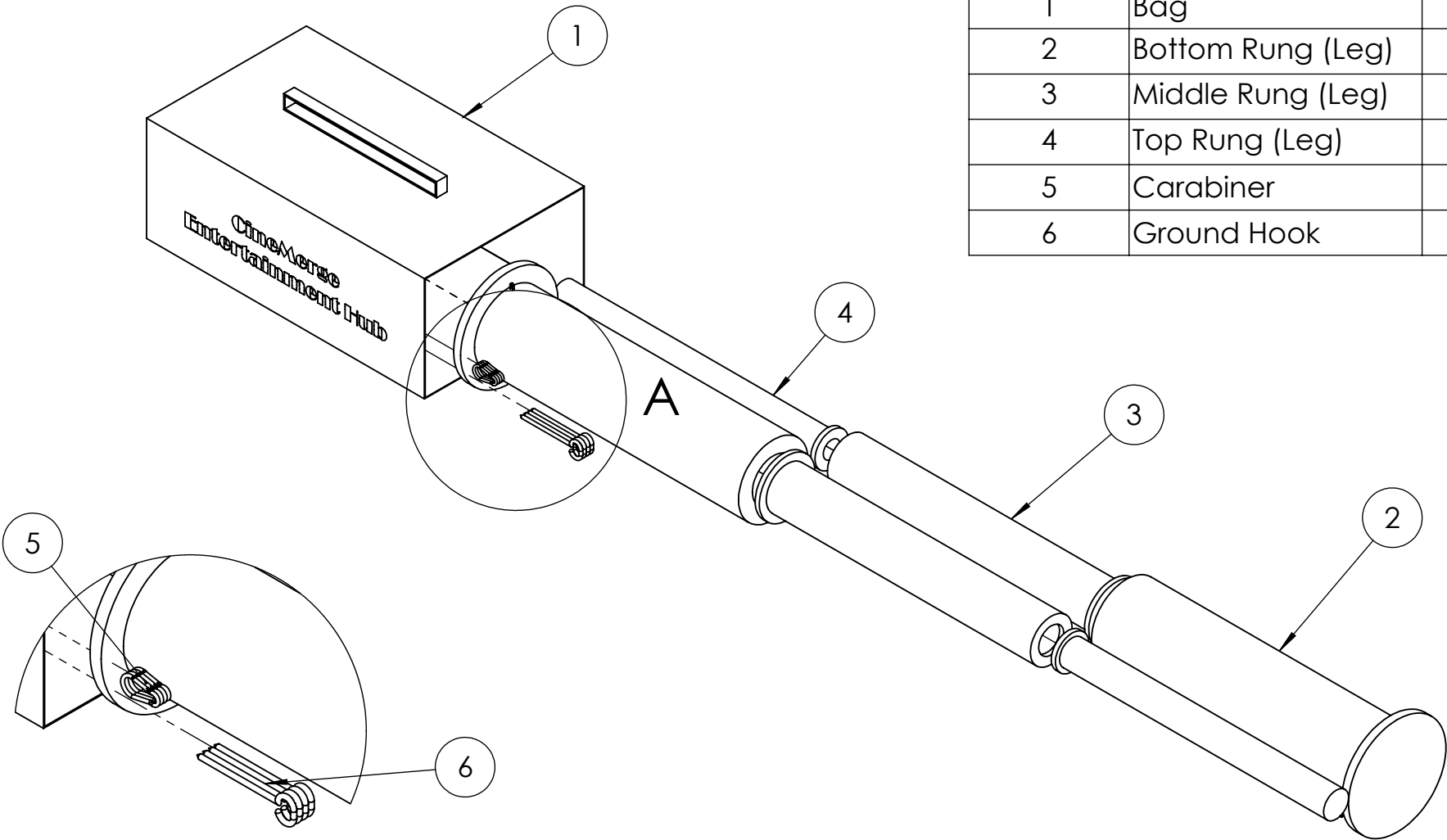
DETAIL G
SCALE 5 : 4



DETAIL H
SCALE 5 : 4

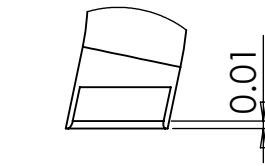
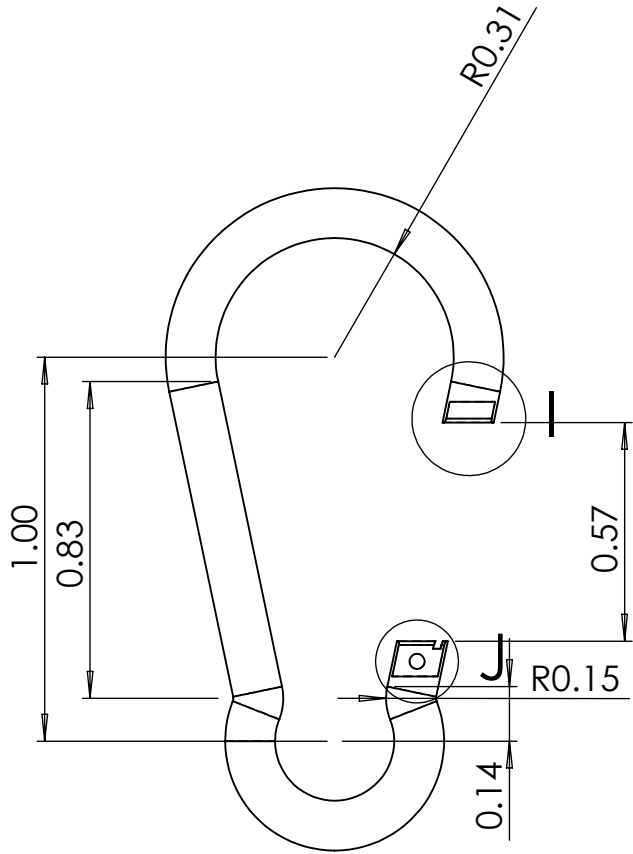


ITEM NO.	PART NUMBER	QTY.
1	Bag	1
2	Bottom Rung (Leg)	2
3	Middle Rung (Leg)	2
4	Top Rung (Leg)	2
5	Carabiner	4
6	Ground Hook	4

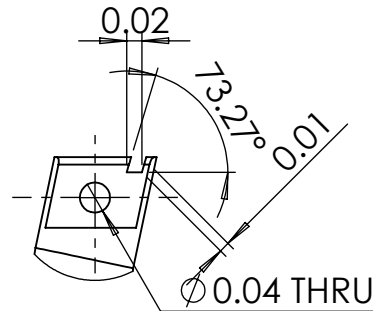
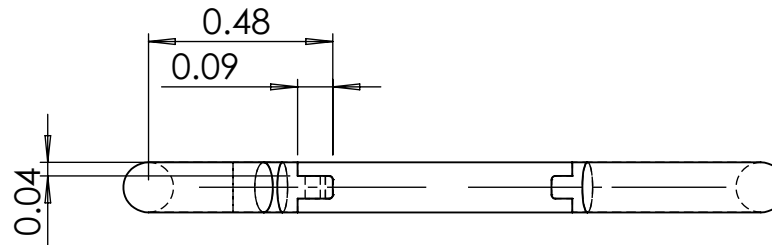
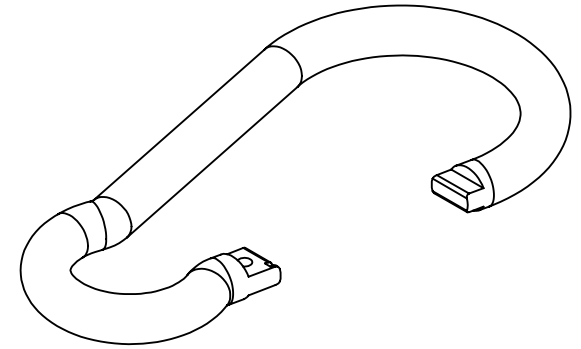


DETAIL A
SCALE 1 : 5

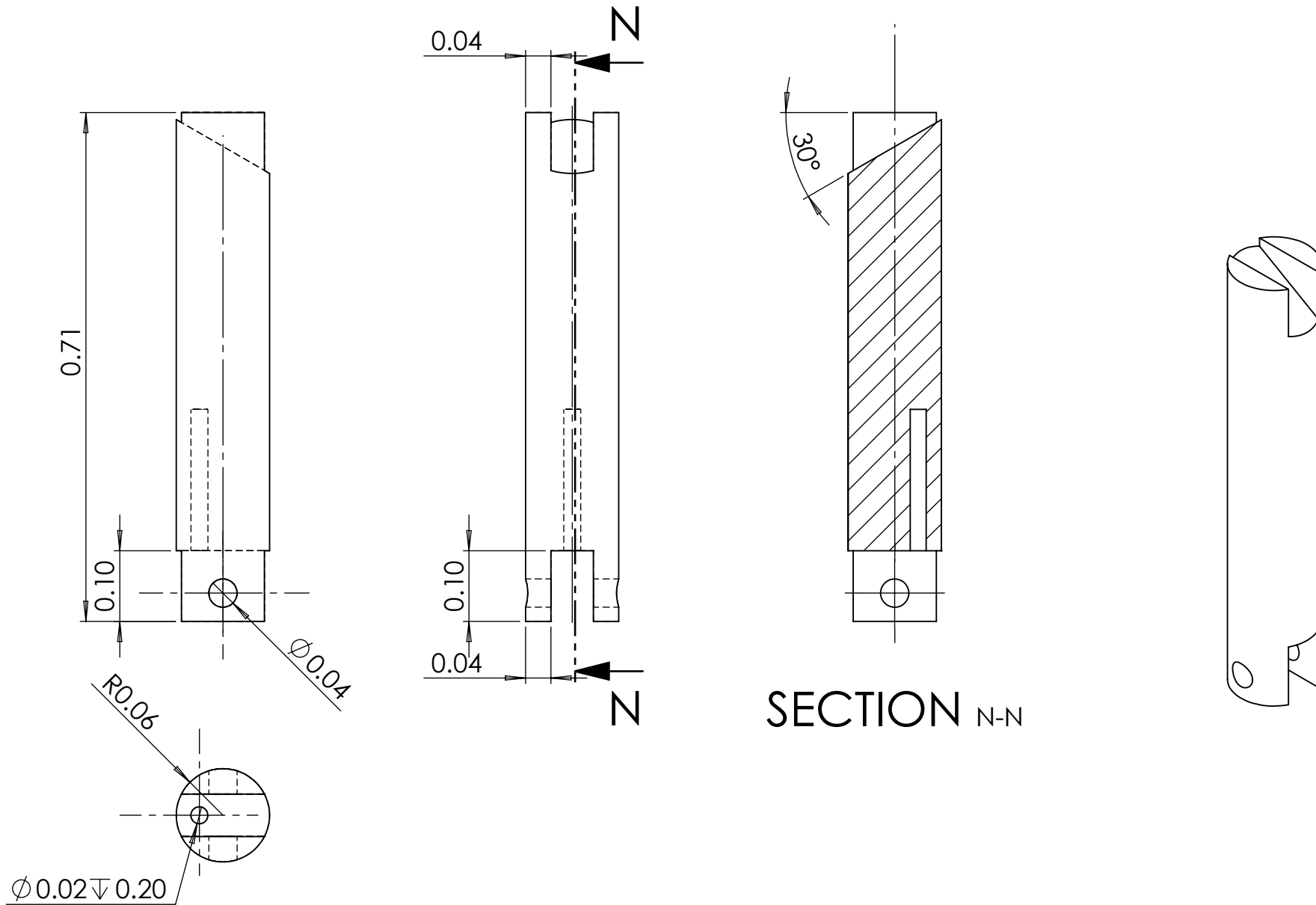
	TITLE		DATE		NAME		
	Bag Assembly		12/9/2024		Tallulah Selhorst		
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES	
	ME 1670	FALL 2024	A			SCALE 1:8	SIZE SHEET 5 OF 12 A

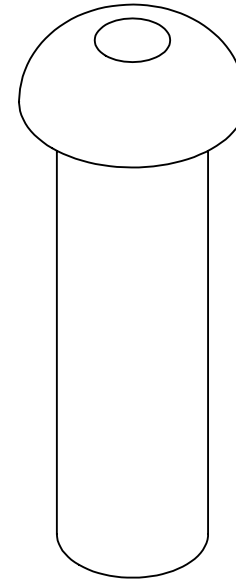
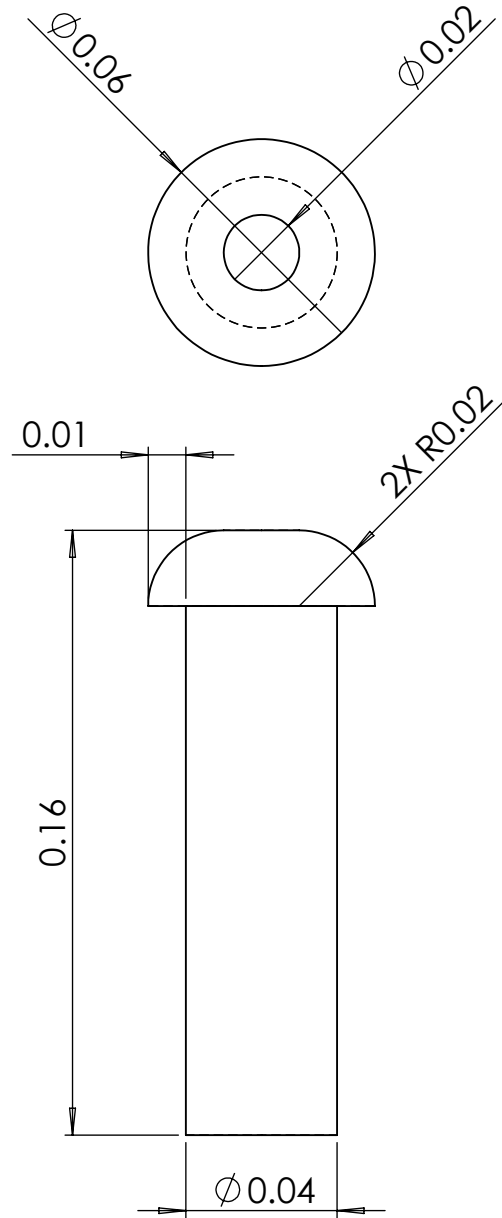


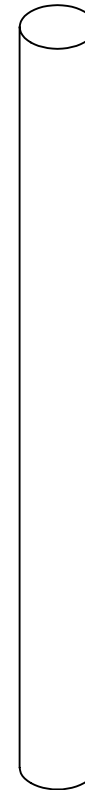
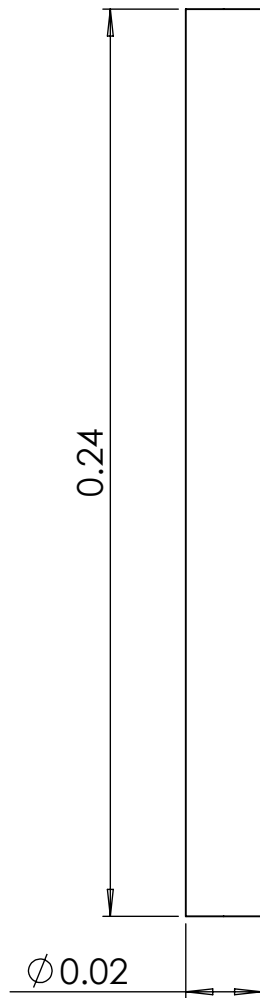
DETAIL I
SCALE 4 : 1



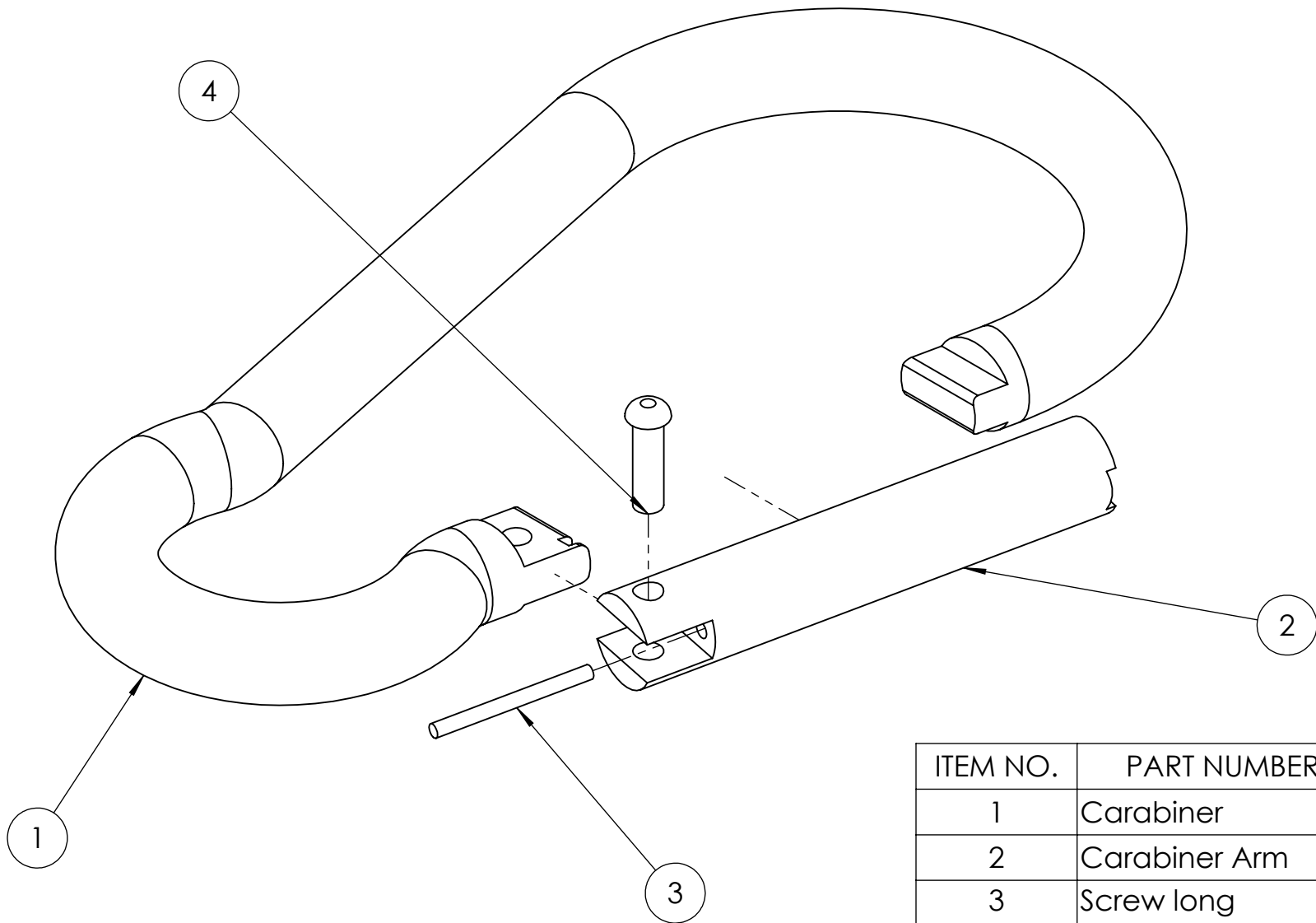
DETAIL J
SCALE 4 : 1







	TITLE		DATE		NAME			
	Carabiner Screw Long		12/9/2024		Tallulah Selhorst			
	COURSE	Carabiner	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES		SIZE
	ME 1670	FALL 2024	A			ANGLES ARE IN DEGREES		A
						SCALE 20:1	SHEET 9 OF 12	
5	4	3		2				1



ITEM NO.	PART NUMBER	QTY.
1	Carabiner	1
2	Carabiner Arm	1
3	Screw long	1
4	Screw short	1



**Georgia Institute
of Technology**

TITLE

Carabiner Assembly

COURSE

ME 1670

Carabiner

FALL 2024

DATE

12/9/2024

NAME

Tallulah Selhorst

SECTION

A

LAB

ACTIVITY

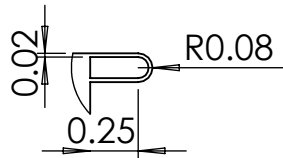
DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

SCALE 20:1 SHEET 10 OF 12

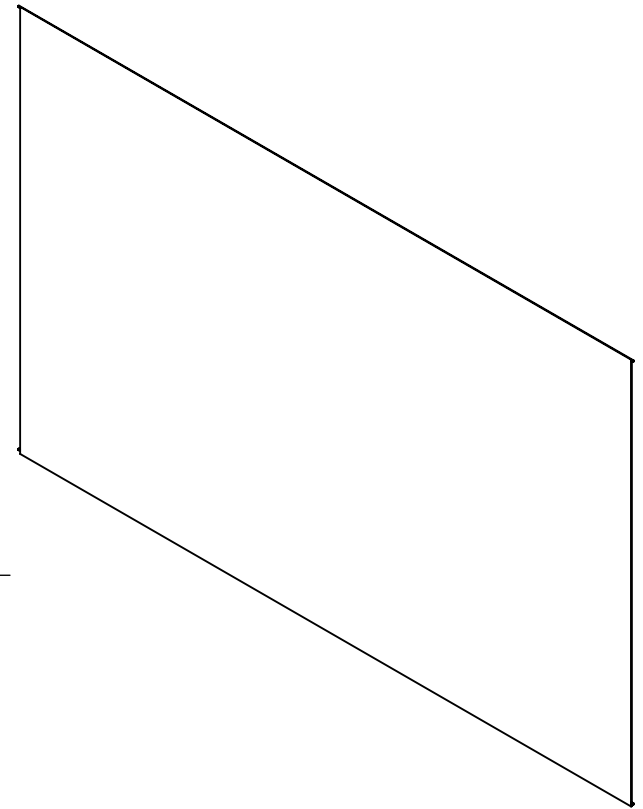
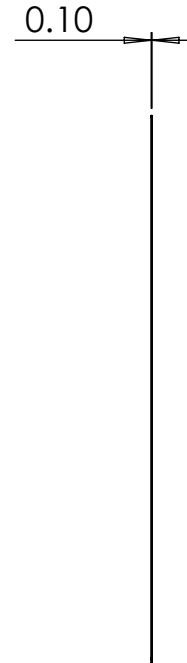
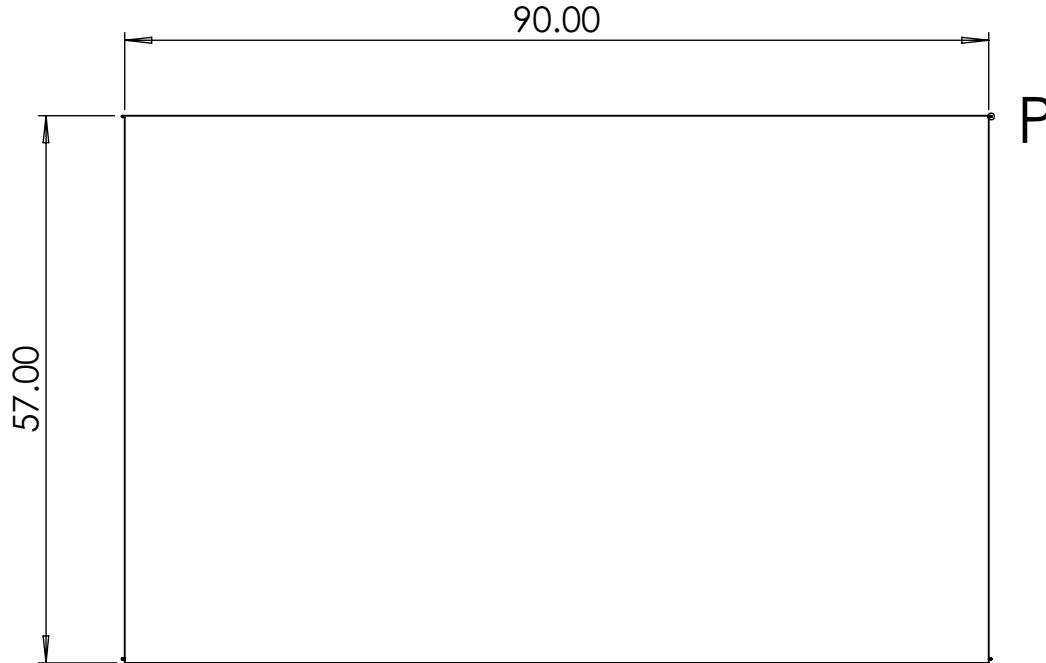
SIZE

A

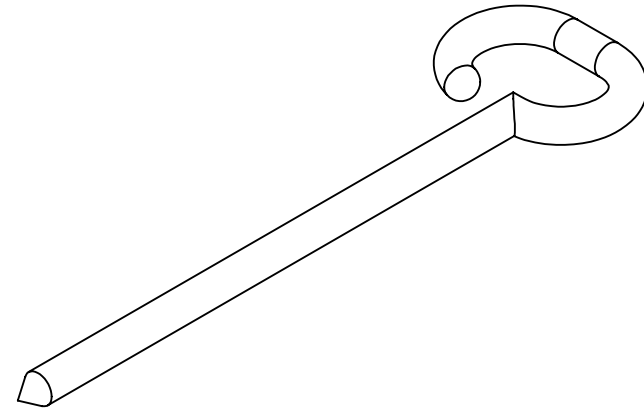
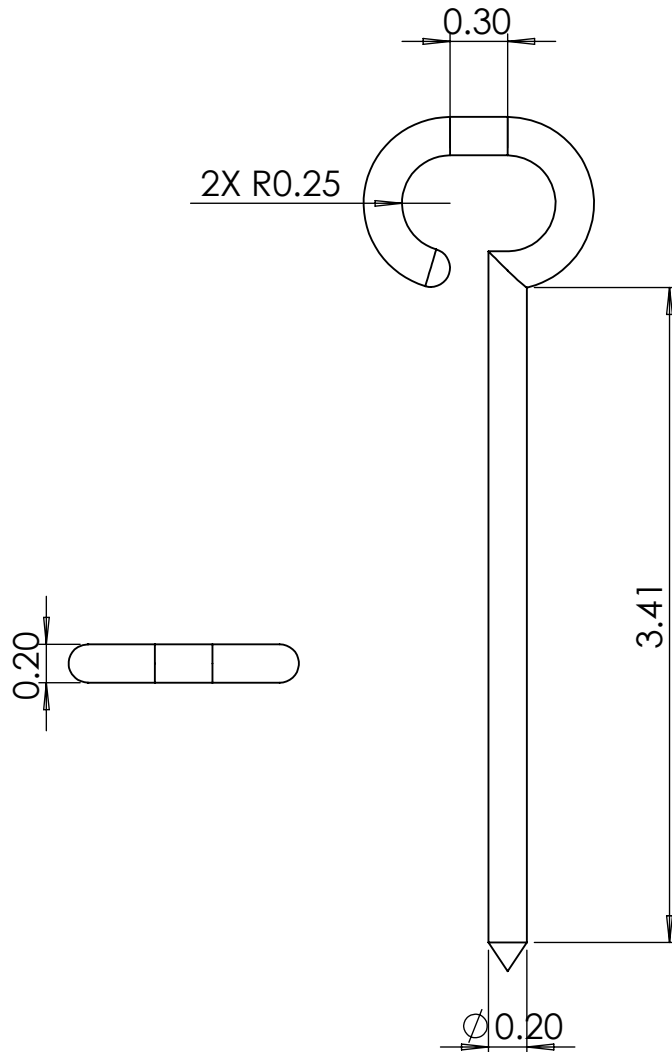
Detail P is repeated four times at every corner of the screen.



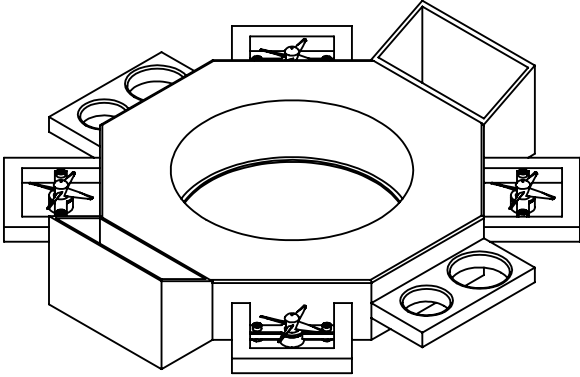
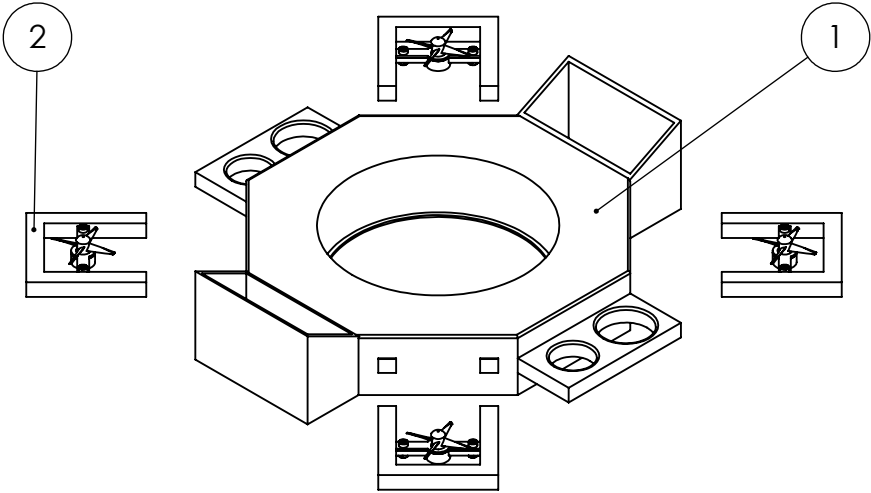
DETAIL P
SCALE 1 : 1



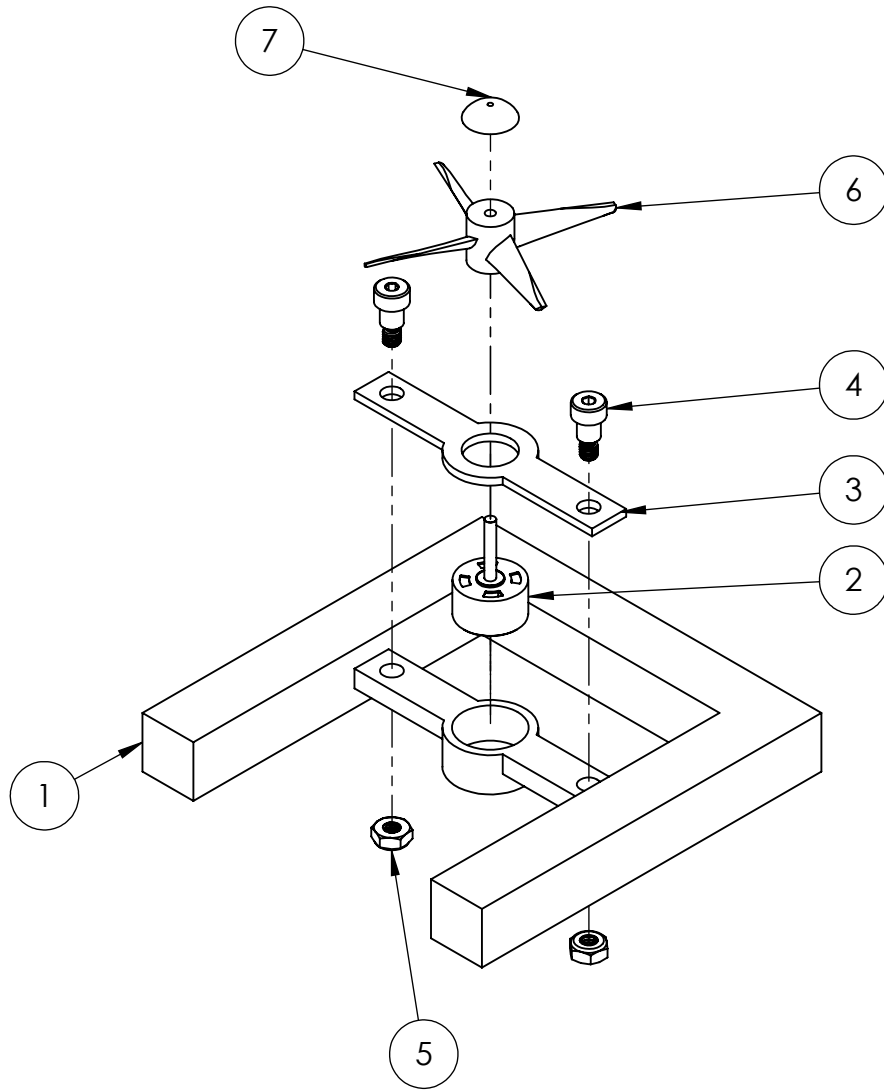
This screen is made of fabric.



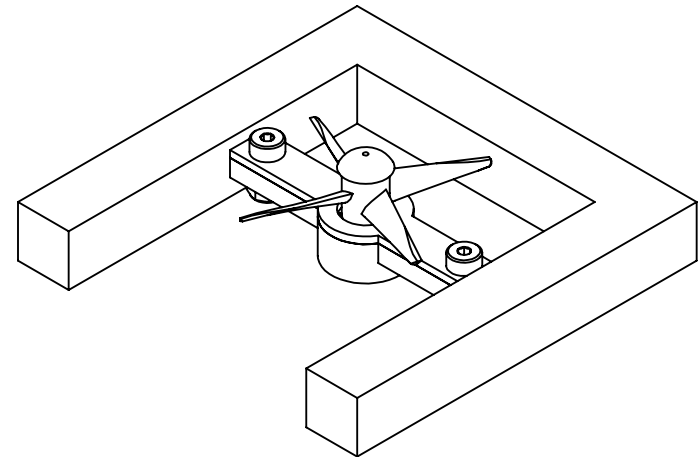
ITEM NO.	PART NUMBER	QTY.
1	DroneBase	1
2	FanAssembly	4

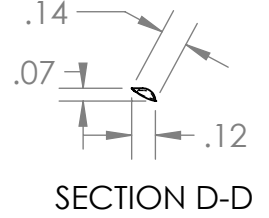
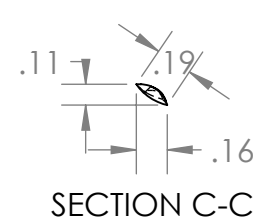
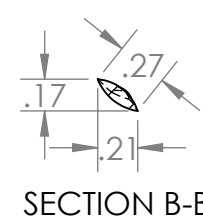
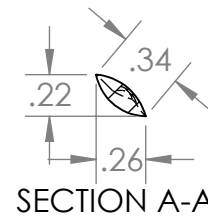
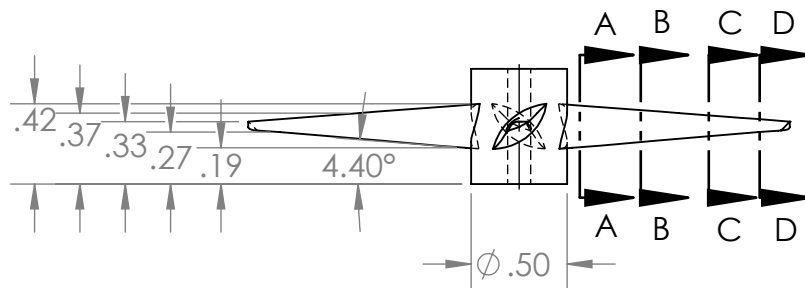
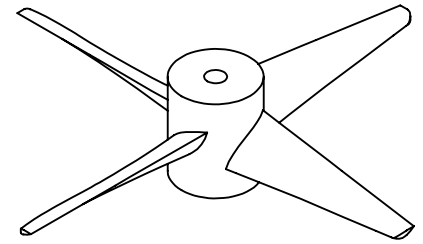
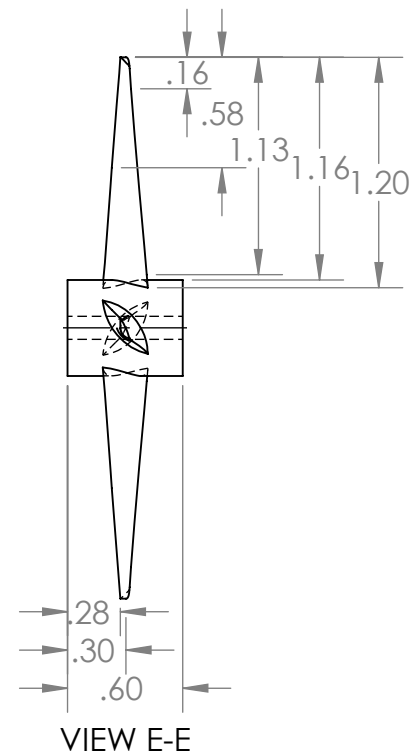
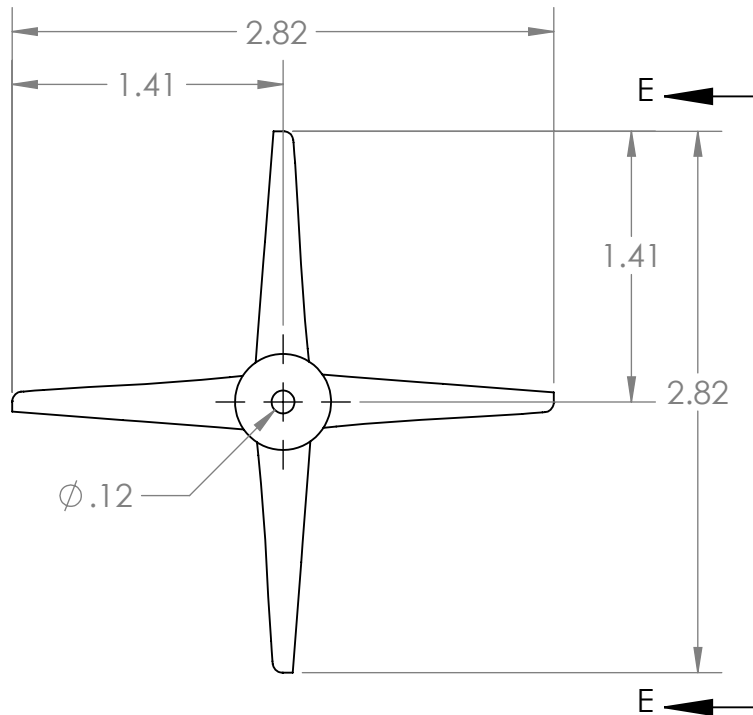


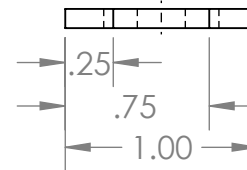
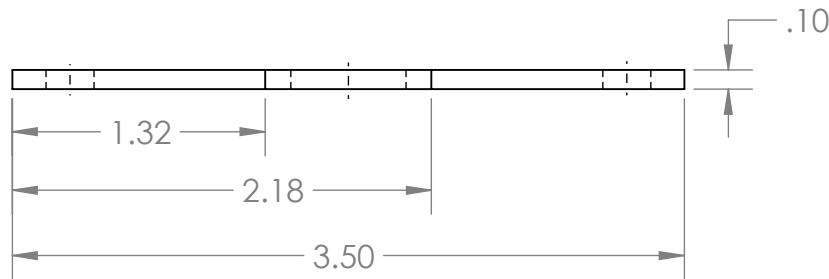
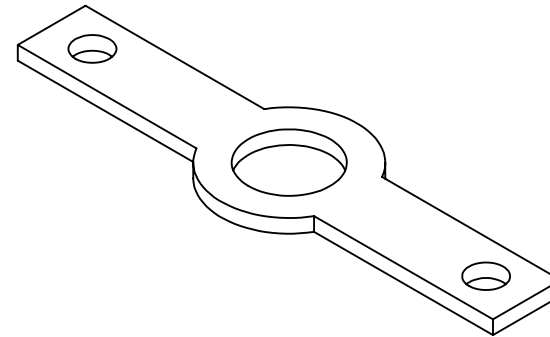
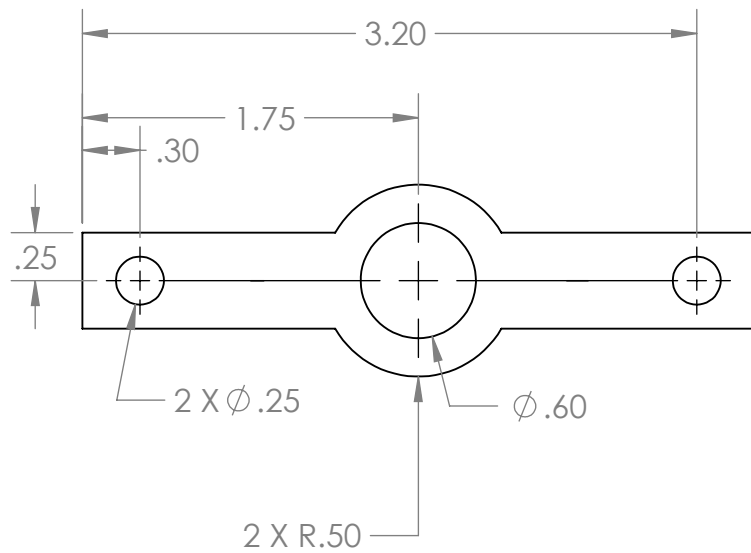
	TITLE		DATE		NAME			
	DRONE ASSEMBLY		11/29/2024		JOY D'SOUZA			
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE
	ME 1670	FALL 2024	A			SCALE 1:8	SHEET 1 OF 1	A



ITEM NO.	PART NUMBER	QTY.
1	FanBase	1
2	Motor	1
3	MotorCover	1
4	94035A178	2
5	97367A112	2
6	Fan	1
7	FanCap	1







TITLE

MOTOR COVER

COURSE
ME 1670

SEMESTER

FALL 2024

DATE

11/29/2024

NAME

LAB

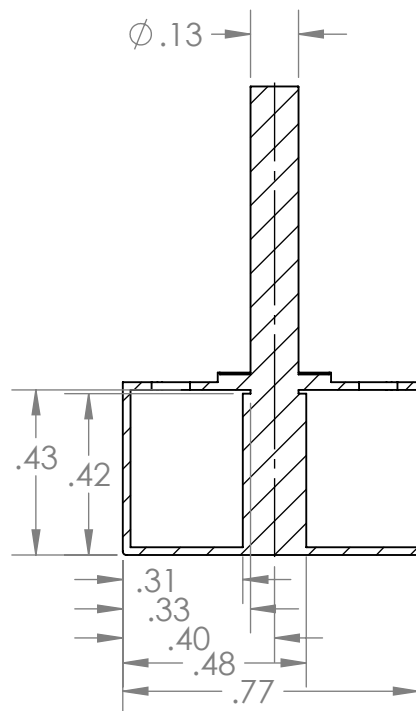
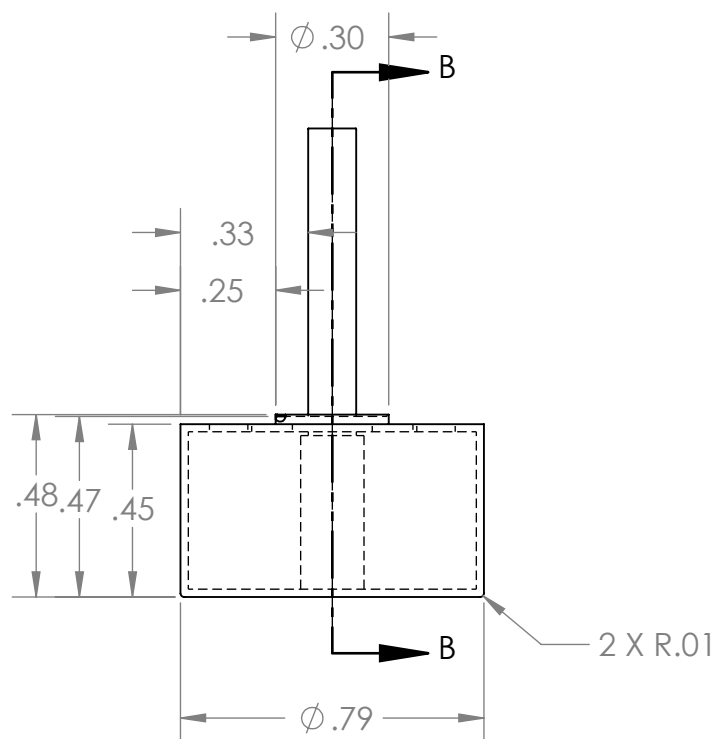
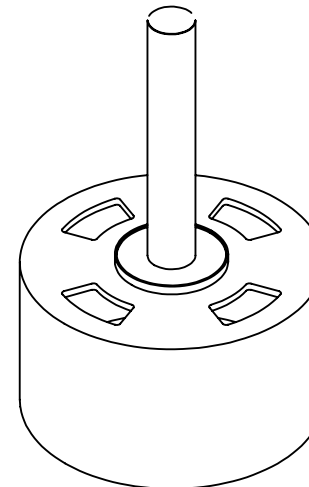
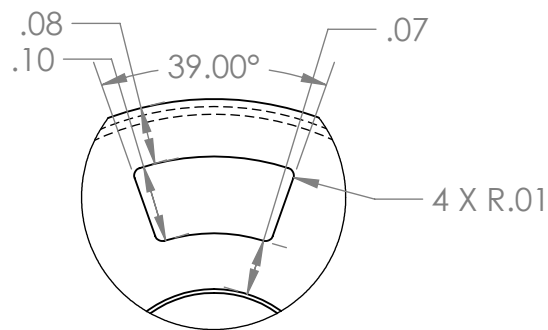
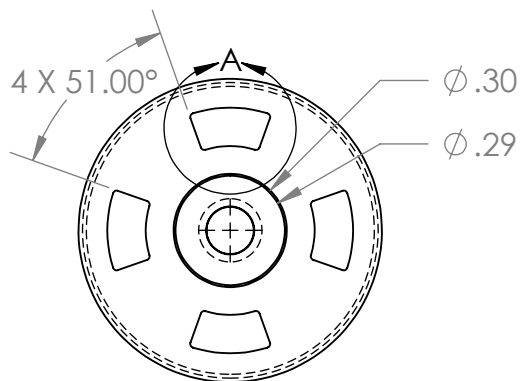
ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

SCALE 1:1 SHEET 1 OF 1

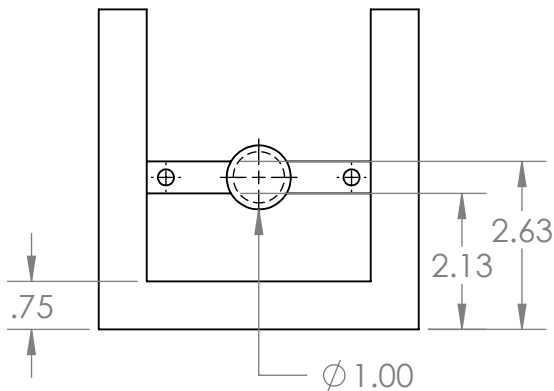
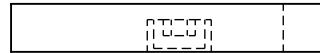
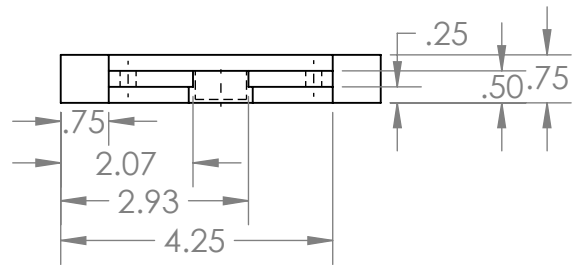
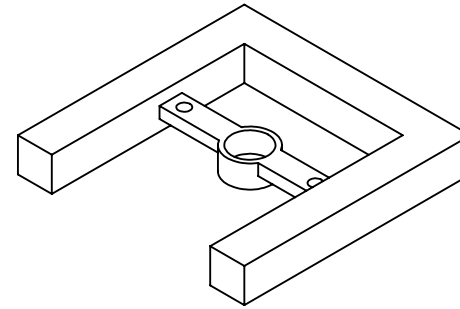
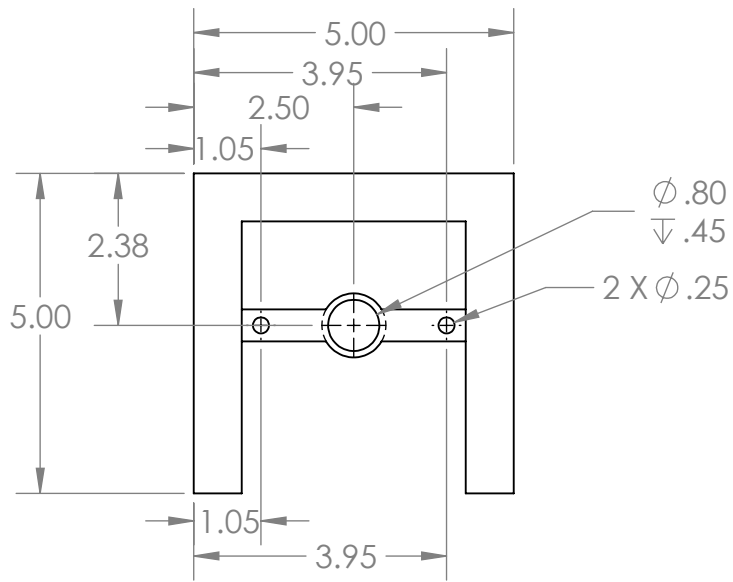
SIZE

A



UNIFORM WALL THICKNESS OF 0.02 INCHES,
UNLESS OTHERWISE SPECIFIED

CUT OUTS ON TOP FACE ONLY GO THROUGH
TOP OF MOTOR



TITLE

FAN BASE

DATE

12/9/2024

NAME

JOY D'SOUZA

COURSE

ME 1670

SEMESTER

FALL 2024

SECTION

A

LAB

ACTIVITY

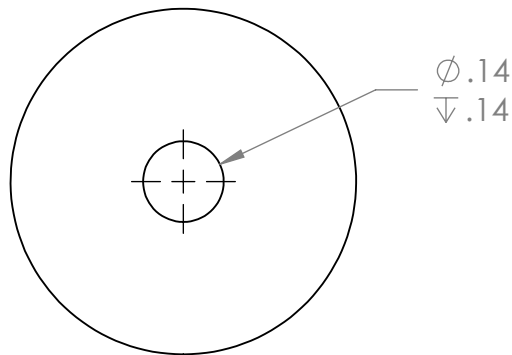
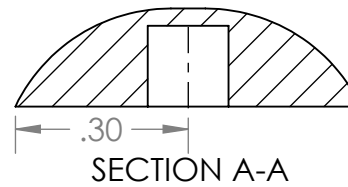
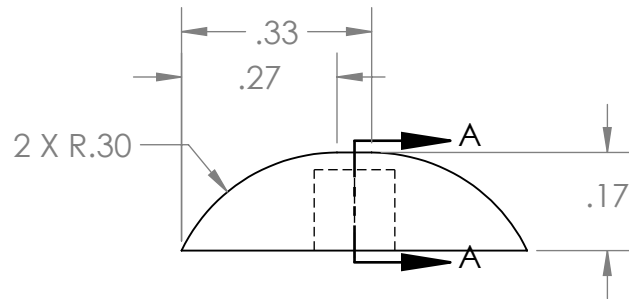
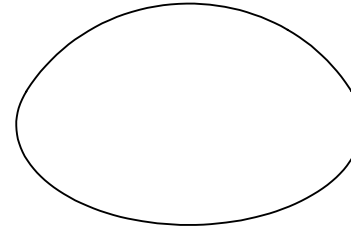
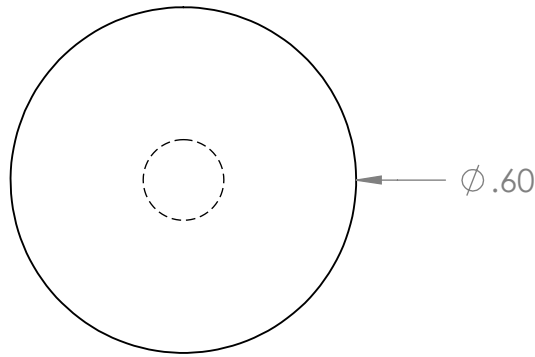
DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

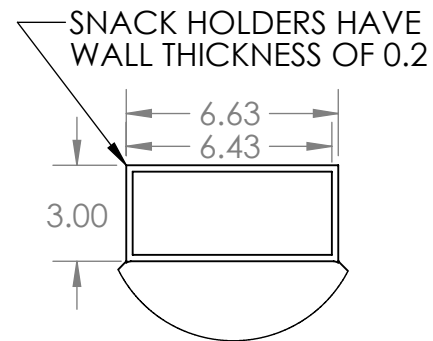
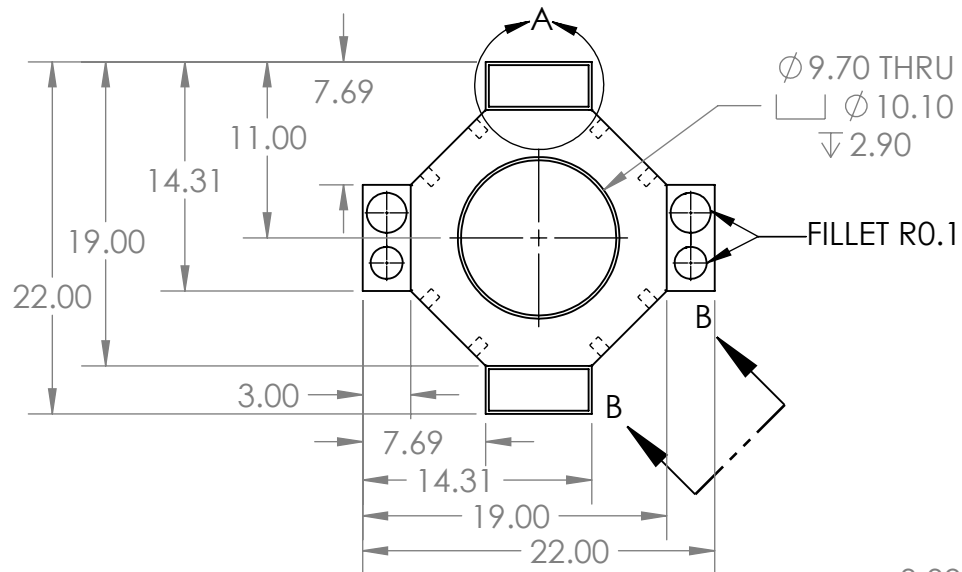
SCALE 1:3

SHEET 1 OF 1

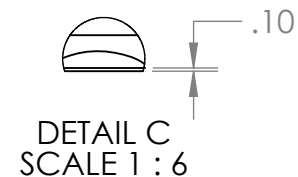
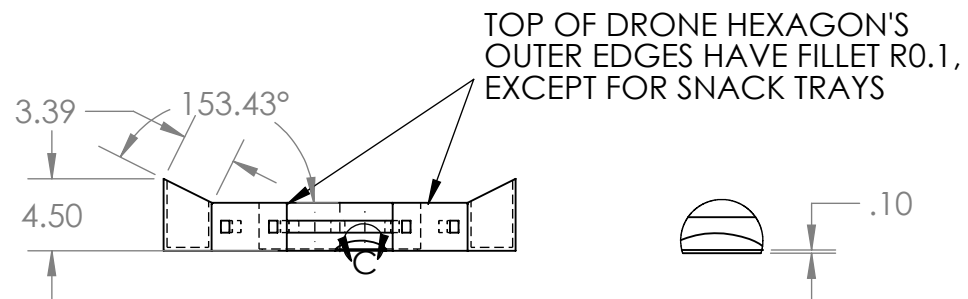
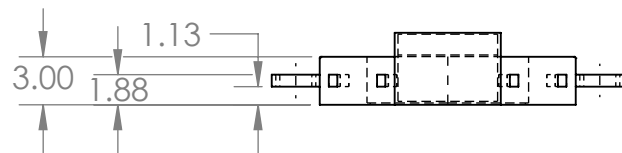
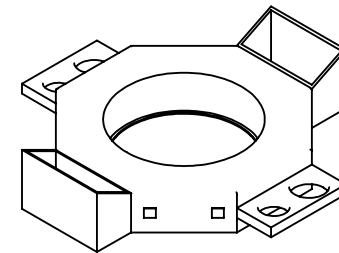
SIZE

A

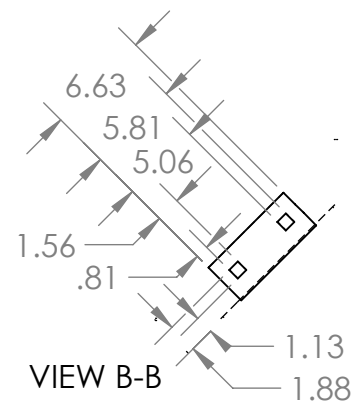
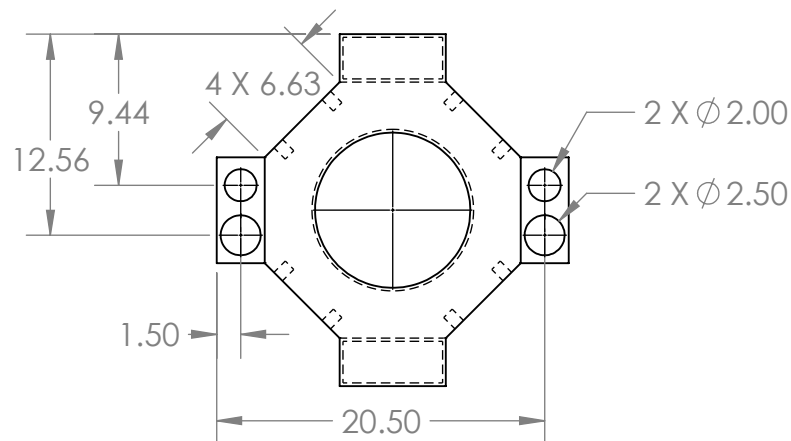




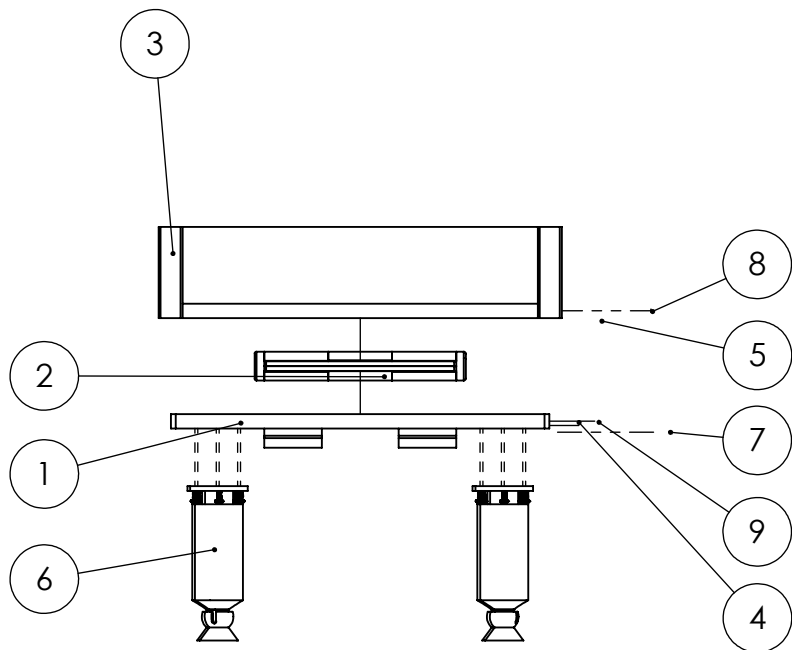
DETAIL A
SCALE 1 : 6



DETAIL C
SCALE 1 : 6



VIEW B-B



ITEM NO.	PART NUMBER	QTY.
1	bottom plate	1
2	hinge assem	1
3	Top Plate	1
4	anchor	1
5	adjustable plate circle	1
6	leg assem	4
7	pin	1
8	92470A294	18
9	90053A110	2



**Georgia Institute
of Technology**

TITLE

EXPLODED STAND

DATE

12/9/2024

NAME

DIEGO APONTE

COURSE

ME 1670

SEMESTER

FALL 2024

SECTION

A

LAB

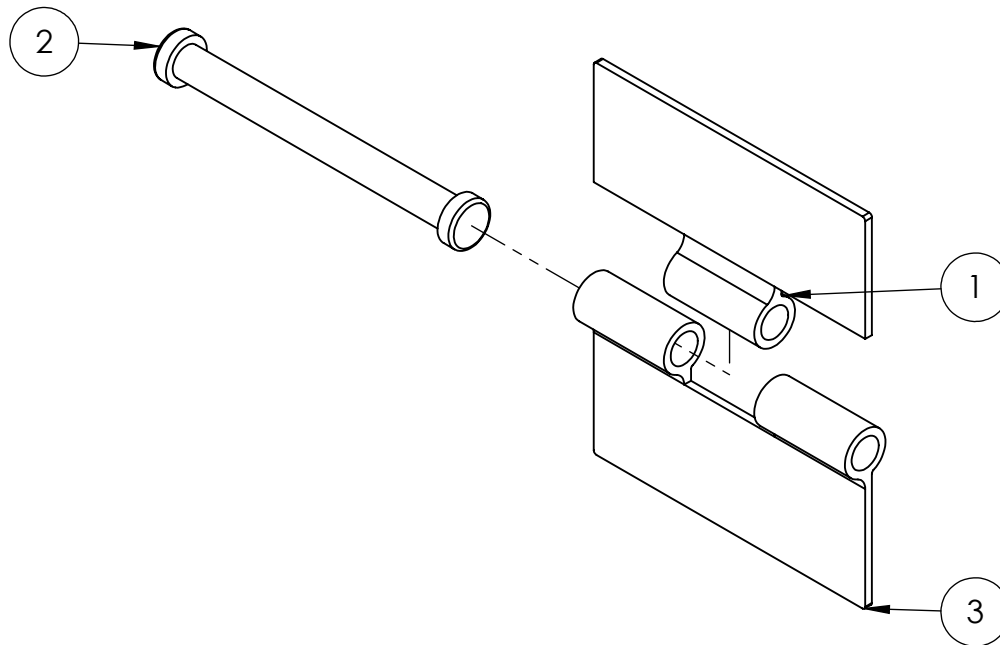
ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

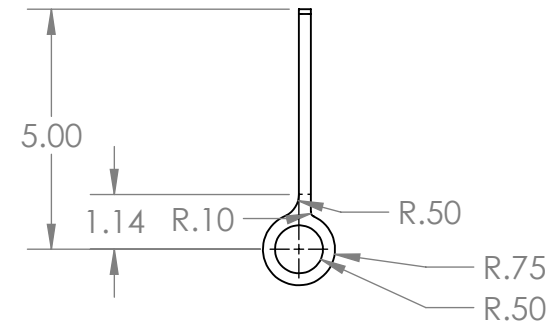
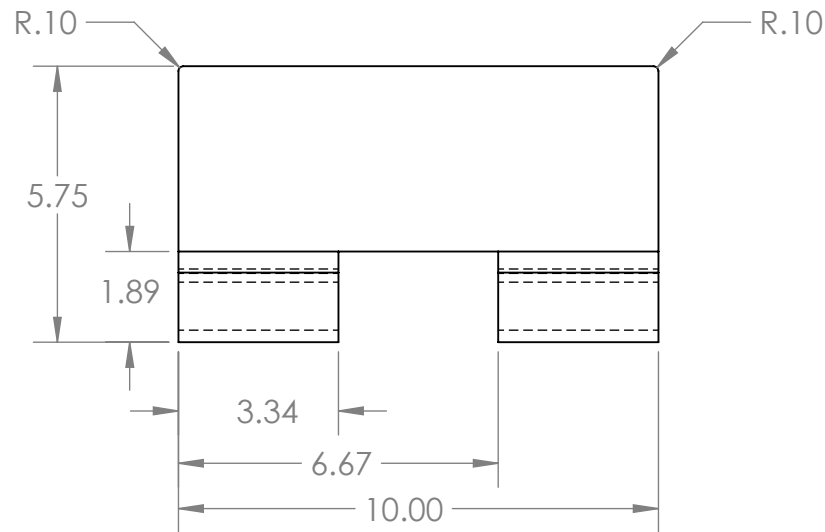
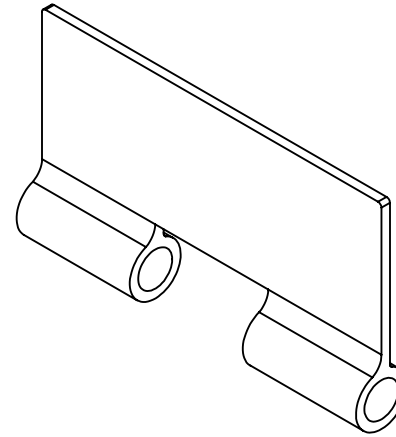
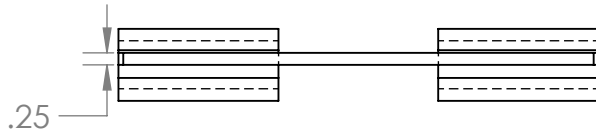
SCALE 1:10 SHEET 1 OF 1

SIZE

A



ITEM NO.	PART NUMBER	QTY.
1	hinge inside1	1
2	hinge pin	1
3	hinge outside	1



TITLE
HINGE OUTSIDE

DATE
12/9/2024

NAME
DIEGO APONTE

COURSE
ME 1670

SEMESTER
FALL 2024

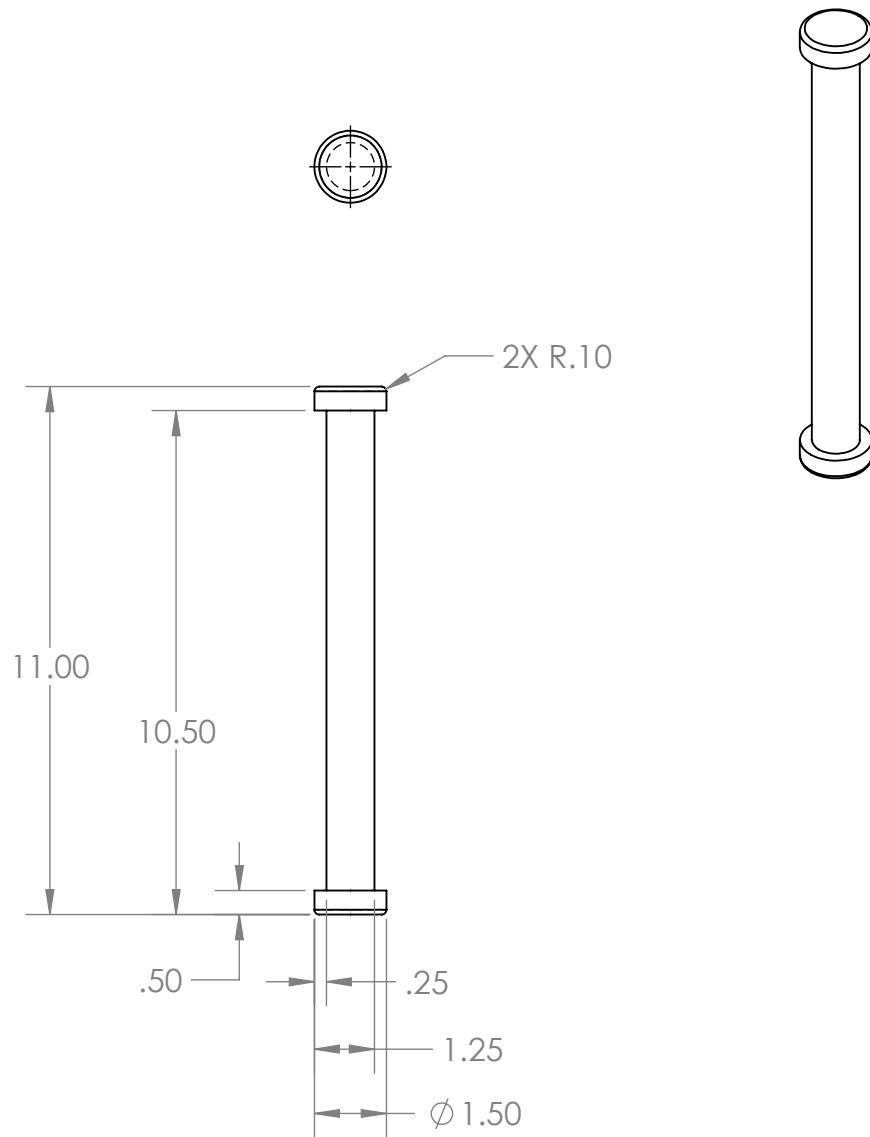
SECTION
A

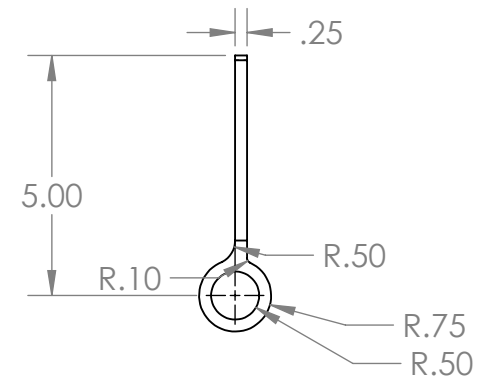
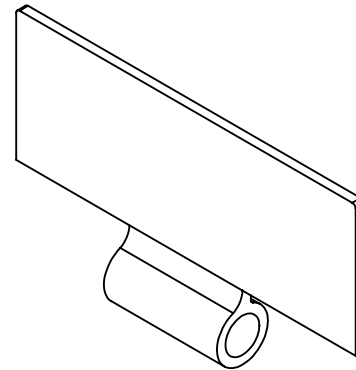
LAB

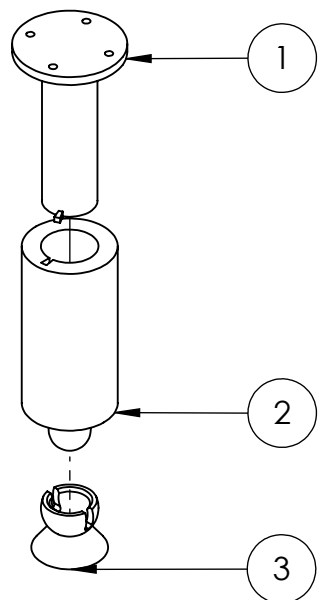
ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES
SCALE 1:4 SHEET 1 OF 1

SIZE
A



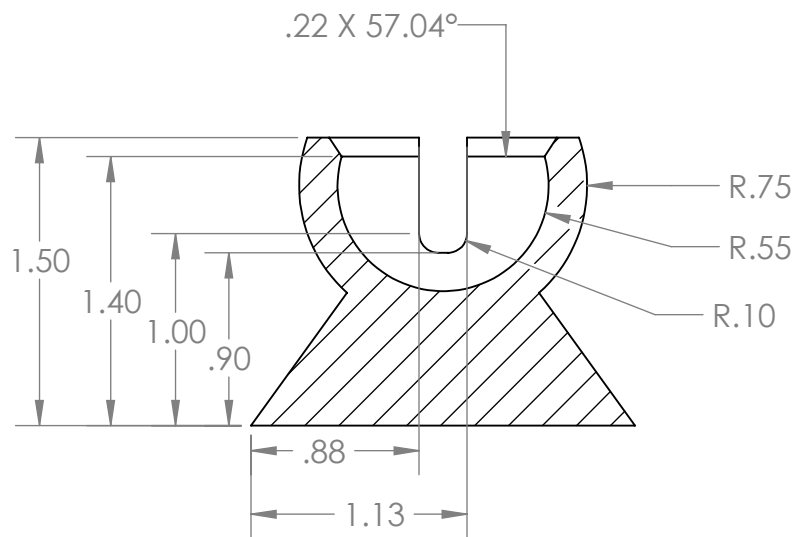
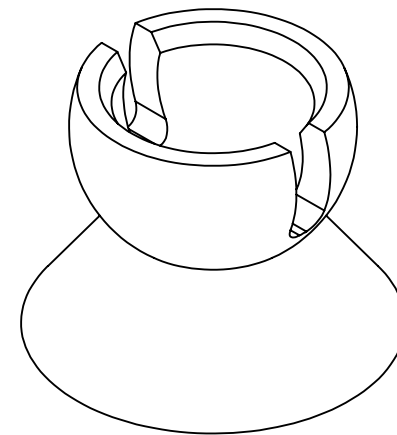
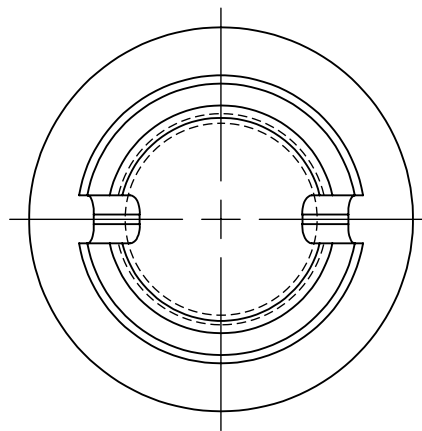




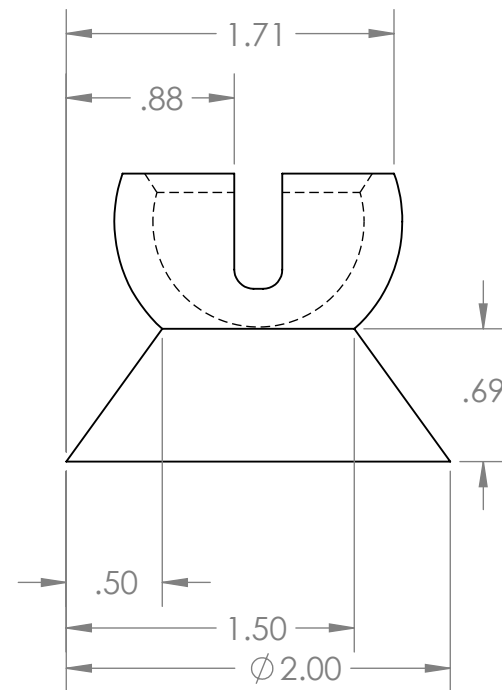
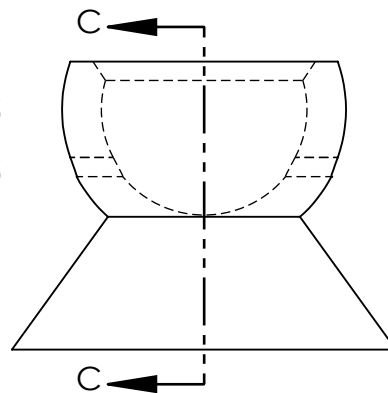
ITEM NO.	PART NUMBER	QTY.
1	leg inside	1
2	leg outside	1
3	foot	1



TITLE		DATE		NAME			
EXPLODED LEG		12/9/2024		DIEGO APONTE			
COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES		SIZE
ME 1670	FALL 2024	A			SCALE 1:4	SHEET 1 OF 1	A



SECTION C-C



TITLE
FOOT

COURSE
ME 1670

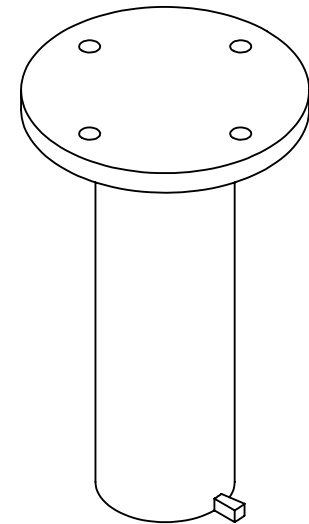
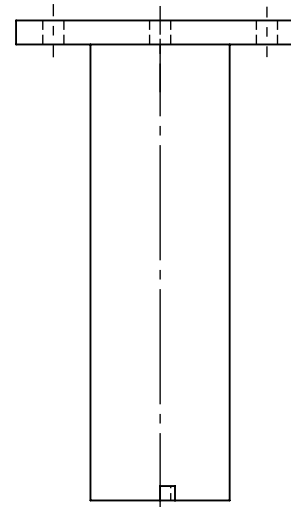
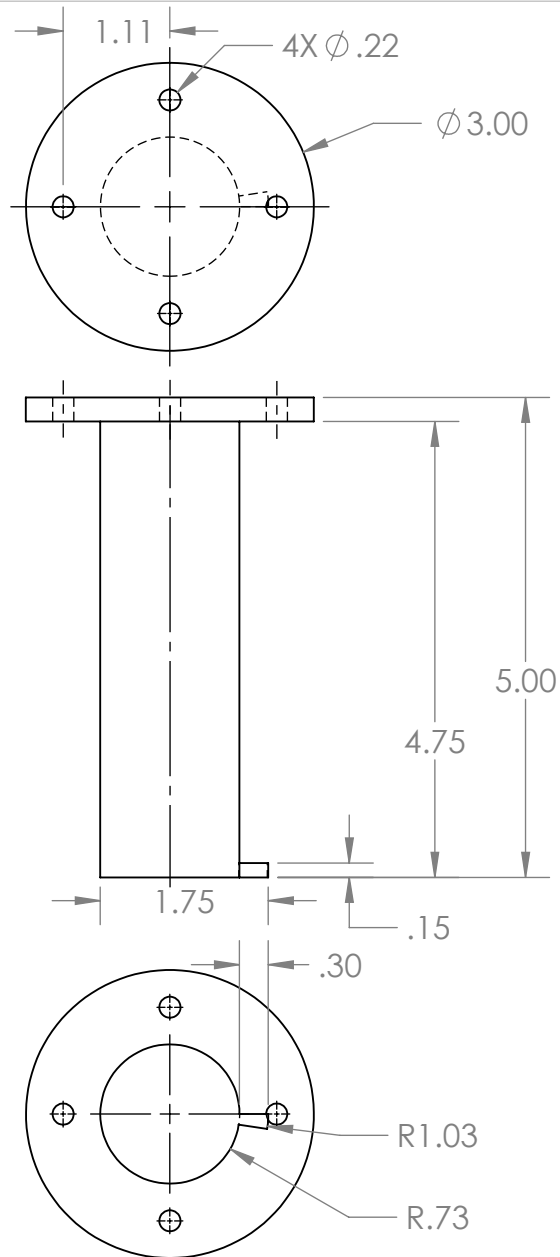
SEMESTER
FALL 2024

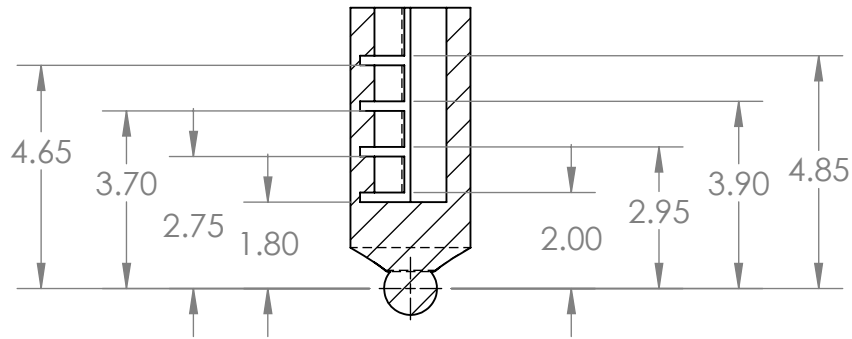
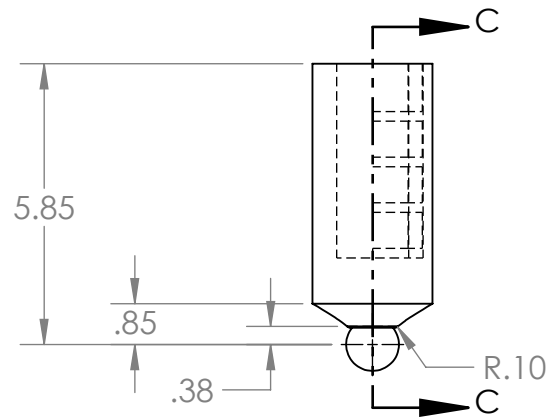
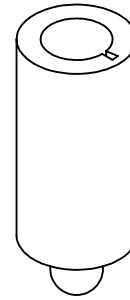
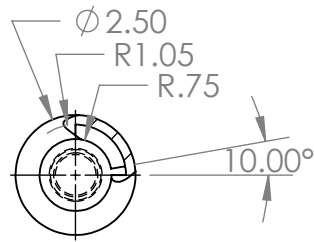
DATE
12/9/2024

SECTION
A

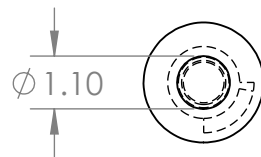
NAME
DIEGO APONTE

LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES	SIZE A
		SCALE 1:1	SHEET 1 OF 1





SECTION C-C



**Georgia Institute
of Technology**

TITLE
LEG OUTSIDE

COURSE
ME 1670

SEMESTER
FALL 2024

DATE
12/9/2024

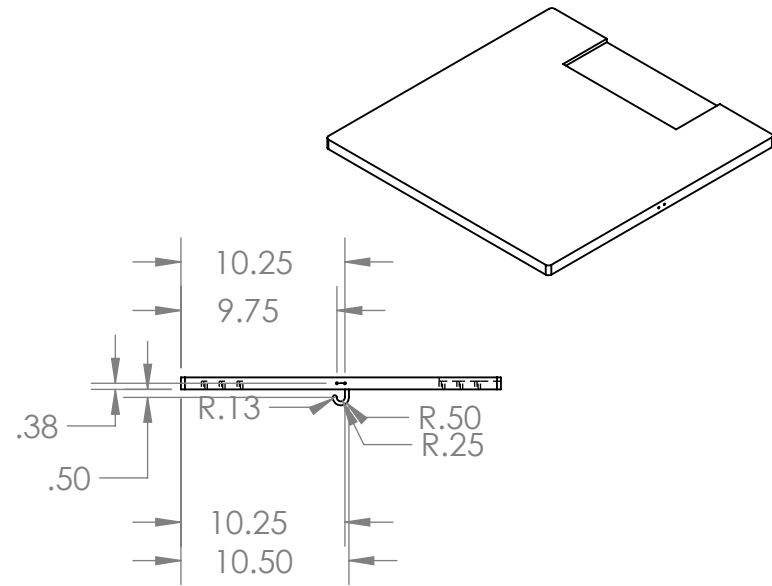
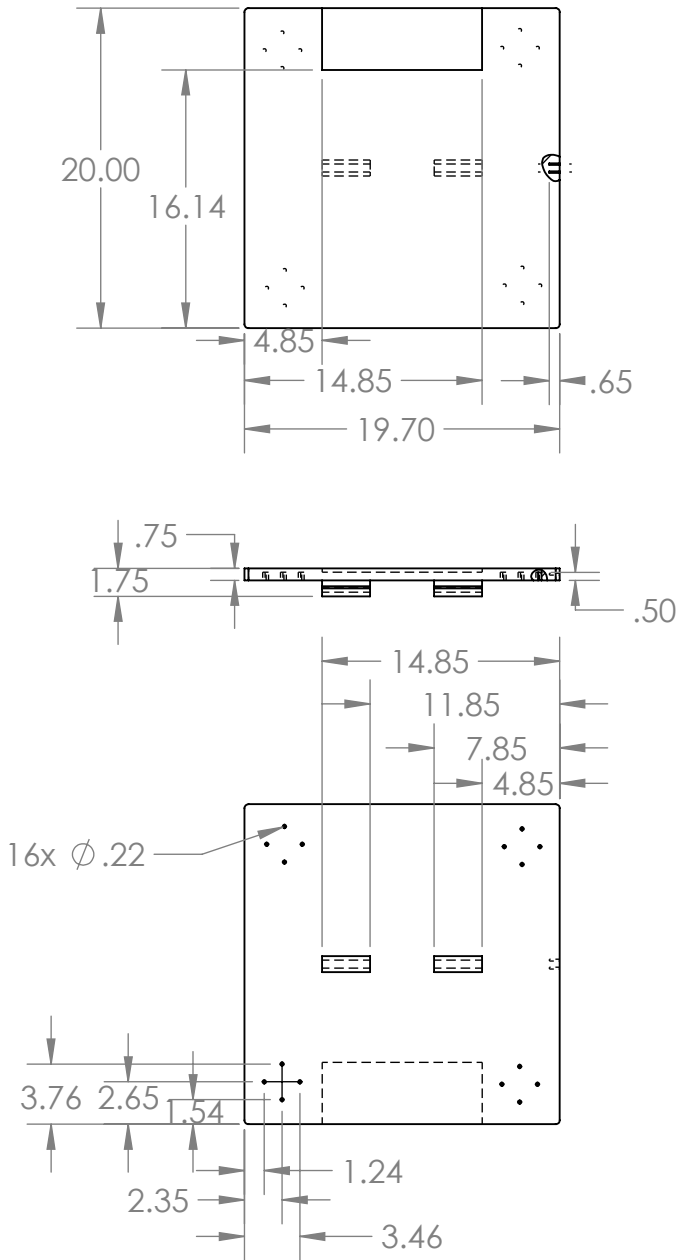
SECTION
A

NAME
DIEGO APONTE

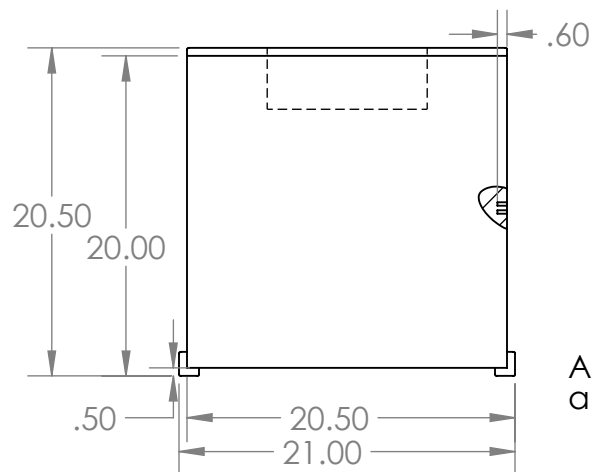
LAB ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES
SCALE 1:4 SHEET 1 OF 1

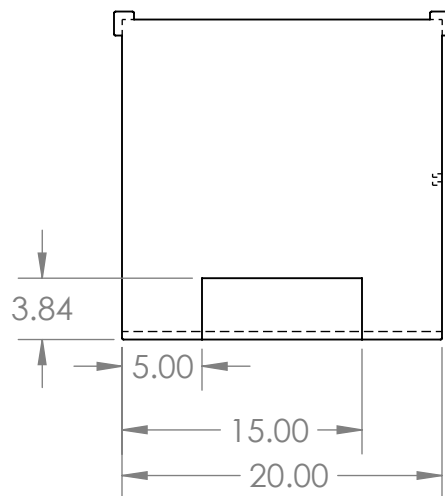
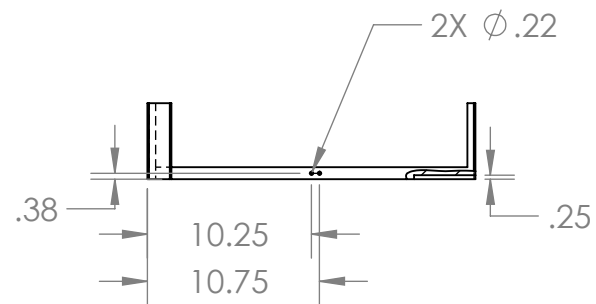
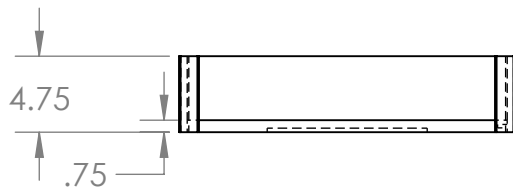
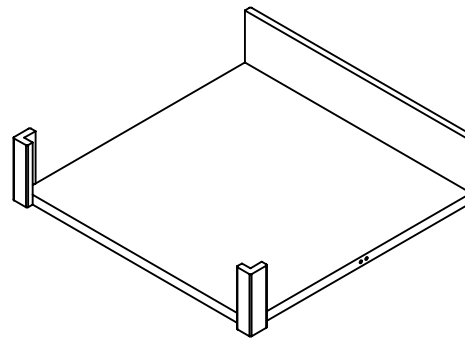
SIZE
A



	TITLE		DATE		NAME		
	BOTTOM PLATE		12/9/2024		DIEGO APONTE		
	COURSE	SEMESTER	SECTION	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES	
	ME 1670	FALL 2024	A			SCALE 1:12	SIZE A
						SHEET 1 OF 1	



All fillets have a radius of 0.1



TITLE

TOP PLATE

COURSE

ME 1670

SEMESTER

FALL 2024

DATE

12/9/2024

SECTION

A

NAME

DIEGO APONTE

LAB

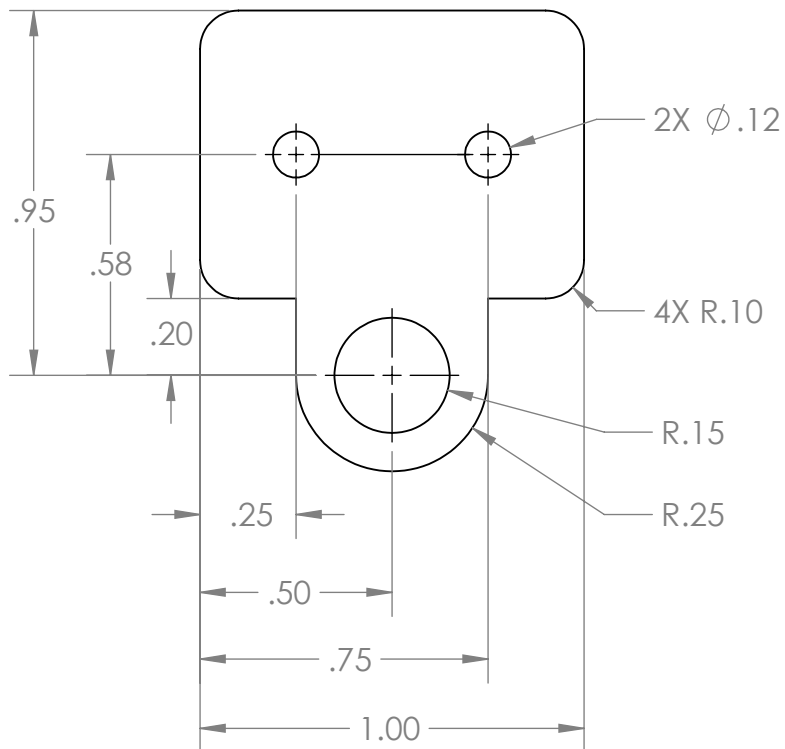
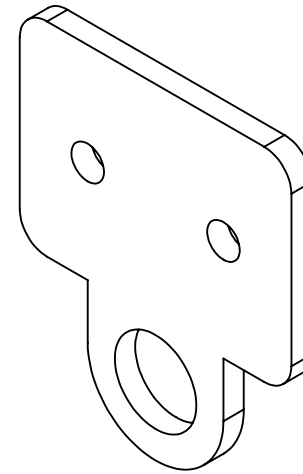
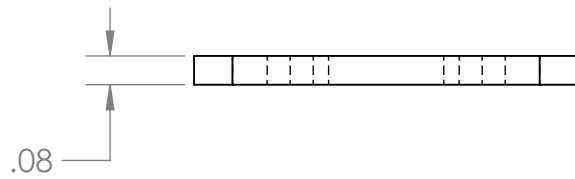
ACTIVITY

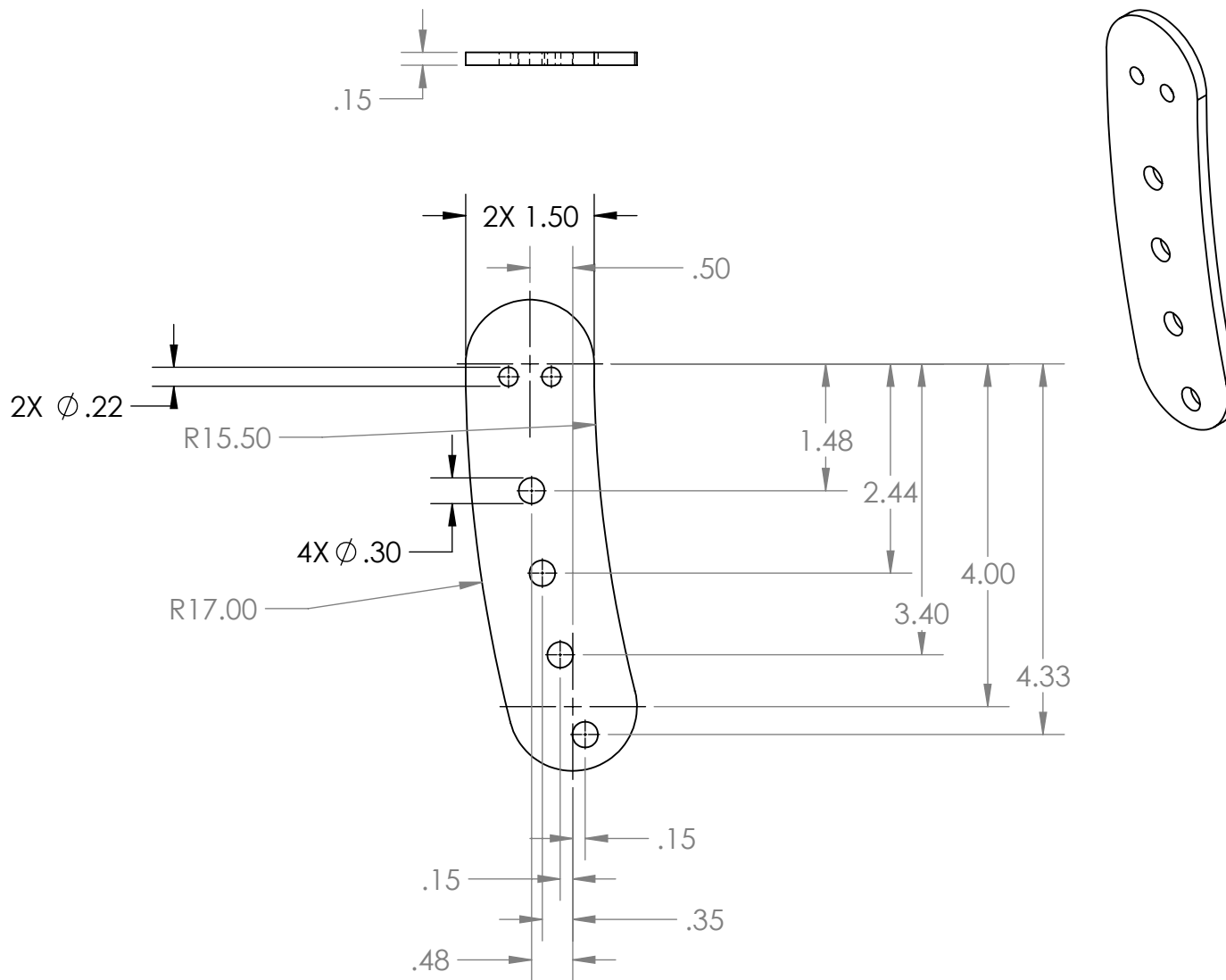
DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES

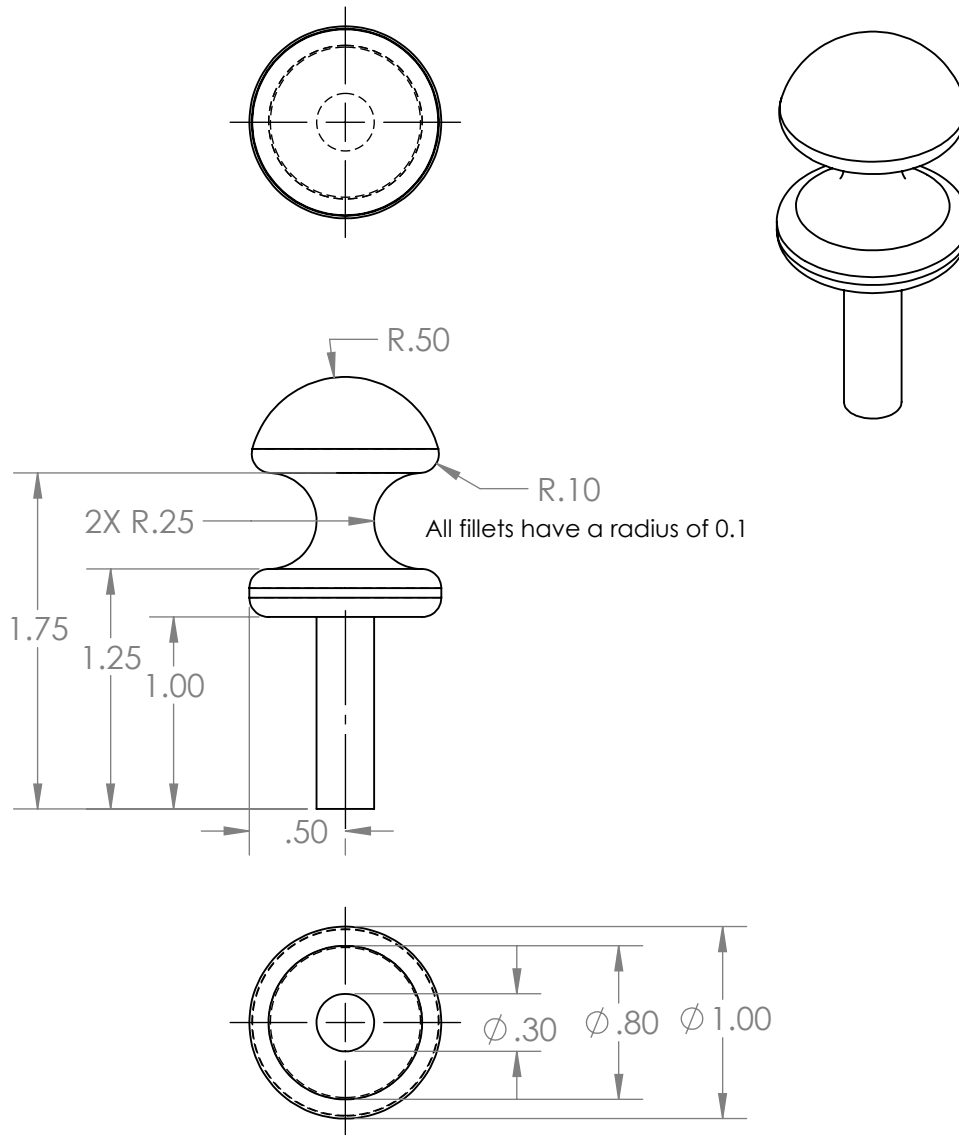
SCALE 1:12 SHEET 1 OF 1

SIZE

A







TITLE
PIN

COURSE
ME 1670

SEMESTER
FALL 2024

DATE
12/9/2024

SECTION
A

NAME
DIEGO APONTE

LAB

ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES
SCALE 1:1 SHEET 1 OF 1

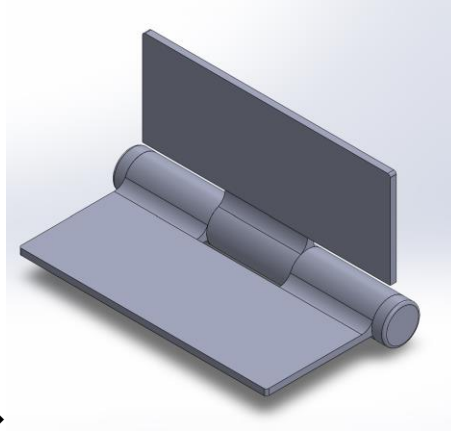
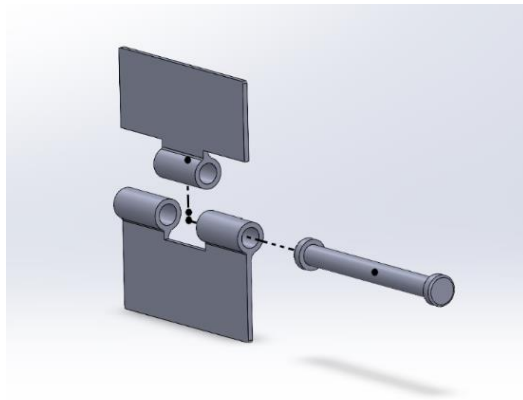
SIZE
A

Assembly Instructions

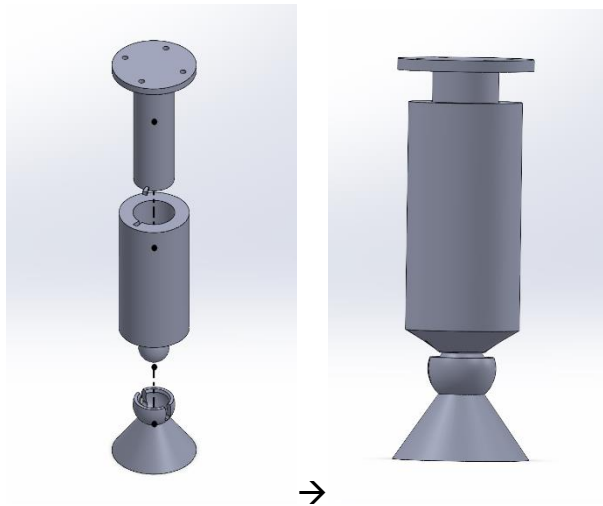
Stand Assembly

Parts List
• Foot (x4) • Outside Leg (x4) • Inside Leg (x4) • Bottom Plate • Top Plate • Outside Hinge • Inside Hinge • Hinge Pin • Pin • Adjustable Plate • Anchor • 90053A110_Slotted Rounded Head Screws for Sheet Metal (x2) • 92470A294_Phillips Rounded Head Screws for Sheet Metal (x18)

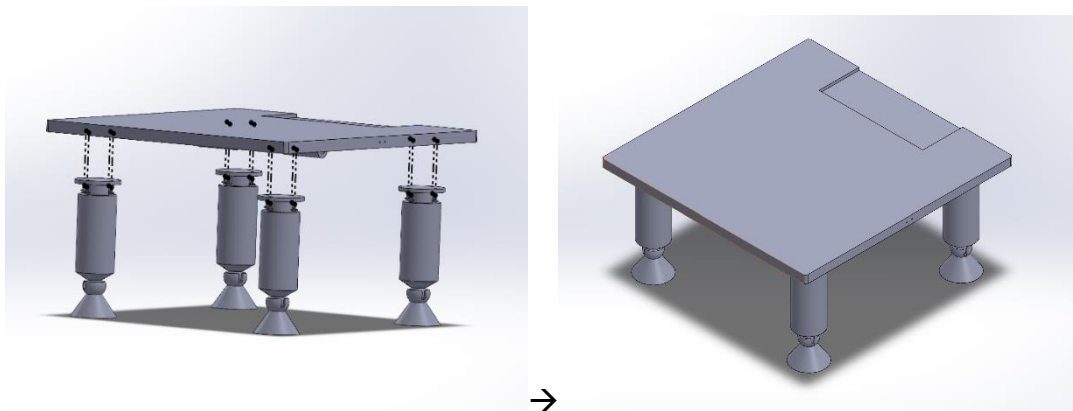
Step 1: Assemble the hinge by aligning the holes of the inside and outside hinge pieces and placing the pin inside with the face flush on the outside.



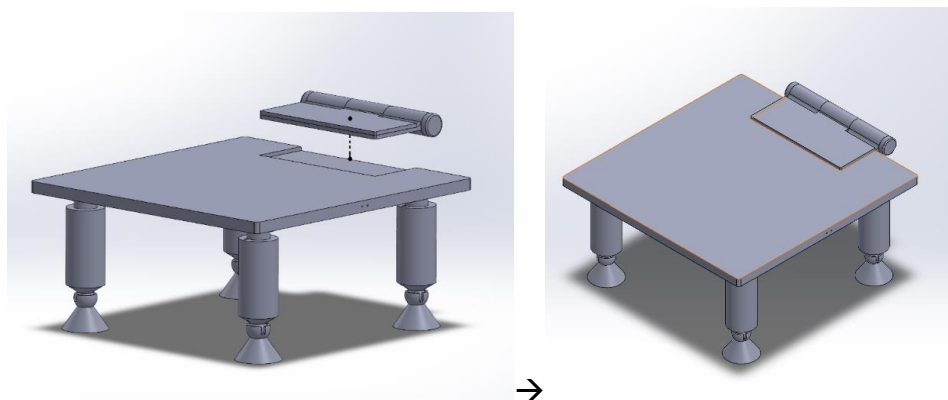
Step 2: Assemble the leg by aligning the lip of the inside leg with the rectangular hole on the outside leg sliding down to the desired height and twisting. Then place the ball of the outside leg into the socket of the foot.



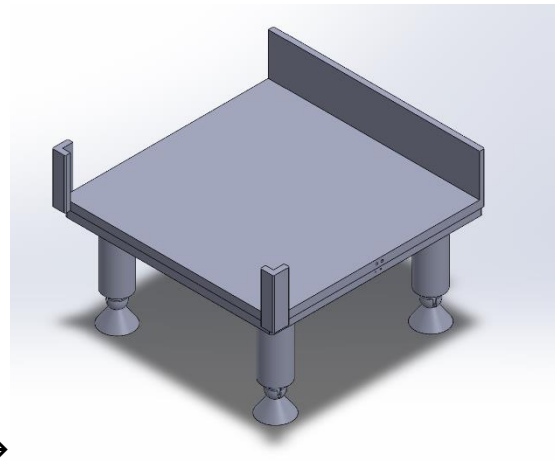
Step 3: Using 16 of the Phillips rounded head screws, screw each leg into the bottom plate using the screw holes provided. Each leg had 4 screw holes.



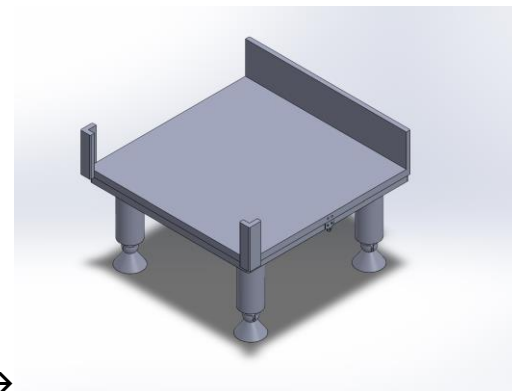
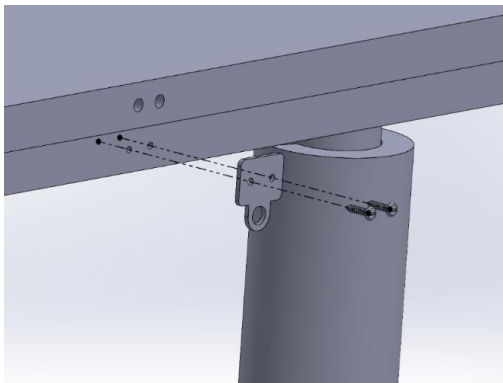
Step 4: Place one face of the hinge into the rectangular cutout on the top of the bottom plate



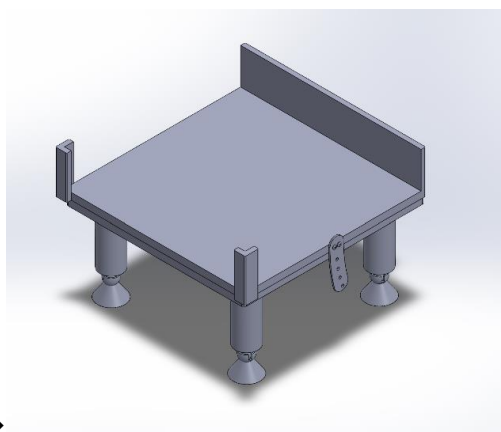
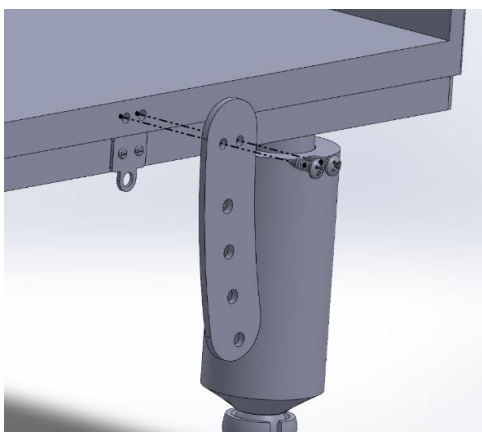
Step 5: Place the top plate of the stand onto the bottom plate and hinge, sliding the hinge into the rectangular shaped cutout on the bottom of the top plate. The bottom and top plate should be level.



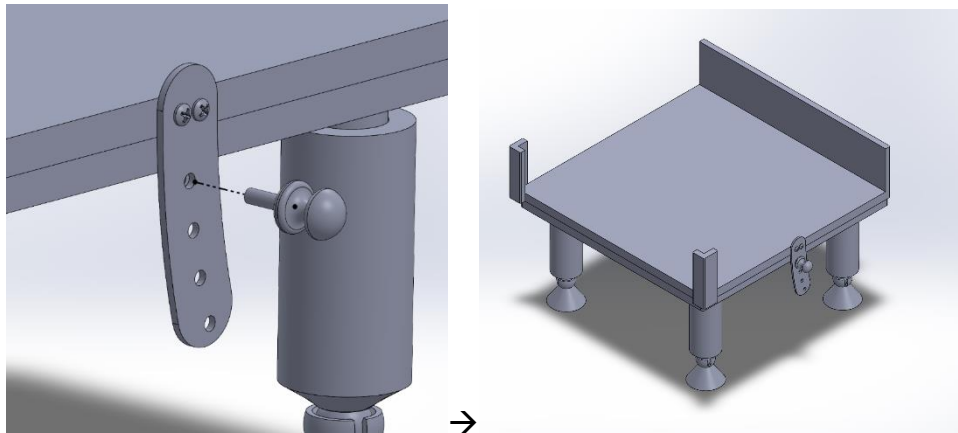
Step 6: Using the two slotted rounded head screws, attach the anchor plate to the side of the bottom plate using the provided screw holes.



Step 7: Using the last two Phillips rounded head screws, screw the adjustable plate into the top plate using the provided screw holes on the side.



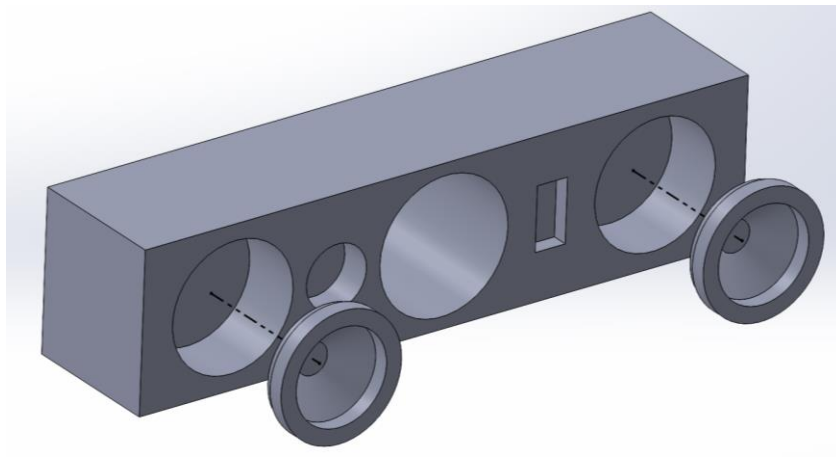
Step 8: With the hinge fully closed, the holes on the adjustable plate and anchor should line up. Finally place the pin through both the adjustable plate and the anchor to keep everything in place. Make sure the pin is all the way in, up against the adjustable plate.



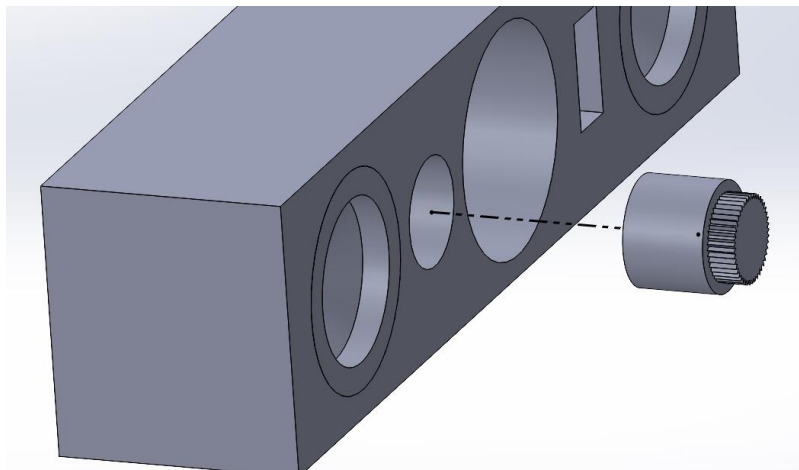
Projector Assembly

Parts List
• Projector Base (x1) • Speaker Driver (x2) • Focus Knob (x1) • Lens (x1) • Lens Adjuster (x1) • Switch Case (x1) • Switch (x1)

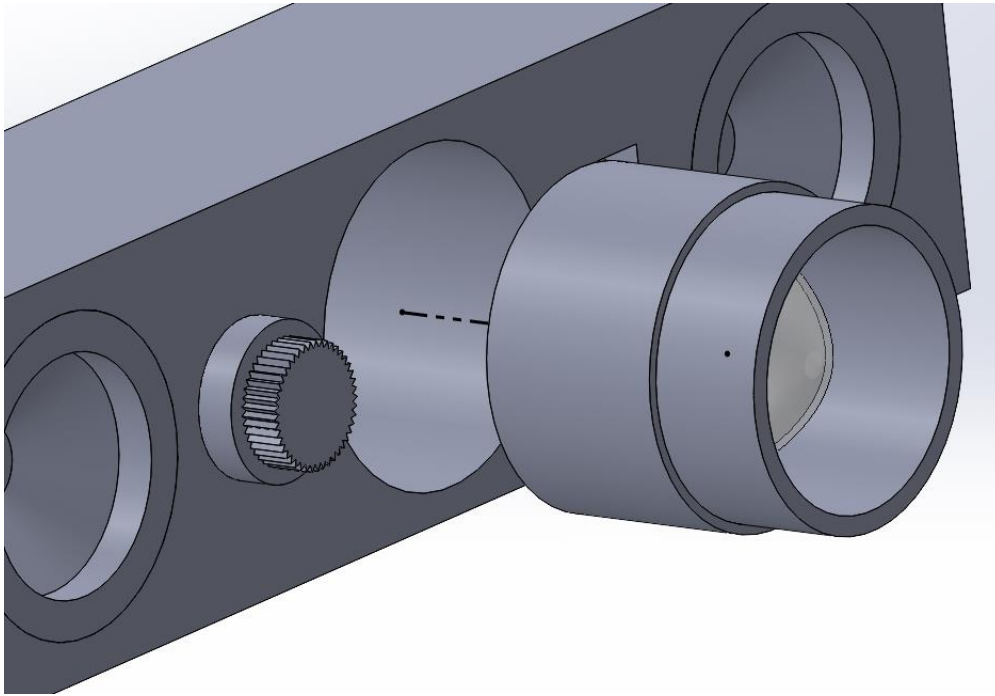
Step 1: Insert the two speaker drivers into the outermost holes of the projector box until they are fully pressed in, glue around the outside of the speaker drivers to keep them in place.



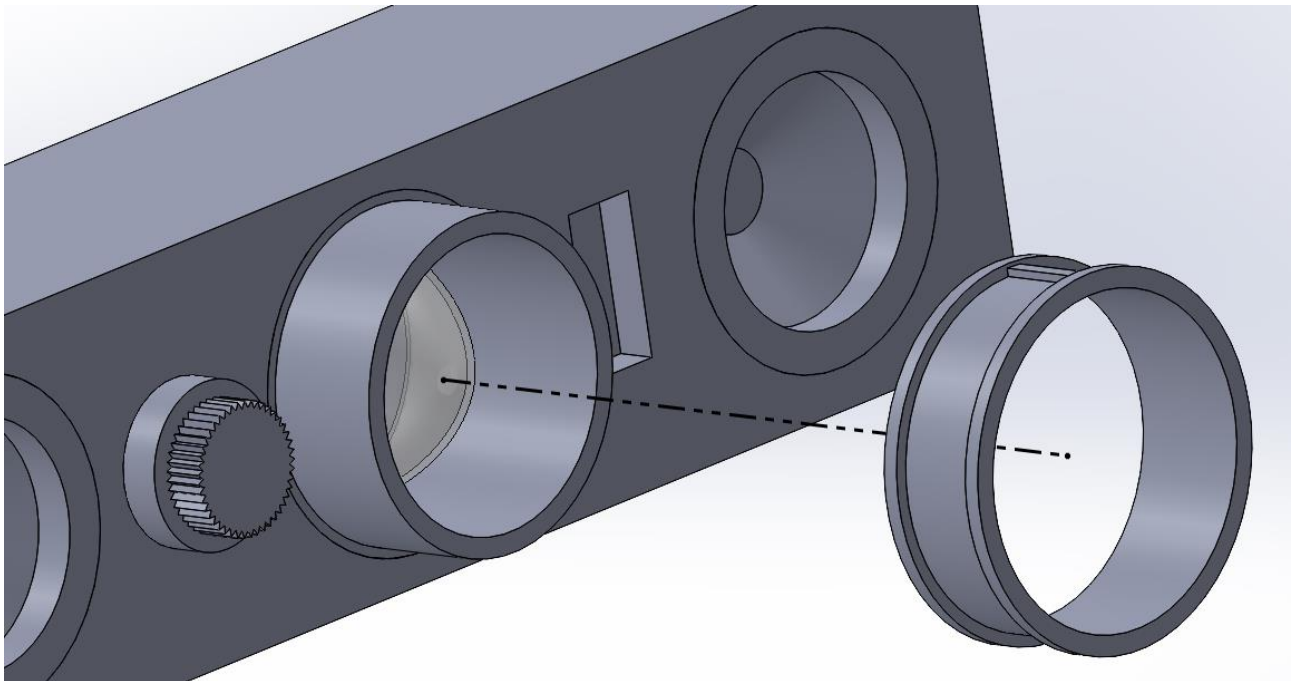
Step 2: Glue around the portion of the focus knob with the bigger diameter and insert it fully into the second hole from the left.



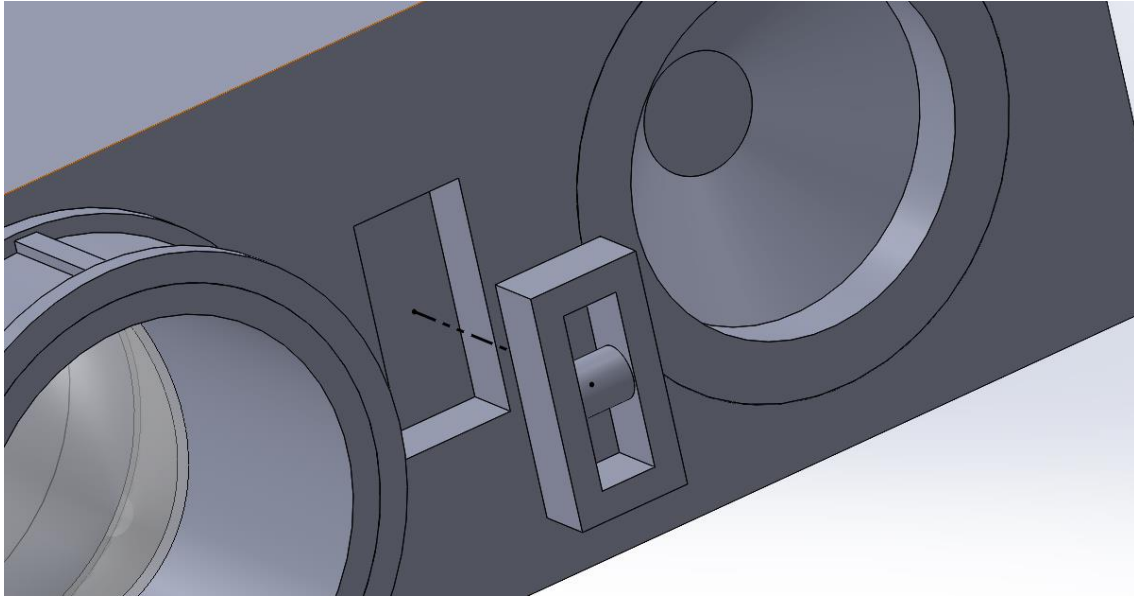
Step 3: Glue around the outside of the thicker diameter part of the projector lens, and push it into the hole in the middle of the projector box until the back of both parts are flush with each other.



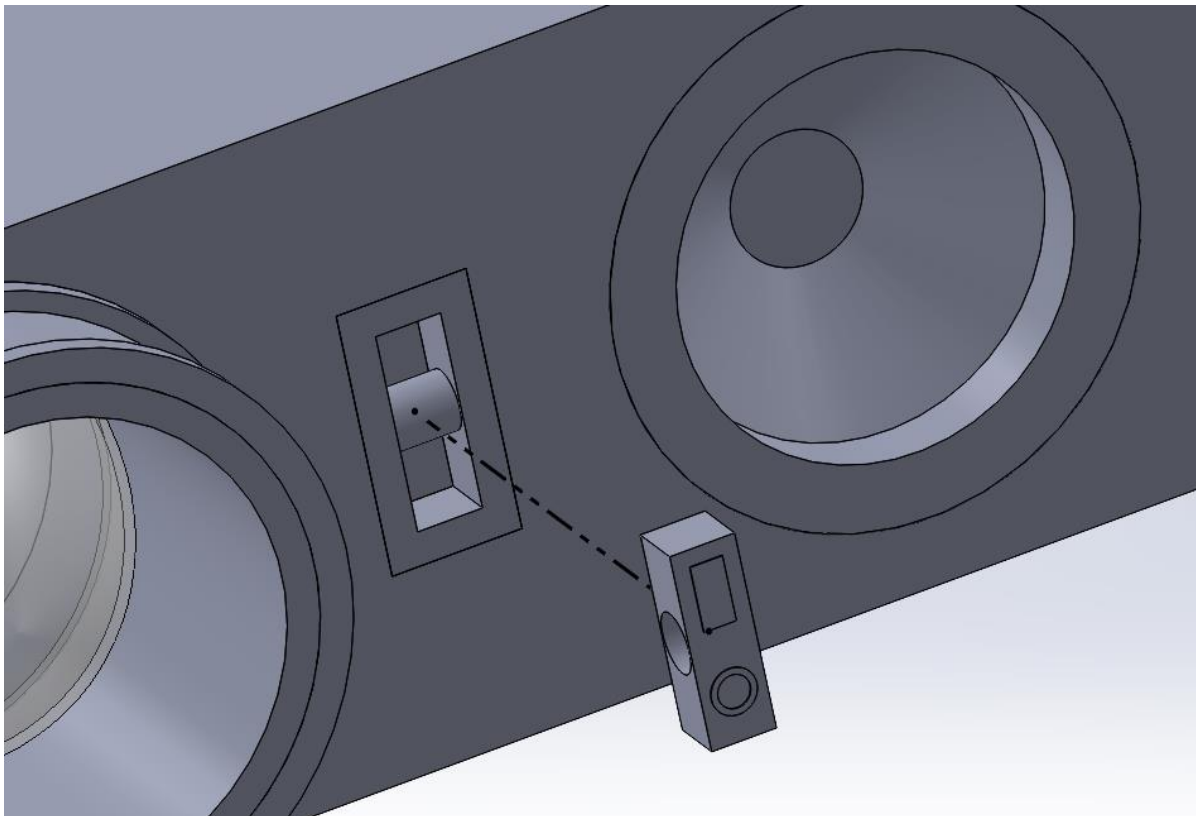
Step 4: Push the projector lens adjuster onto the projector lens part until it cannot be pushed further back, it will fit with friction.



Step 5: Glue the switch case onto the 2nd hole from the right of the projector box.



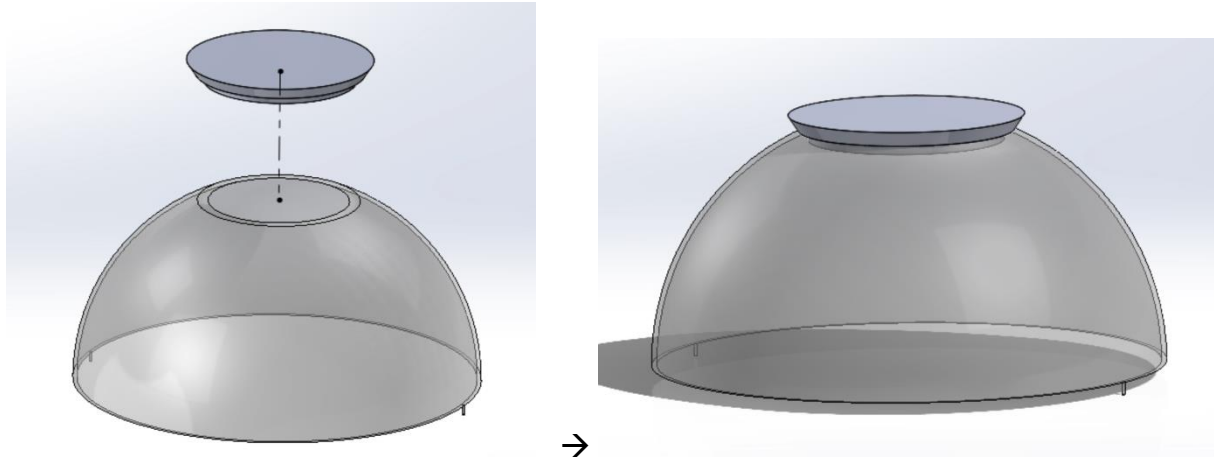
Step 6: Connect the switch to the switch casing, the cylinder in the middle will be a flexible rubber, which can be pushed to the side to allow the switch to fit in.



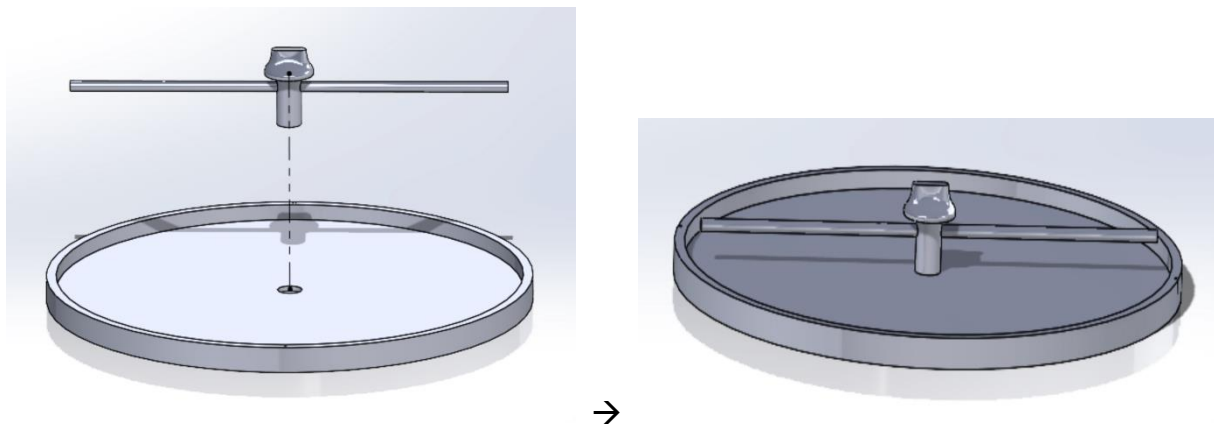
Popcorn Maker Assembly

Parts List
• Glass Dome • Lid • Stirring Rod • Hot Plate • Tray

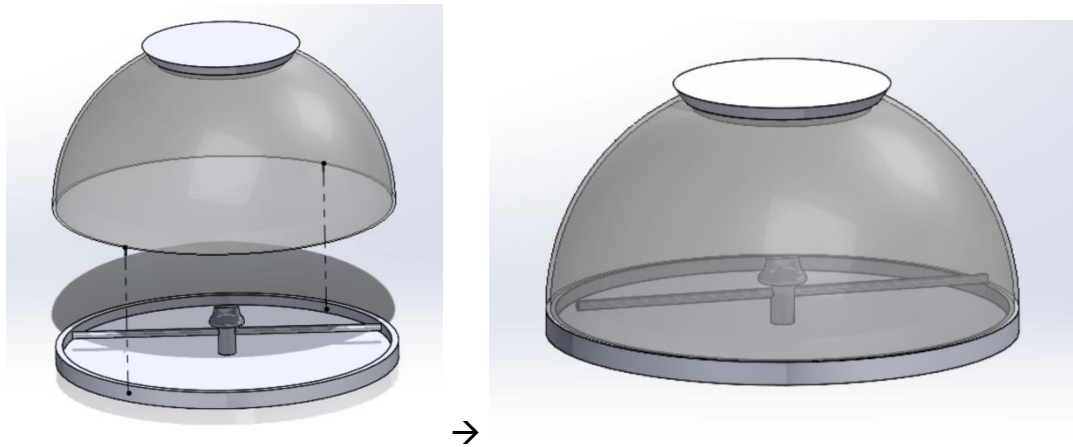
Step 1: Take the Lid and insert it into the top opening of the Glass Dome.



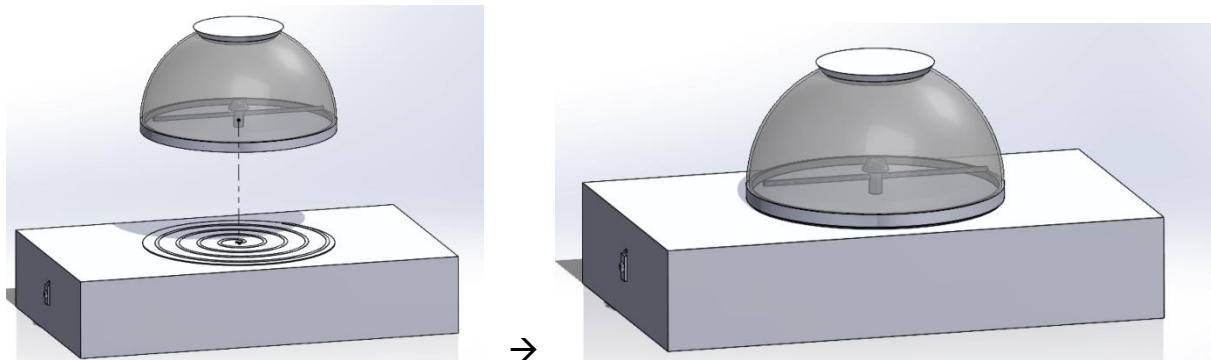
Step 2: Take the Stirring Rod and insert it into the hole in the center of the Tray. The Tray and the Stirring Rod magnetically attach to allow rotation of the Rod.



Step 3: Align the holes of the Tray with the pegs of the Glass Dome and insert.



Step 4: Align the assembled pieces with the center of the Hot Plate. These parts sit on top of one another, so they do not lock into place.



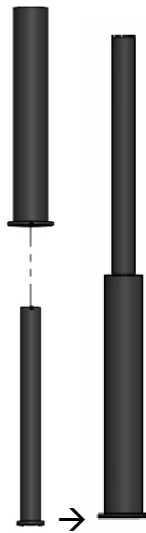
Screen Assembly

Parts List

- Fabric Screen
- Legs (x2)
 - Top rung
 - Middle rung
 - Base rung
- Carabiner (x4)
 - Carabiner Body
 - Carabiner Stint Long
 - Carabiner Screw Short
- Ground Hooks (x2)
- String (not modeled)

1. Prepare the leg rungs

- (1) Pull the top rung through the middle rung until the ledge portion meets the top of the middle rungs



- (2) Then slide the middle rung (with the top rung still attached) through the bottom of the base rung until fully extended (about 60 inches)



- (3) Attach the bottom face of the bottom rung, so that all rungs remain attached and secure

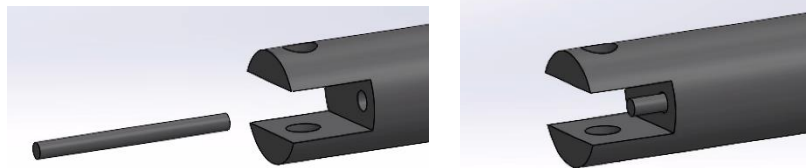


- (4) Collapse the rungs to their smallest height (about 20 inches) and insert into bag for storage

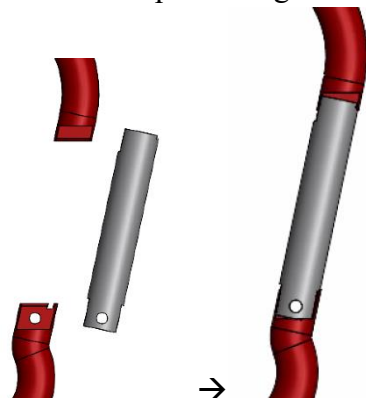


2. Build the carabiner

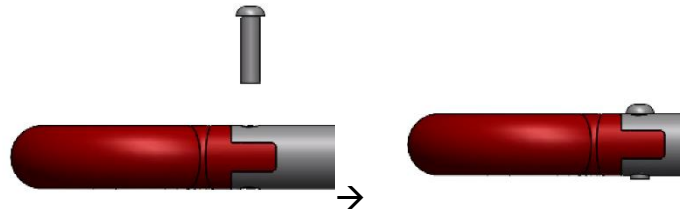
(1) Insert the long clip screw into the hole cutout in the clip arm



(2) Aligning the clip arm to the angled portion of the carabiner body. Slide the arm onto to line up with edge

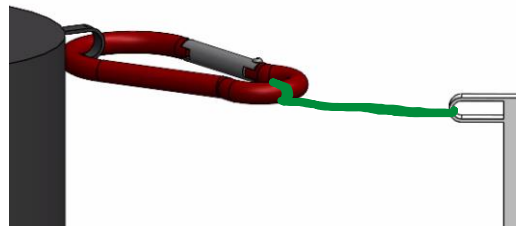


(3) Insert short screw into hole to secure all parts together



3. Open Screen Assemblage

- (1) Extend the legs to their full length
- (2) Attach the screen to the legs by looping the provided string and carabiner through the hoops on each corner of the screen and legs. Pull each corner of the screen taut with the string to ensure the best viewing experience (string indicated by green line)



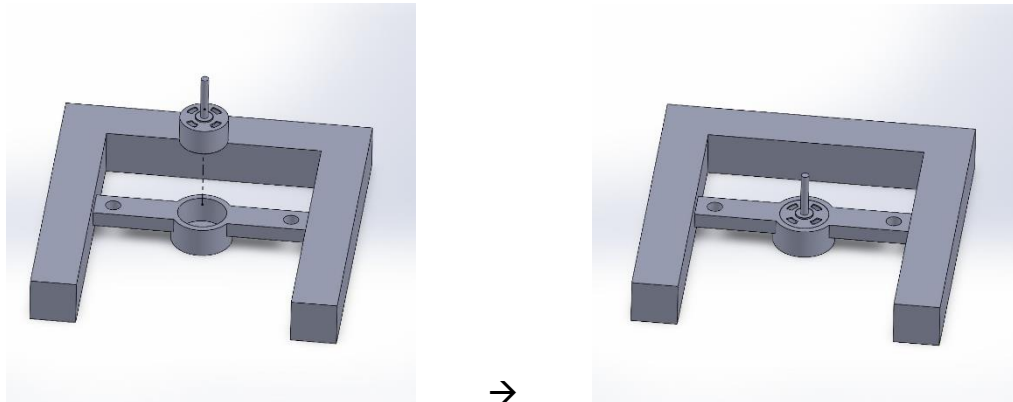
- (3) (For extra support) Secure the ground stakes into the ground and use extra string to stretch from the top corner of the screen to the rounded portion of the ground hook

Drone Assembly

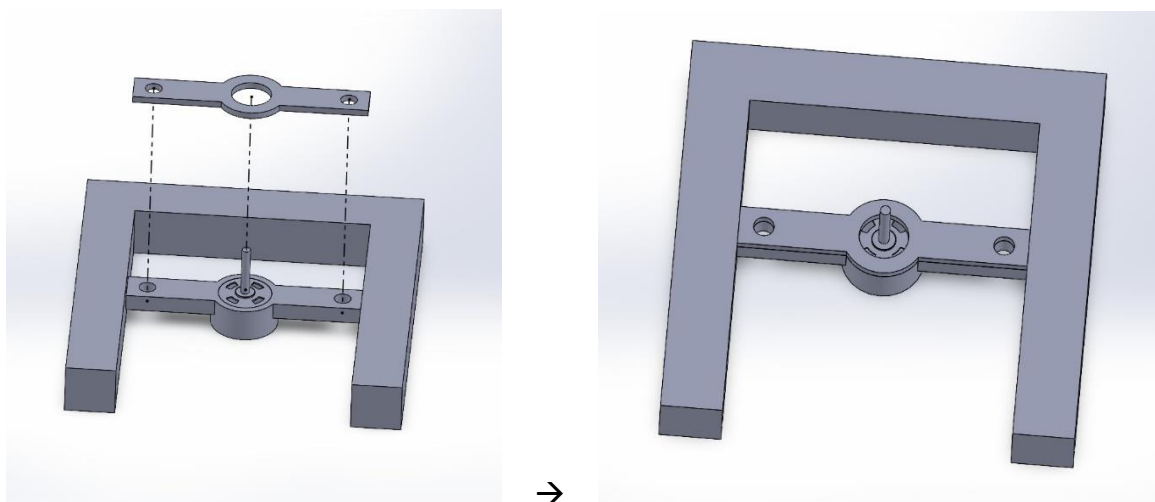
Parts List

- Drone Base
- Fan Base (x4)
- Motor (x4)
- Motor Cover (x4)
- 94035A178_18-8 Stainless Steel Precision Shoulder Screw (x8)
- 97367A112_316 Stainless Steel Thin Nylon-Insert Locknut (x8)
- Fan (x4)
- Fan Cap (x4)

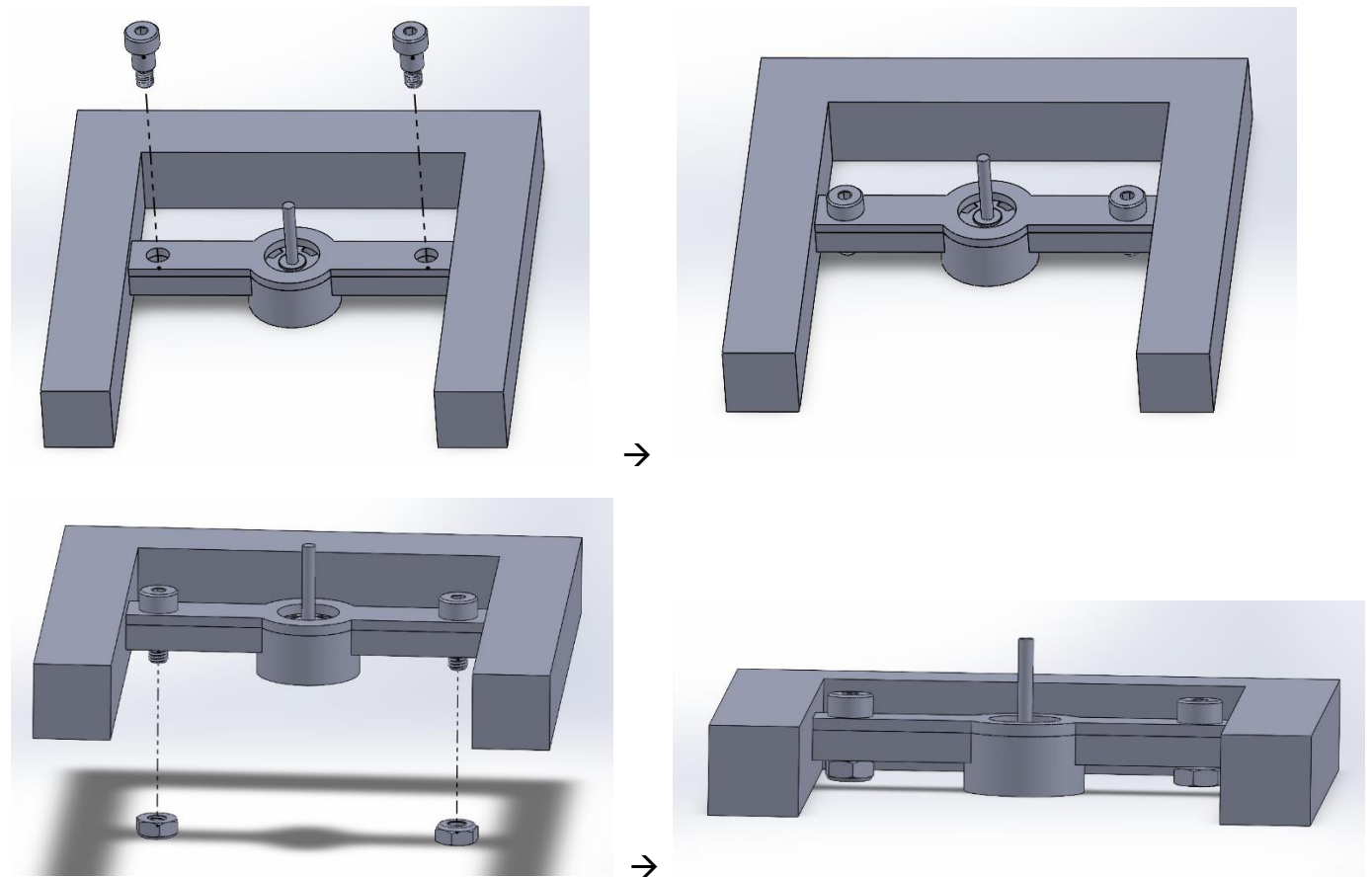
Step 1. Take one Motor with the rod side up into the motor insert in one Fan Base.



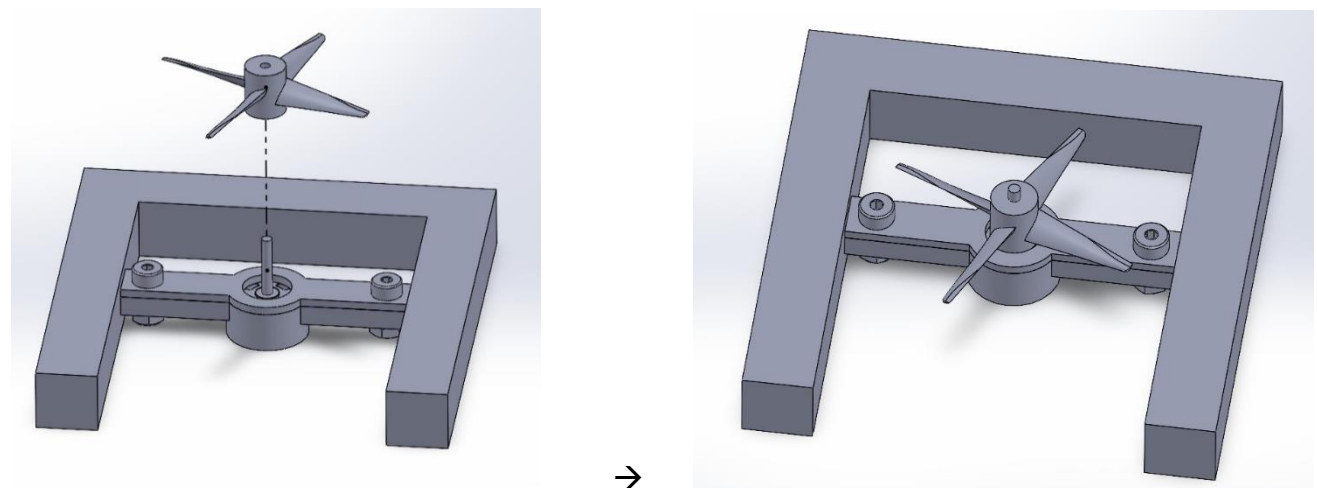
Step 2. Place one Motor Cover over the middle part of the Fan Base so that the holes in the Motor Cover and the Fan Base align.



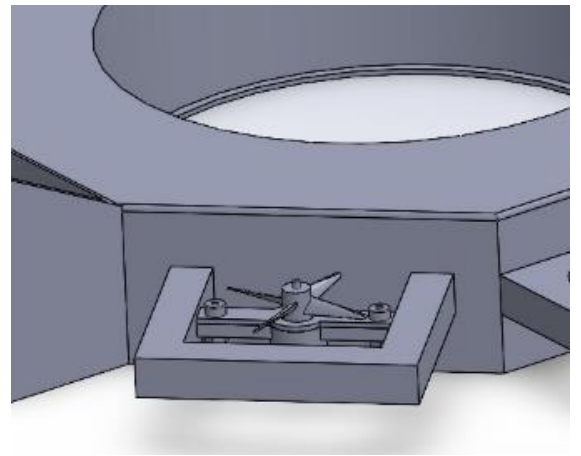
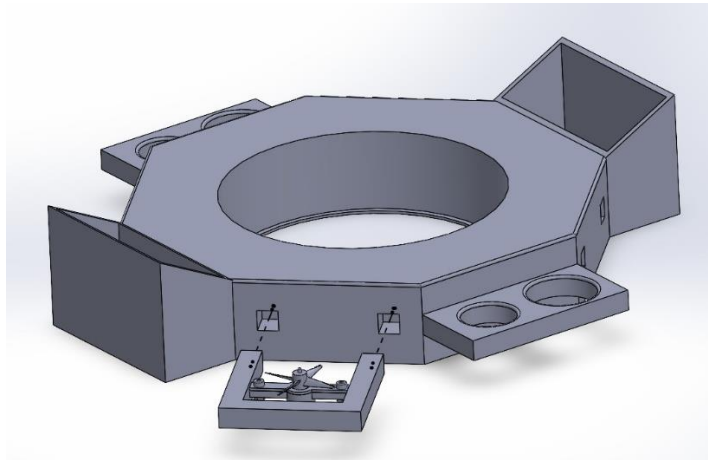
Step 3. Take two Screws and place them in the left and right holes in the Fan Base so that the top of each screw is on top of the Motor Cover. Then take two Nuts and fasten one to the bottom of each Screw until the Motor Cover cannot move.



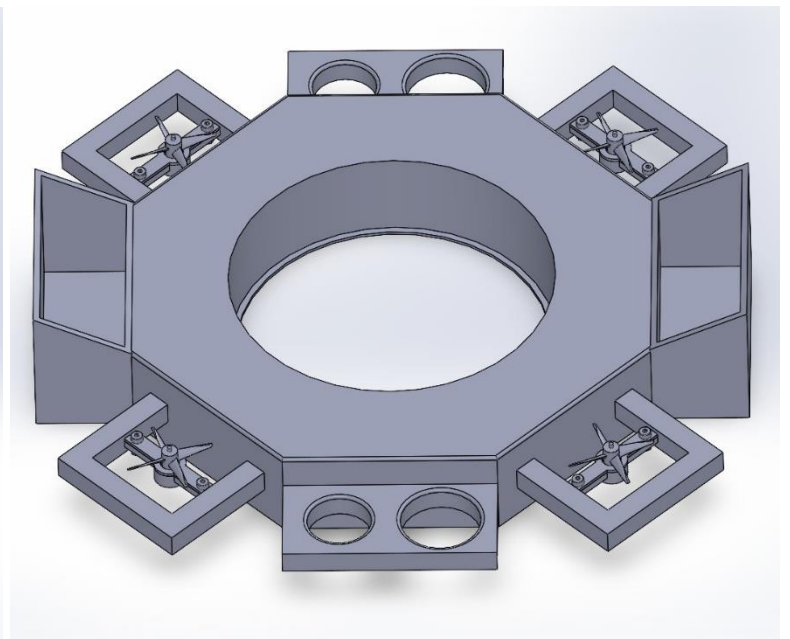
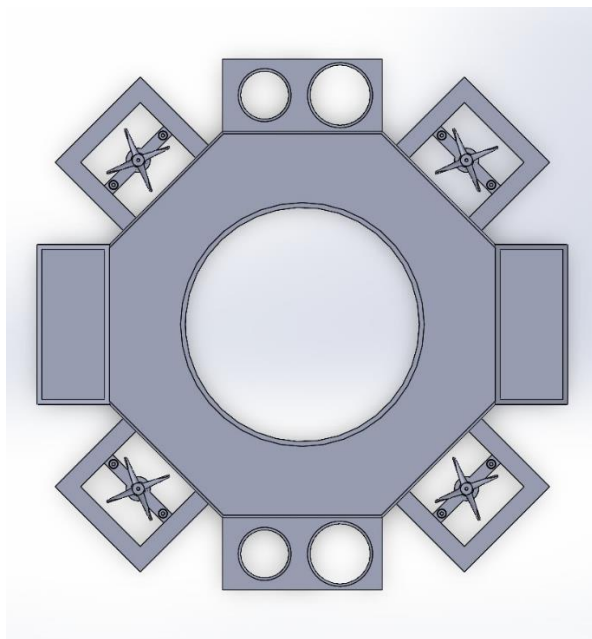
Step 4. Take the Fan and line the hole in the center with the Motor's rod. Slide the fan down until the fan rests on the motor with the blades orientated upwards. The fan should be free to spin.



Step 5: Take the whole Fan Assembly and align the ends of the Fan Base with one of the pairs of square holes in the Drone Base. Slide the Fan Assembly all the way in.



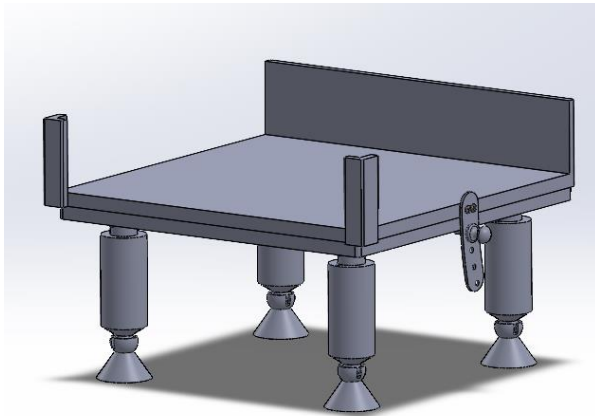
Step 6: Repeat steps 1 through 5 to make the other 3 Fan Assemblies and attach them to the Drone Base.



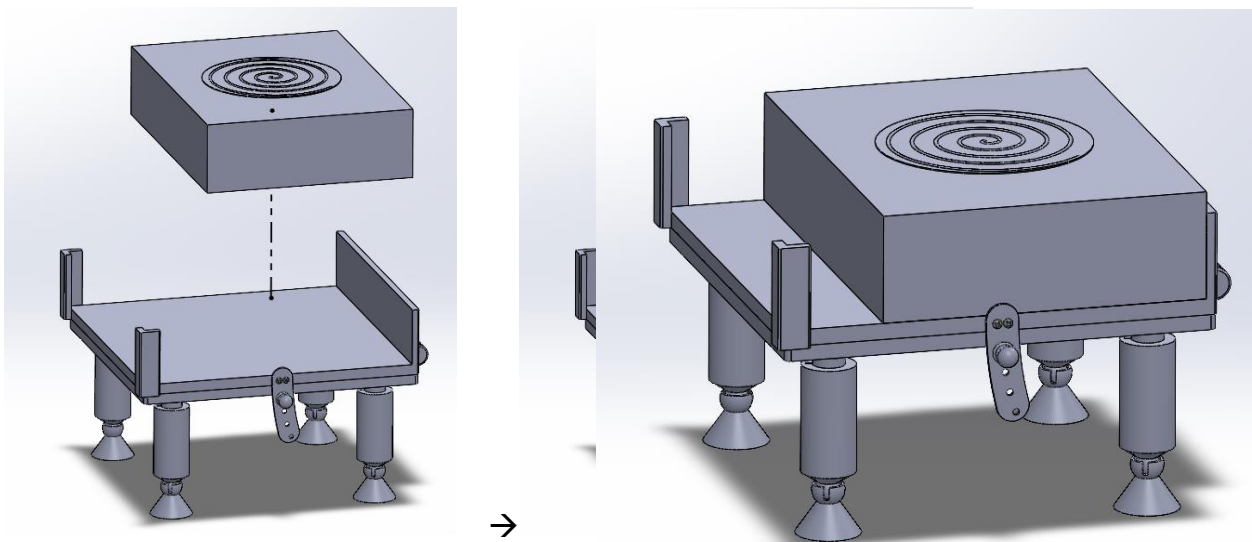
Full Assembly

Parts List
• Base Assembly • Projector Assembly • Popcorn Maker Assembly • Screen Assembly • Drone Assembly

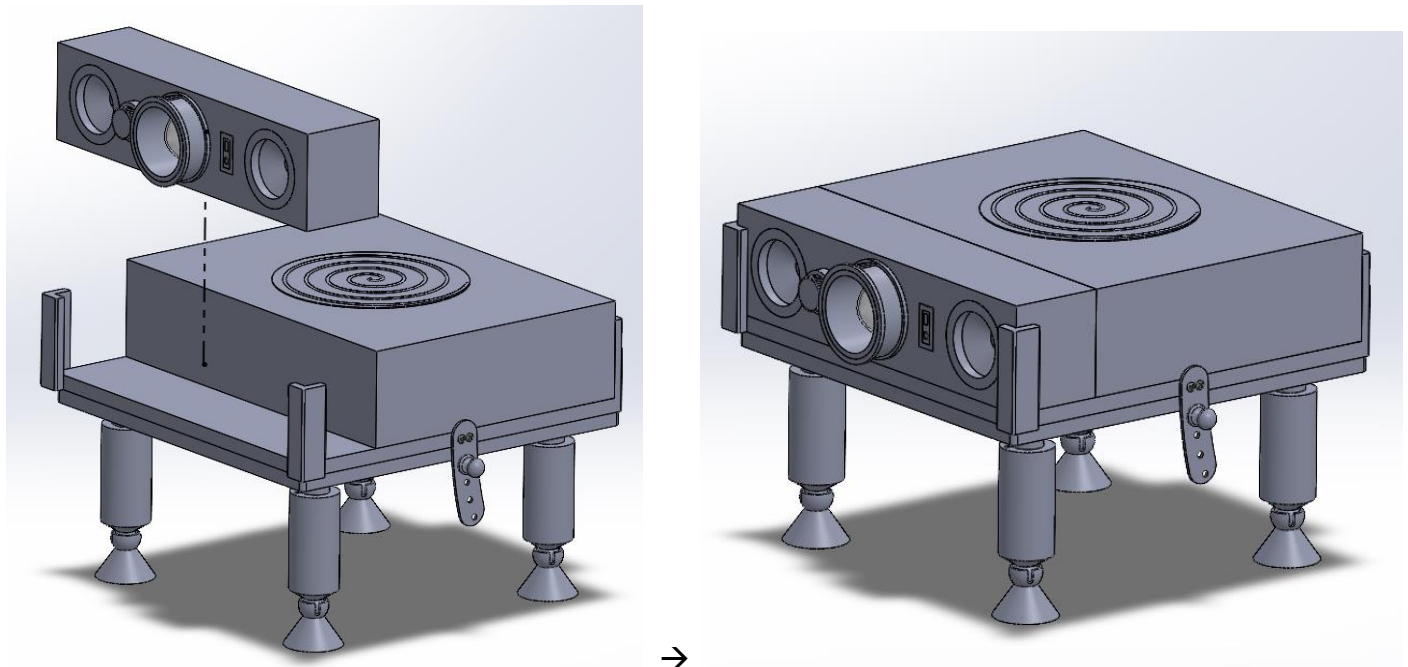
Step 1: Start with the Base level on the ground.



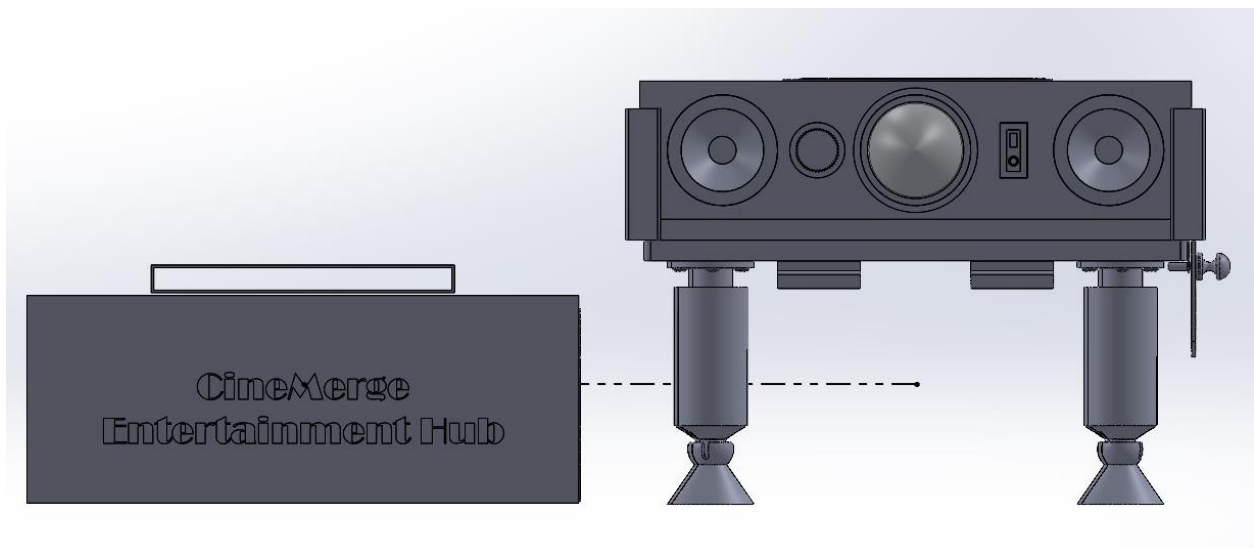
Step 2: Place the Popcorn Maker with just the Hot Plate inside of the top of the Base so that the back of the Hot Plate is flush with the back of the Base.

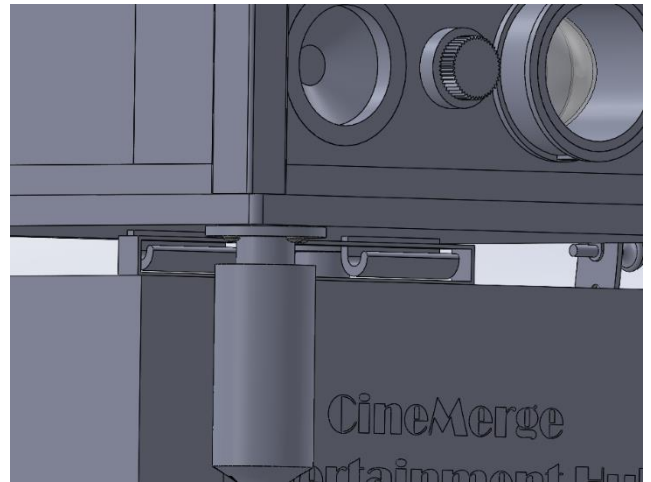


Step 3: Place the Projector inside the top of the base in front of the Popcorn Maker with the projector lens facing the front of the Base.

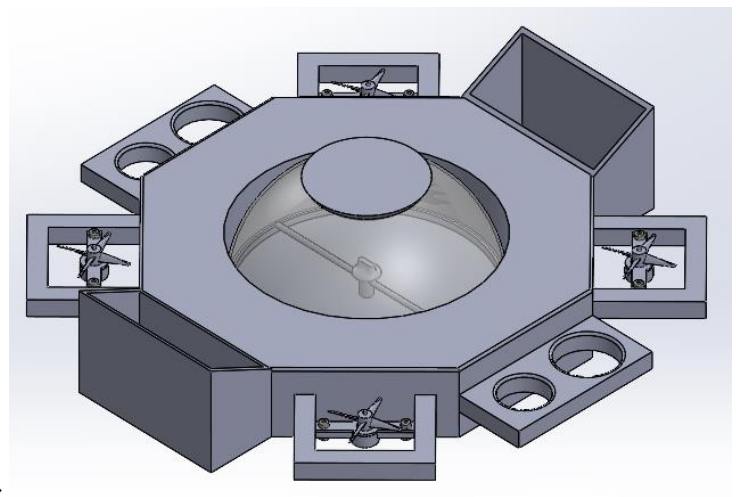
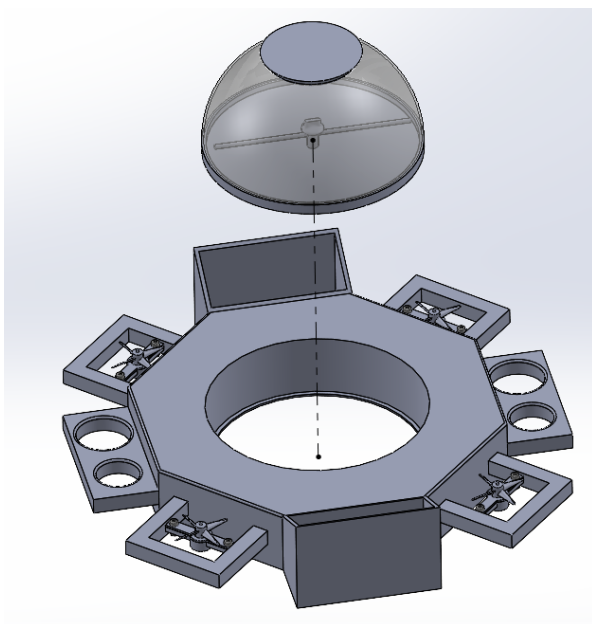


Step 4: Attach the Screen assembly (with parts inside) to the bottom of the Base, using the hooks on the very bottom of the Base.

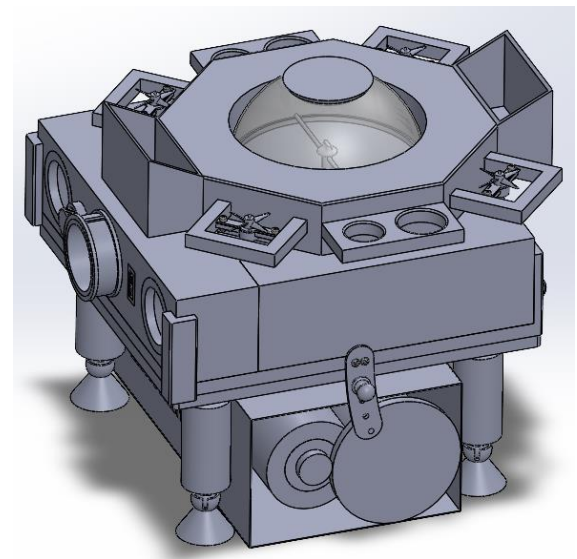
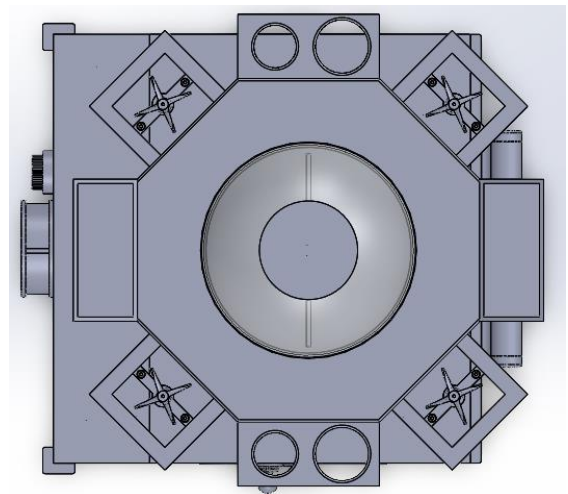
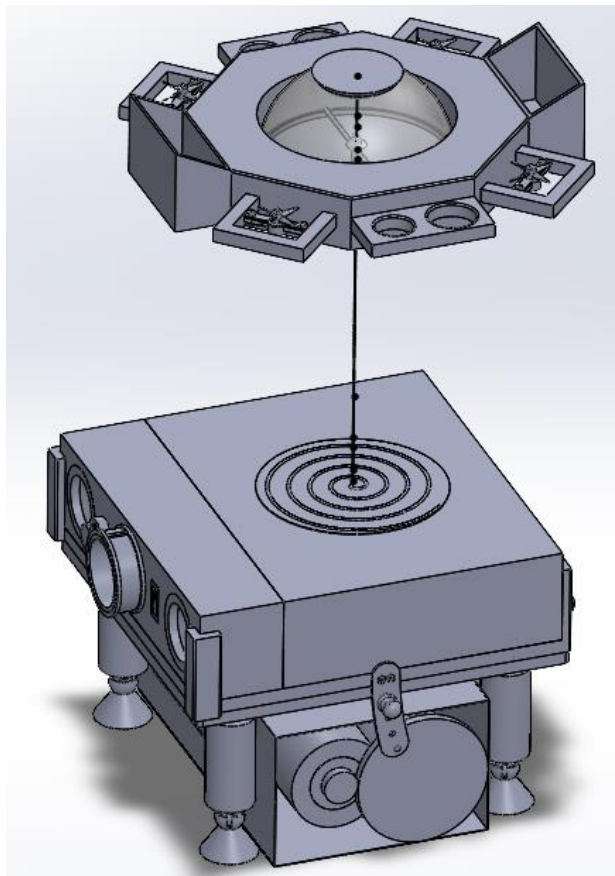




Step 5: Take the remaining parts of the Popcorn Maker and put them inside the circular hole in the Drone.



Step 6: Then, put the Drone plus these parts over the Hot Plate so that the hole in the drone is centered on the Hot Plate.



Renderings







